

Conduct and Scale Economies: Evaluating Tariffs in the US Automobile Market

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Abstract

We develop a data-driven procedure for testing firm conduct under non-constant marginal costs and apply it to the US automobile market. The data select Cournot quantity-setting with economies of scale as the best-fitting supply model; we estimate that a 10% production increase lowers marginal cost by 1.3%. We then evaluate medium-run automobile tariffs in a 2018 counterfactual, accounting for global value chains. The model implies that 25% tariffs on imported cars raise foreign-assembled prices by 26.2% while lowering US-assembled prices by 1.0%; adding parts tariffs raises US-assembled prices by 6.6% and raises total surplus losses from \$18.6 to \$28.4 billion.

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1 Introduction

In the last decade, trade policy has reemerged as a dominant form of economic strategy, with tariffs now serving as its primary instrument. In the US, both Trump administrations have proposed and enacted tariffs on imported goods, prompting retaliatory measures that further complicate international commerce. Analyzing these interventions requires understanding how tariffs propagate through complex global value chains ([Antràs and Chor, 2022](#)), particularly in industries where components may cross borders multiple times before reaching consumers.

The automobile industry exemplifies this complexity. Targeted for tariff intervention by both Trump administrations, this sector features intricate production networks crossing national borders. Foreign components constitute over 40% of US-assembled vehicles, while US-made parts feature prominently in foreign-made cars. This web of interdependencies, fostered by trade agreements like the North American Free Trade Agreement (NAFTA) and its successor, the United States-Mexico-Canada Agreement (USMCA), creates layered effects when tariffs are applied. In particular, both Trump administrations have threatened stacked tariffs (tariffs applied to both final goods and intermediate inputs) on foreign-assembled cars and foreign car parts used in domestic assembly, magnifying potential impacts throughout the supply chain. The stated goals of such policies often involve reshaping the geography of production to boost domestic employment. However, the high degree of associated policy uncertainty tends to slow down investment (see, e.g., [Caldara, Iacoviello, Molligo, Prestipino, and Raffo, 2020](#); [Handley and Limão, 2022](#)), reducing firms' ability to relocate production. In turn, this makes it useful to assess the medium-run (i.e., a one-year time frame) effects of tariffs on consumers and firms while taking as given the locations of production and market structure.

Such evaluation requires a methodological framework that accounts for two critical but often neglected features of industry structure. First, the literature has long recognized that firm conduct matters significantly for how trade policy affects market outcomes ([Dixit, 1984](#); [Brander and Spencer, 1985](#); [Dixit and Grossman, 1986](#)),¹ as the degree of strategic substitutability or complementarity guides domestic firms' responses to foreign competitors' tariff-induced price changes. Second, cost structures, particularly returns to scale, shape these responses (see, e.g., [Antràs, Fort, Gutiérrez, and Tintelnot, 2024](#)), potentially allowing domestic producers to reduce their costs as they gain market share following tariff implementation. Crucially, neither conduct nor cost structure is known a priori by researchers or policymakers, calling for a data-driven approach to accurately predict the effects of policy.

¹See [Head and Spencer \(2017\)](#) for a discussion of how oligopoly models have been used in the international trade literature over the last few decades. [Bian, Head, and Orr \(2026\)](#) show in simulations that oligopoly conduct affects optimal trade policy.

We address these challenges by developing a new empirical framework and applying it to the US automobile market using a uniquely comprehensive dataset. Building on a long line of studies that use industrial organization (IO) tools to study industry-specific trade policy (e.g., [Goldberg, 1995](#); [Berry, Levinsohn, and Pakes, 1999](#)), our approach makes two key contributions that underpin the empirical analysis.

First, we assemble novel data that not only records equilibrium outcomes but also tracks both country-model level production and the assembly location and parts origin of every vehicle model sold in the US market. This granular information on global value chains allows us to precisely model how stacked tariffs cascade through production networks, affecting prices and welfare.

Second, we develop a general econometric procedure for testing firm conduct under non-constant marginal costs. Existing methods for testing conduct ([Backus, Conlon, and Sinkinson, 2021](#); [Duarte, Magnolfi, Sølvesten, and Sullivan, 2024](#)) assume constant marginal costs. Our procedure accommodates non-constant marginal costs in the [Rivers and Vuong \(2002\)](#) framework,² and is portable to any differentiated products industry where conduct and cost structure are empirically relevant, including, e.g., pharmaceuticals, airlines, and other manufacturing sectors characterized by economies of scale. We also provide results on instrument relevance for testing conduct under non-constant costs. [Berry and Haile \(2014\)](#) show that, in differentiated products markets, combining sources of variation that operate through different economic channels can falsify models of conduct. Building on the framework in [Dearing, Magnolfi, Quint, Sullivan, and Waldfogel \(2024\)](#), we characterize the precise ways in which instruments must be *economically distinct*, and therefore able to both test models of conduct and estimate the cost structure, in parametric settings like our application. The cross-firm and cross-product variation in product characteristics and cost shifters highlighted by [Berry and Haile \(2014\)](#) naturally provides economically distinct instruments, making our testing procedure feasible with standard industry data.

We apply this framework to the US automobile market, leveraging state-of-the-art demand estimates from [Grieco, Murry, and Yurukoglu \(2024\)](#) that credibly capture consumers’ substitution patterns. We find that Nash-Cournot quantity-setting competition (hereafter Cournot) with substantial economies of scale best characterizes the US automobile market. The finding that Cournot best fits the data aligns with institutional features of the industry, where production targets are set well in advance and used in negotiations with suppliers. In terms of scale economies, we estimate that a 10% increase in production reduces marginal costs by about 1.3%. This estimate is somewhat larger than most prior estimates and has a meaningful effect on our counterfactual predictions.

When we evaluate tariff scenarios under the conduct model and scale economies selected

²[Bian et al. \(2026\)](#) propose an alternative testing approach, also under constant marginal costs.

by our framework, we find striking results that underscore the importance of both accounting for global value chains and correctly modeling conduct and costs. Specifically, in a medium-term equilibrium of the US market, holding production locations and the product set fixed, we perform three counterfactuals in the 2018 US car market where we levy a 25% tariff on (i) imported cars alone, (ii) imported cars and parts, and (iii) cars and parts imported to the US as well as parts exported from the US for foreign assembly. With car-only tariffs, foreign-assembled car prices increase by 26.2% while US-assembled car prices *decrease* by 1.0%, reflecting economies of scale and strategic substitution under Cournot competition. Adding tariffs on imported parts fundamentally reverses these patterns: US-assembled cars, which initially benefited from car-only tariffs, see prices rise by 6.6% rather than fall, as imported components constitute a large share of their production cost. Reciprocal (retaliatory) tariffs levied by foreign countries on parts produced in the US lead to further price increases, driven by higher marginal costs for foreign manufacturers.

Under our preferred model, pass-through of tariffs to foreign car prices exceeds one, broadly consistent with the empirical literature on the 2018 Trump administration tariffs (Amiti, Redding, and Weinstein, 2019; Fajgelbaum, Goldberg, Kennedy, and Khandelwal, 2020; Flaaen, Hortaçsu, and Tintelnot, 2020; Cavallo, Gopinath, Neiman, and Tang, 2021). Welfare losses are \$18.6 billion with car-only tariffs, and sharply increase to \$28.4 billion when tariffs on foreign parts are added. One mitigating factor is the potential for increased domestic employment. Under the car-only tariff, these gains come at a welfare cost of about \$172,000 per job, roughly twice the sector’s average annual compensation; under stacked tariffs the cost is far higher, roughly \$717,000 to \$761,000 per job, in line with previous estimates of the cost of policy-created jobs in US manufacturing (Hufbauer and Lowry, 2012; Flaaen et al., 2020; Montag, 2026). Many of these predictions depend critically on the supply model. The standard specification in the literature (Bertrand with constant marginal cost) predicts sub-unit pass-through for foreign cars and, crucially, *positive* pass-through to domestic car prices. Similarly, it predicts that parts manufacturing employment *declines* rather than increases under stacked tariffs, reversing the sign of the effect.

The United States enacted automobile tariffs in 2025–2026. While the enacted policy differs from our counterfactual experiment and the market has likely not yet reached a medium-run equilibrium, we can compare our predictions against realized outcomes. Manufacturers’ stock prices moved sharply in the direction the model implies when the tariffs were announced, and import quantities fell by much of the predicted amount. Retail prices, however, rose far less than our stylized counterfactual predicts. Recomputing that counterfactual under the considerably milder policy as enacted closes most of the gap. The remainder is consistent with firms, restrained by forces outside the model (e.g., political pressure or the expectation of further negotiated cuts), absorbing the tariff in the short run

rather than pricing in line with the model’s medium-run response.

Our study is related to an important paper by [Head and Mayer \(2019\)](#), which also analyzes counterfactual tariffs in the US auto market. Our approach differs in two key dimensions, offering complementary evaluations of this policy. First, while they study long-run effects allowing for global production reallocation and product entry or exit, we examine medium-term impacts where market outcomes adjust but production locations, product portfolios, and parts sourcing remain fixed. Second, while they model the global industry structure, we focus exclusively on the US market with granular data on parts content, enabling a detailed account of how stacked tariffs propagate through value chains.

Our medium-term approach follows a broader empirical literature that evaluates trade policy in the automobile industry using IO methods; see [Van Biesebroeck and Verboven \(2026\)](#) for a comprehensive survey. This literature includes analyses of voluntary export restraints ([Goldberg, 1995](#); [Berry et al., 1999](#)), tariff reductions ([Fershtman, Gandal, and Markovich, 1999](#); [Tovar, 2012](#)), and related questions of international price discrimination and home-market advantage ([Verboven, 1996](#); [Goldberg and Verboven, 2001](#); [Coşar, Grieco, Li, and Tintelnot, 2018](#)). We contribute to this literature by pursuing greater flexibility on the supply side, specifically addressing questions of firm conduct and cost structure raised by earlier pioneering studies such as [Feenstra and Levinsohn \(1995\)](#) and [Verboven \(1996\)](#). Rather than assuming a particular form of competition, we develop new methods to test alternative models of conduct under non-constant marginal costs.

Other approaches to incorporating variable markups or increasing returns in trade models work at a higher level of aggregation; for instance, [Atkeson and Burstein \(2008\)](#) introduce oligopolistic markups at the sector level, while [Bartelme, Costinot, Donaldson, and Rodríguez-Clare \(2025\)](#) estimate external economies of scale across sectors. Our approach is complementary: by working at the product level with a fine-grained demand system, we can test rather than assume the model of conduct and estimate firm-level returns to scale, both of which we show to be quantitatively important for policy predictions.

The paper proceeds as follows. In the next section, we present facts illustrating the dependence of the US auto market on global value chains and our data sources. In [Section 3](#), we introduce our empirical model of demand and supply. In [Sections 4 and 5](#), we develop a general method for distinguishing models of conduct allowing for non-constant marginal costs and implement our testing procedure in the auto industry. [Section 6](#) presents our counterfactual analysis of tariff policies. [Section 7](#) concludes.

2 Data

We bring together three data sources. To estimate demand and test firm conduct, we require data on equilibrium market outcomes and on the scale of production; to evaluate tariffs after estimation, we additionally require, for each car model, its country of assembly and the origin of its parts.

For equilibrium outcomes in the US automobile market and car model characteristics, we rely on the dataset constructed by [Grieco et al. \(2024\)](#). The dataset contains 5,046 car model-year observations from 2002 to 2018 and includes information on the manufacturer’s suggested retail price (MSRP), which we refer to throughout as price,³ sales, and product characteristics, including vehicle dimensions (height, width, and length), curb weight, horsepower, and fuel efficiency measured as miles per dollar (MPD) and miles per gallon (MPG). We also observe the number of trim levels available for each model,⁴ the number of years since the current design was introduced, and the real exchange rate (RXR) for the country of production. The largest vehicle segments in our data are sedans (54% of observations), sport utility vehicles (34%), trucks (6%), and vans (6%).

To capture economies of scale more comprehensively than previous studies of this market, we need reliable measures of production in car plants that serve the US market. We obtain annual production data at the car model-country level from MarkLines, an automotive industry data provider, from 2002 onward.

To perform our counterfactuals and understand how stacked tariffs would propagate through automotive value chains, we use data collected by the National Highway Traffic Safety Administration (NHTSA) under the American Automobile Labeling Act (AALA), a source also used by [Klier and Rubenstein \(2007\)](#) and [Head, Mayer, and Melitz \(2024\)](#). The AALA requires vehicle window stickers to display assembly location and parts sourcing information, including the percentage of parts value produced domestically versus abroad. Because the NHTSA treats US and Canadian parts as domestic, we manually separate their contributions for each car model produced in 2018. We do so by identifying all Canadian plants manufacturing transmissions and engines, the two highest-value automotive parts, and tracing which car models use their output. We merge the AALA data for 2018 to the [Grieco et al. \(2024\)](#) and MarkLines datasets; the final sample contains 3,929 car model-year observations, matched from 5,046 observations in [Grieco et al. \(2024\)](#). The unmatched

³Dealer-negotiated transaction prices typically fall below MSRP, and discounts could vary systematically across segments or firms, through price discrimination tied to foreign competition or differences in dealer-network density. Simulations in Appendix F.8 suggest that our test results are robust to moderate discounts of these kinds.

⁴Car models may have multiple “trim levels.” For example, the 2018 Toyota Camry was available in L, LE, SE, XLE, and XSE trims, with progressively more options and higher prices. The [Grieco et al. \(2024\)](#) dataset aggregates observations across trims.

models have relatively small market shares and cumulatively account for roughly 10% of US car sales, on average, across years.⁵

TABLE 1: Summary Statistics

	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
Prices (thousands \$)	39	16	13	27	46	100
Sales (thousands)	60	91	0.01	8.3	74	891
RXR	0.98	0.17	0.39	0.9	1.1	1.6
Height (inches)	64	8.1	45	57	69	107
Width (inches)	74	4	61	71	76	89
Length (inches)	190	18	106	179	200	274
Curb weight (lbs)	3,966	900	1,808	3,362	4,456	7,230
Horsepower	240	82	66	173	296	645
MPD	7.6	3.4	2.8	5.3	8.9	28
MPG	21	7.8	10	17	23	50
Number of trims	6.7	14	1	2	7	210
Years since design	4.2	1.9	3	3	5	12
Production by car model (thousands) - US made	134	162	0.67	26	163	1,023
Production by car model (thousands) - Foreign made	83	97	0.11	13	118	579
Trade exposure from automotive labeling (2018)						
Domestic content, US-assembled (% of value)	50.3	13.8	1.0	43.0	60.0	75.0
Domestic content, foreign-assembled (% of value)	8.5	15.4	0.0	0.0	9.0	70.0
Number of car models per category			2002	2007	2013	2018
Sedan			51	89	146	150
SUV			39	82	94	106
Truck			17	18	12	10
Van			12	17	12	15

The table reports summary statistics for our new car sales sample with 3,929 car model-year observations from 2002–2018. RXR is the real exchange rate for the country of production. MPD is miles per dollar, calculated using contemporaneous local gas prices. MPG is miles per gallon. Domestic content is the share of parts value sourced from the United States after separating Canadian content (engines and transmissions).

Table 1 presents summary statistics, which point to two features of the data that shape our modeling approach. The first is the scale of production. Annual sales range from 8,300 units at the 25th percentile to 74,000 at the 75th, and model-level production volumes vary just as widely: US-assembled models range from 26,000 units at the 25th percentile to 163,000 at the 75th, and foreign-assembled models from 13,000 to 118,000. Because economies of scale could be an essential determinant of production costs, we must account for scale when modeling firm behavior and predicting the effects of trade policy.

The second is product differentiation. Prices average \$39,000 with wide variation that likely reflects not only cost differences but differentiation across many dimensions: horse-

⁵Our conduct test results are robust to using the full sample under constant marginal cost; see Table 25 in Appendix F.7.

power ranges from 173 to 296 between the 25th and 75th percentiles, and fuel economy from 17 to 23 miles per gallon. The average model offers seven trim levels, and some popular models are available in more than 20 configurations. The set of competing models also shifts over time, with the sport utility vehicle (SUV) category growing sharply in both models offered and sales while the truck category contracts. This multidimensional differentiation can confer market power on manufacturers within their products’ segments.

Table 1 also shows that foreign production is substantial and uneven; foreign-assembled vehicles make up 39.5% of US sales and parts sourcing varies widely across models. Given this exposure, together with the variation in scale and product differentiation in our dataset, different assumptions on cost functions and conduct, for example whether firms compete on prices or quantities, can lead to very different predictions about pass-through, production reallocation, and consumer welfare after tariffs are levied. Our empirical approach therefore develops a testing procedure that evaluates alternative models of supply rather than imposing one in advance. To learn cost and conduct, we rely on variation in the [Grieco et al. \(2024\)](#) and MarkLines data; the data on assembly location and parts sourcing from the AALA then enable the counterfactual exercises in Section 6.

3 Model

To capture the medium-run effects of tariffs on the US automobile market, we develop a static equilibrium model of the industry. We discuss demand and supply in turn.

3.1 Demand

We use the state-of-the-art demand system estimated in [Grieco et al. \(2024\)](#), which builds on seminal work on the auto industry (e.g., [Berry, Levinsohn, and Pakes, 1995](#); [Goldberg, 1995](#)). The model allows for flexible substitution patterns, informed by detailed microdata on consumer choices and second-choice survey responses.⁶ We provide a brief summary here, which allows us to develop the notation used throughout the paper. In what follows, for a generic variable a_{jt} , a_t denotes a'_{jt} stacked across products j within market t and a denotes a_t stacked across markets t .

Firms offer a set of products $\mathcal{G}_t$ in each market t . Consumers in market t , indexed by i receive indirect utility from each new car model j according to:

$$u_{ijt} = \alpha_{it}p_{jt} + x'_{jt}\beta_{it} + \xi_{jt} + \epsilon_{ijt}.$$

⁶[Head and Mayer \(2026\)](#) find that CES and monopolistic competition can approximate a DGP with BLP demand, Bertrand conduct, and constant cost when the implied pass-through is close to one. In this paper, we want to allow for flexible substitution patterns under alternative supply models.

Here, consumer utility depends on the product’s price p_{jt} , a vector of observed car characteristics x_{jt} , an unobserved car attribute ξ_{jt} and an idiosyncratic shock ϵ_{ijt} which is assumed to follow a Type 1 extreme value distribution. The model permits rich and time-varying consumer heterogeneity by allowing the preference parameters α_{it} and β_{it} to depend on observed demographics and consumer-level shocks. To concisely describe this specification and report the demand estimates we use throughout, we reproduce Table IV of [Grieco et al. \(2024\)](#) in Appendix D.1. Importantly, α_i contains both consumer-level income and income squared. This flexibility in the mixing distribution helps alleviate the curvature concerns raised by [Birchall, Mohapatra, and Verboven \(2024\)](#) and [Miravete, Seim, and Thurk \(2025\)](#).

Consumers maximize their utility by choosing either a single car model or the outside option of no new car purchase, with the utility of the latter varying over time (η_t will capture the average utility of the outside good). The market share of each product in each market s_{jt} takes on the familiar form:

$$s_{jt} = \int \frac{\exp(\alpha_{it}p_{jt} + x'_{jt}\beta_{it} + \xi_{jt})}{\exp \eta_t + \sum_{k \in \mathcal{G}_t} \exp(\alpha_{it}p_{kt} + x'_{kt}\beta_{it} + \xi_{kt})} dF_t(i),$$

where F_t is the distribution of the random coefficients. Thus market shares stacked across products in market t can be expressed as the function $s_t = \mathfrak{s}(p_t, x_t, \xi_t, \theta_0^D)$ where θ_0^D is the vector of demand parameters. For a market t of size M_t , the equilibrium quantity of model j is given as $q_{jt} = M_t s_{jt}$.

This demand specification produces realistic substitution patterns, with similar vehicle styles serving as close substitutes and strong correlations between purchased and second-choice vehicles in characteristics such as size, horsepower, and fuel economy. Given the rigorous estimation strategy developed in [Grieco et al. \(2024\)](#) and their extensive microdata, we adopt their estimate of θ_0^D without modification. Because we do not observe their microdata, our inference cannot accurately account for the joint uncertainty in their estimates and ours. Instead, we treat their estimate as fixed.⁷

3.2 Supply

The trade literature has shown that assumptions on the supply side—the imposed model of conduct (see e.g., [Dixit, 1984](#); [Brander and Spencer, 1985](#); [Dixit and Grossman, 1986](#); [Eaton and Grossman, 1986](#)) and functional form of marginal costs (see e.g., [Antràs et al., 2024](#))—have ramifications for the effects of tariffs. Simulations in [Bian et al. \(2026\)](#) also show that the oligopoly model crucially affects conclusions about optimal trade policy. In empirical IO, researchers typically assume that firms face constant marginal costs and

⁷While we lack the full covariance of the demand parameters, Appendix F.6 shows the robustness of our supply-side results to demand uncertainty by performing parametric bootstraps under a range of correlation structures.

compete according to Bertrand pricing. While these assumptions may hold in many settings, there is reason to question whether they are appropriate for studying the US automobile market. In particular, [Berry et al. \(1995\)](#) provide suggestive evidence that economies of scale may exist in the assembly of cars. Furthermore, the long production lead times and capacity constraints in automobile manufacturing suggest that output decisions may be more rigid than prices. In the US market, [Feenstra and Levinsohn \(1995\)](#) and [Berry et al. \(1999\)](#) consider alternatives to Bertrand pricing, which include Cournot quantity setting and mixed models where some firms set prices and others set quantities.

Thus, instead of assuming a particular parametric model and using it to measure the effect of tariffs, we seek to use the data to guide our assumptions on conduct and cost. To do so, we begin with a general framework where the data in each market t are generated by equilibrium play in some static model of supply in which prices and quantities are endogenous. A system of first-order conditions characterizes the true supply model,

$$p_t = \Delta_{0t} + c_{0t},$$

where $\Delta_{0t} = \Delta_0(p_t, s_t, \theta_0^D)$ is the true vector of markups in market t and c_{0t} is the true vector of marginal costs. We specify our model for the log of marginal cost, and assume a Cobb-Douglas production function (used in, e.g., [Berry et al. 1995](#); [Verboven 1996](#); [Goldberg and Verboven 2001](#), and nesting the constant-marginal-cost specification in, e.g., [Goldberg 1995](#); [Berry et al. 1999](#); [Van Biesebroeck, Gao, and Verboven 2012](#)):

$$\log(c_{0jt}) = \gamma_0 \log(q_{jt}^p) + w_{jt}'\tau_0 + \omega_{0jt}, \quad (1)$$

where w_{jt} is a vector of observed cost shifters that affects the product’s marginal cost, q_{jt}^p reflects the production quantity corresponding to the level at which economies of scale accrue, and ω_{0jt} is an unobserved shock.

The choice of q_{jt}^p deserves some discussion. In our main specification, we assume q_{jt}^p corresponds to the total production of model j in the country that supplies the US market in year t . In this industry, this is similar to the assumption that economies of scale accrue at the plant-model level. In the vast majority of cases, car models sold in the US are sourced from one country ([Head and Mayer, 2019](#)), and produced in a single plant in that country. Our assumption stands between the approaches that scale economies accrue from global production across countries for a model ([Verboven, 1996](#); [Goldberg and Verboven, 2001](#)), and those that assume “external” scale economies across all models produced by a firm within a country ([Head and Mayer, 2019](#)). We explore the robustness of our results to alternative assumptions in Appendix F.2, including platform-level economies of scale (which captures economies of scope⁸ among car models using the same engineering platform) and

⁸See [Khmelnitskaya, Marshall, and Orr \(2025\)](#) for a recent empirical approach that also allows for

a log-linear and quadratic functional form.

Our specification captures economies of scale in the total marginal cost of producing a vehicle, which encompasses both assembly and the procurement of parts. Higher production volumes may reduce per-unit costs through more efficient assembly operations, lower input prices as parts suppliers themselves benefit from scale, or improved bargaining leverage with suppliers. Our scale elasticity reflects the net effect of these channels without distinguishing among them. We do not, however, separately model parts production, for which firm-level data are not available.

Because the true markups (or, equivalently, true costs) are unobserved in the market for cars, we want to determine the supply model that best fits the data from a menu of candidates motivated by the literature. Distinguishing supply models requires instruments (see e.g., [Berry and Haile, 2014](#); [Dearing et al., 2024](#)). We therefore maintain that instrumental variables z_{jt} exist such that, for the true model, the exclusion restriction $E[\omega_{0jt} \mid w_{jt}, z_{jt}] = 0$ holds. This assumption requires that the instruments are exogenous for supply, and therefore uncorrelated with the unobserved cost shifters for the true model. [Berry and Haile \(2014\)](#) provides standard sources of exogenous variation, such as variation in the set of rival firms and rival products, own and rival product characteristics, rival cost, and market demographics. These variables are available in our dataset and we now explore how they can be used to distinguish models of conduct under economies of scale.

4 Testing Conduct With Non-Constant Marginal Cost

Our application requires distinguishing between models of firm conduct while accounting for economies of scale in production. [Berry and Haile \(2014\)](#) show that, in differentiated products markets, positing a model of firm conduct yields falsifiable restrictions, and that a wide range of cross-product and cross-market variation can discriminate between alternative models, even under non-constant marginal costs. However, operationalizing these falsification results into a valid testing procedure requires addressing two limitations in the IO literature: the existing procedures for testing conduct in [Backus et al. \(2021\)](#) and [Duarte et al. \(2024\)](#), and the framework for understanding instrument relevance in [Dearing et al. \(2024\)](#), both assume constant marginal costs.

We address these two concerns in this section. First, we propose a general procedure to test models of conduct while also estimating economies of scale, which we apply to the car market in Section 5. Our procedure extends the methodology of [Duarte et al. \(2024\)](#), which adapts the RV non-nested model selection test ([Rivers and Vuong, 2002](#)) to implement the falsifiable restrictions of [Berry and Haile \(2014\)](#). This approach is appropriate because our

economies of scope.

candidate supply models are non-nested⁹ and the RV test offers advantages over model assessment alternatives. Second, we examine instrument relevance for the RV test under economies of scale. [Berry and Haile \(2021\)](#) emphasize that instruments can discriminate between conduct models by differentially altering the models’ implied markups, an idea which underpins instrument relevance for testing under constant marginal costs in [Duarte et al. \(2024\)](#). By extending the framework in [Dearing et al. \(2024\)](#) to the non-constant marginal cost case, we show that relevant instruments for testing must also move markup differences and equilibrium quantities non-proportionally, a condition we term *economic distinctness*. However, the cross-firm and cross-product variation in characteristics and cost shifters highlighted by [Berry and Haile \(2014\)](#) naturally provides economically distinct instruments, making our testing procedure feasible with standard industry data. While we focus on the specific cost function in Equation (1), adopted for our application, Appendices A and B broaden our procedure and guidance for instrument relevance beyond this specific case.

4.1 Our General Procedure

Step 0. Construct Menu of Models The researcher specifies plausible candidate models of conduct based on institutional knowledge. Our procedure applies to static conduct models where, for each model m , prices and quantities in market t satisfy model-specific first-order conditions:

$$p_t = \Delta_{mt} + c_{mt} \quad (2)$$

where Δ_{mt} is the vector of *implied markups* under model m in market t and c_{mt} is the vector of *implied marginal costs*.¹⁰

Step 1. Obtain Implied Markups and Marginal Costs Under Each Model For each model we consider, Δ_{mt} can be expressed as a function of data and demand primitives. Thus, given data on equilibrium outcomes and the demand estimates (such as those in [Grieco et al. \(2024\)](#) for our application), Δ_{mt} can be computed in each market and the implied costs c_{mt} can be recovered as $c_{mt} = p_t - \Delta_{mt}$.

As examples, consider the canonical Bertrand pricing model ($m = B$) and the Cournot quantity setting model ($m = C$). The first-order conditions, stacked across products j in market t , allow us to express implied markups for the two models as

$$\Delta_{Bt} = -(\Omega_t \odot D_t')^{-1} s_t \quad (3)$$

⁹Even with nested models, testing can be preferred to estimation; see [Magnolfi and Sullivan \(2022\)](#).

¹⁰For the first-order conditions of any model m to characterize a well-defined empirical model, we require there exists a unique equilibrium, or the equilibrium selection rule is such that the same p_t arises whenever the vector (c_{mt}, x_t, ξ_t) is the same, analogous to Assumption 13 in [Berry and Haile \(2014\)](#).

and

$$\Delta_{Ct} = -(\Omega_t \odot (D_t^{-1})')s_t, \quad (4)$$

respectively, where Ω_t is the ownership matrix in market t , D_t is the matrix of demand derivatives, s_t the vector of market shares, and $\odot$ denotes Hadamard (element-wise) multiplication.¹¹

Step 2. Estimate Model-Implied Scale Economies and Cost Shocks Given the vector of implied markups Δ_{mt} and implied marginal costs $c_{mt} = p_t - \Delta_{mt}$, we can estimate the marginal cost function implied by model m ,

$$\log(c_{mjt}) = \gamma_m \log(q_{jt}^p) + w'_{jt}\tau_m + \omega_{mjt}. \quad (5)$$

Compared to the case of constant marginal costs, the right-hand side of Equation (5) contains *endogenous* log quantities. Thus, estimating γ_m and τ_m requires the use of instruments and estimation via two-stage least squares (2SLS). We pin down the parameters in marginal cost as solutions to the standard 2SLS moment equations $E[w_{jt}\omega_{mjt}] = 0$ and $E[\tilde{q}_{jt}\omega_{mjt}] = 0$ where $\tilde{q}_{jt}$ is the best linear predictor of log quantities using cost shifters and instruments,

$$\tilde{q}_{jt} = (z'_{jt}, w'_{jt})E[(z, w)'(z, w)]^{-1}E[(z, w)' \log(q^p)].$$

This leads to standard 2SLS definitions of the parameters γ_m and τ_m ,

$$\begin{pmatrix} \gamma_m \\ \tau_m \end{pmatrix} = E[(\tilde{q}, w)'(\log(q^p), w)]^{-1}E[(\tilde{q}, w)' \log(p - \Delta_m)].$$

The implied cost shocks ω_{mjt} , recovered as the 2SLS projection errors, rationalize the observed equilibria (p_t, s_t) under model m in market t .

Step 3. Construct Measure of Lack-of-Fit for Each Model For the true model, we know that $E[\omega_{0jt} | w_{jt}, z_{jt}] = 0$. Having recovered the model-implied cost shocks ω_{mjt} , we can therefore define a measure of lack-of-fit based on the moment condition $E[z_{jt}\omega_{mjt}] = 0$, which will hold if model m is the correct model.

Pinning down the economies of scale parameter uses part of the variation in z_{jt} . Thus, the key deviation from the constant marginal cost case is that the effective instrument only contains $d_z - 1$ (instead of d_z) sources of linearly independent variation. For example, if z_{jt} is a scalar, then the cost function would be exactly identified and, by definition, the variation in z_{jt} would be orthogonal to ω_{mjt} for any model. We therefore base our measure

¹¹That is, $(\Omega_t)_{j,k} = 1$ if products j and k are produced by the same firm and 0 otherwise; $(D_t)_{j,k} = \frac{\partial s_{jt}}{\partial p_{kt}}$; and for two matrices A and B of the same size, $(A \odot B)_{j,k} = A_{j,k}B_{j,k}$.

of lack-of-fit only on the variation that remains in the instruments *after* pinning down the cost function. This remaining variation in the instruments, denoted z_{jt}^e , is the error in z_{jt} from a population projection on w_{jt} and $\tilde{q}_{jt}$,

$$z_{jt}^e = z_{jt} - E[z'(\tilde{q}, w)] E[(\tilde{q}, w)'(\tilde{q}, w)]^{-1} \begin{pmatrix} \tilde{q}_{jt} \\ w_{jt} \end{pmatrix}.$$

We consider the following generalized method of moments (GMM) objective function as our measure of lack-of-fit for a model m ,

$$Q_m = g_m' W g_m,$$

where $g_m = E[z_{jt}^e \omega_{mjt}]$. In analogy with [Duarte et al. \(2024\)](#), the weight matrix is based on the $d_z \times d_z$ covariance matrix $E[z_{jt}^e z_{jt}^{e'}]$. However, z^e is obtained by residualizing z against a linear combination of its elements, i.e., $\tilde{q}$. As the rank of $E[z_{jt}^e z_{jt}^{e'}]$ is $d_z - 1$, we use as the weight matrix the Moore-Penrose inverse,¹² $W = E[z_{jt}^e z_{jt}^{e'}]^+$. Notice that, if $E[z_{jt}^e \omega_{mjt}] = 0$, model m is indistinguishable from the true model and $Q_m = 0$; if $E[z_{jt}^e \omega_{mjt}] \neq 0$, the instruments distinguish model m from the truth and $Q_m > 0$.

Step 4. Run RV Test, Obtain Model Confidence Set For each model m , we now have a measure of fit $Q_m = E[z_{jt}^e \omega_{mjt}]' W E[z_{jt}^e \omega_{mjt}]$. To make inferences about conduct, we adopt the non-nested model selection test in [Rivers and Vuong \(2002\)](#). We consider all pairs of models from our menu and, for each, we run the pairwise RV test of models $m = 1, 2$. The null hypothesis for the test is that the two competing models of conduct have the same fit,

$$H_0^{\text{RV}} : Q_1 = Q_2.$$

Relative to this null, we define two alternative hypotheses corresponding to cases of better fit of one of the two models:

$$H_1^{\text{RV}} : Q_1 < Q_2 \quad \text{and} \quad H_2^{\text{RV}} : Q_2 < Q_1.$$

With this formulation of the null and alternative hypotheses, the statistical problem is to determine which of the two models has the best fit, or equivalently, the smallest lack of fit.

For the GMM measure of fit, the RV test statistic is then

$$T^{\text{RV}} = \frac{\sqrt{n}(\hat{Q}_1 - \hat{Q}_2)}{\hat{\sigma}_{\text{RV}}},$$

where $\hat{Q}_m$ is a sample analog of Q_m and $\hat{\sigma}_{\text{RV}}^2$ is an estimator for the asymptotic variance of the scaled difference in the measures of fit appearing in the numerator of the test statistic.

¹²Equivalently, one may redefine z^e to only include $d_z - 1$ of the residualized instruments, where the dropped instrument can be any instrument with a non-zero coefficient in $\tilde{q}$. In doing so, one obtains full rank of $E[z_{jt}^e z_{jt}^{e'}]$ and can let $W = E[z_{jt}^e z_{jt}^{e'}]^{-1}$. Derivations in [Appendix B](#) use this full-rank representation.

We denote this asymptotic variance by σ_{RV}^2 .

Estimation of γ_m is already incorporated into $\hat{Q}_m$ and therefore requires no separate adjustment to $\hat{\sigma}_{\text{RV}}^2$. Although estimation of the linear predictor $\tilde{q}$ affects $\hat{g}_m$ and $\hat{W}$ separately, these effects cancel in the first-order influence function for $\hat{Q}_m$, so estimating $\tilde{q}$ likewise requires no additional adjustment to $\hat{\sigma}_{\text{RV}}^2$. Appendix B provides the details.

For a menu of two models, there is one RV test statistic, and a researcher can conclude for a model if the test rejects in favor of that model. For more than two models, the researcher obtains multiple RV test statistics. For instance, in our application, the menu contains seven candidate models, meaning that we have twenty-one unique pairs, leading to twenty-one RV test statistics. To adjust for multiple testing, Duarte et al. (2024) shows how all the pairwise T^{RV} statistics can be used to construct a model confidence set.

Step 5. Construct F -statistic to Diagnose Weak Instruments for Testing For each pair of models, the RV test statistic, T^{RV} , is asymptotically standard normal under the null as long as the estimator of the asymptotic variance converges in probability to $\sigma_{\text{RV}}^2 > 0$. If instead $\sigma_{\text{RV}}^2 = 0$, the RV test is said to be degenerate, which Duarte et al. (2024) characterizes as a problem of irrelevant instruments for testing. Near the space of degeneracy, instruments may be weak, resulting in size distortions or low power. Duarte et al. (2024) provides a diagnostic for both size distortions and low power. Their diagnostic relies on a joint F -statistic of two “first stage” regressions, $\omega_{mjt} = z'_{jt}\pi_m + e_{mjt}$ for model $m = 1, 2$ in the pair being tested. They then provide critical values for worst-case size and maximal power to which the F -statistic can be compared in order to diagnose instrument strength.

Here, given that our measure of fit is formed with the residual variation in the instruments z^e , we modify the weak instruments diagnostic as a scaled F -statistic for a joint test of the hypotheses $H_{0,m} : \pi_m = 0$ for $m = 1, 2$ in the regressions

$$\omega_{mjt} = z_{jt}^{e'}\pi_m + e_{mjt} \quad \text{with } E[z_{jt}^e e_{mjt}] = 0 \quad \text{for model } m = 1, 2. \quad (6)$$

To appropriately diagnose instrument strength, one can compare our F -statistic to the *same critical values* for size and power as reported in Duarte, Magnolfi, S¸olvsten, Sullivan, and Tarascina (2022) with the value for the number of instruments being $d_z - 1$. This is because our modified RV test statistic T^{RV} and F -statistic converge to the same joint distribution as in Proposition 4 of Duarte et al. (2024), with d_z being adjusted to $d_z - 1$. Appendix B proves this and provides the definition of the F -statistic for our setting.

4.2 Instrument Relevance for Testing Conduct with Economies of Scale

The procedure outlined in Section 4.1 relies on the presence of relevant instruments for testing to obtain sharp inference on conduct under non-constant marginal cost. While

Bresnahan (1982) and Lau (1982) discuss the importance of demand rotator instruments in homogeneous products settings, Berry and Haile (2014) show that differentiated products markets offer a much broader set of exogenous variation that rotates residual marginal revenue, and can therefore discriminate between conduct models under non-constant marginal costs. In this subsection, we specialize their insights to characterize requirements for instruments to be relevant for testing conduct in parametric settings such as our application.

For an intuitive yet practical discussion, we consider a simplified environment where the true model of conduct is Bertrand, and the researcher wants to test Bertrand and Cournot. We further suppress the observed cost shifters, so that for either model $\log(c_{mjt}) = \gamma_m \log(q_{jt}^p) + \omega_{mjt}$. Appendix A develops instead formal results for relevance under general economies of scale, and for a broader range of models.

The literature has considered relevance when testing models under constant marginal cost. For this case, Dearing et al. (2024) establish that instruments are relevant if they target features of the cost pass-through matrix which differ across models. For the case of Bertrand versus Cournot, they illustrate that for logit demand systems, a single instrument formed from standard sources of variation (e.g., product characteristics and rival cost shifters) meets this requirement. Here we show that in the presence of economies of scale, there are two additional requirements for an instrument to be relevant for our testing procedure.

Remark 1. A single instrument $z_{jt}^{(1)}$ is irrelevant for testing any pair of models of conduct with economies of scale.

If the researcher has a single available instrument, then in Step 2 of our procedure the 2SLS estimator for economies of scale would be given by the sample analogue of

$$\gamma_m^{(1)} = E[z_{jt}^{(1)} \log(q_{jt}^p)]^{-1} E[z_{jt}^{(1)} \log(p_{jt} - \Delta_{mjt})].$$

Because γ_m is just-identified, it follows that the implied 2SLS errors satisfy $E[\omega_{mjt} z_{jt}^{(1)}] = 0$. When forming a measure of lack-of-fit in Step 3, the researcher only has one instrument and thus has to form Q_m with the same $z^{(1)}$. Mechanically, then, without an overidentifying restriction, $Q_m = 0$ for any model m , and the RV test is degenerate with $F = 0$. This parallels the observation in Berry and Haile (2014, 2021) that overidentifying restrictions from instruments beyond those needed to identify the cost function provide falsifiable restrictions for distinguishing models.

Remark 2. Two instruments $z_{jt}^{(1)}$ and $z_{jt}^{(2)}$ must be *economically distinct* to be relevant for testing model m against the true model, which we define as

$$\frac{E[z_{jt}^{(1)} \log(p_{jt} - \Delta_{mjt})]}{E[z_{jt}^{(2)} \log(p_{jt} - \Delta_{mjt})]} \neq \frac{E[z_{jt}^{(1)} \log(q_{jt}^p)]}{E[z_{jt}^{(2)} \log(q_{jt}^p)]}. \quad (7)$$

This requirement adapts the results in Lau (1982) to our parametric setting with non-

constant costs. As in the constant marginal cost case, it is necessary that instruments differentially move markups across models to be relevant for testing. However, in our parametric setting with non-constant marginal cost, instruments must also have different correlations with the log of production quantities and the log of implied costs under model m . As long as the two instruments correspond to sources of variation that have different economic effects, this statistical condition will generally hold.¹³

To see where this condition comes from, suppose a researcher estimates economies of scale with one of the instruments $z^{(k)}$ for $k = 1, 2$, recovering $\omega_{mjt}^{(k)}$ as the error. While mechanically $E[z_{jt}^{(k)} \omega_{mjt}^{(k)}] = 0$, the researcher can use $E[z_{jt}^{(-k)} \omega_{mjt}^{(k)}]$ to form a measure of fit $Q_m^{(-k)}$. If each instrument, however, implies the same economies of scale, then $\omega_{mjt}^{(1)} = \omega_{mjt}^{(2)}$. Therefore, $E[z_{jt}^{(-k)} \omega_{mjt}^{(k)}] = E[z_{jt}^{(-k)} \omega_{mjt}^{(-k)}] = 0$, and the instruments are irrelevant for testing any model m . The condition in (7) ensures that the economies of scale separately measured with the two instruments are different, preventing this mechanical channel for irrelevance. It further provides a parametric analog to the argument in Section 6, Case 2 of [Berry and Haile \(2014\)](#), where a model is falsified if two distinct subsets of instruments, each sufficient to identify the marginal cost function under that model, imply different cost shocks.

Remark 3. Standard datasets from differentiated product markets typically contain economically distinct instruments.

In the representative-firm homogeneous goods setting in [Bresnahan \(1982\)](#), there are no rival firms and products, so that variation that both shifts and rotates the inverse demand curve, which can be achieved by a demand rotator, is the only relevant source of variation. As shown in [Berry and Haile \(2014\)](#), with product differentiation, a much broader class of variation is available for discriminating between conduct models. Our economically distinct instruments requirement formalizes what is needed from this variation for our testing procedure under scale economies: observing distinct product characteristics and cost shifters across firms provides sources of economically distinct instruments. To see why, note that for two instruments to fail to be economically distinct, they must shift own quantities and the difference in implied markups across models by exactly proportional amounts (see [Appendix A.5](#) for a formal statement). This proportionality is a knife-edge condition that holds when two instruments affect market equilibrium through the same economic channel. For instance, two cost shifters entering through the same rival’s marginal cost will have effects on all equilibrium objects that differ only by a scalar multiple. Instruments operating through different channels, on the other hand, enter the equilibrium system at different points and break this proportionality.

These observations yield concrete guidance for instrument selection. In a market with

¹³Relatedly, the example in Section 6.1.1 of [Berry and Haile \(2014\)](#) illustrates how two sources of variation operating through different economic channels can discriminate between models.

two single-product firms and logit demand, economically distinct pairs leverage cross-firm or cross-product variation. Pairing an own product characteristic with a rival cost shifter works because the former directly shifts demand for the own product while the latter operates through the rival’s pricing behavior. Similarly, pairing own and rival product characteristics is effective because they shift demand for different products with different equilibrium consequences. Economically indistinct pairs, by contrast, use variation within the same economic channel: two cost shifters of the same rival, two characteristics of the same product, or a characteristic and cost shifter from the same rival all affect the equilibrium through a single pathway, producing the proportional responses that preclude falsification. In richer demand systems with random coefficients, asymmetric substitution patterns further break proportionality, so that even broader sets of instrument combinations are economically distinct.

Taken together, we have developed a general and portable procedure to test conduct under non-constant marginal cost. Appendix A includes a more formal discussion of the economic determinants of instrument relevance for more general models of conduct and cost functions. Appendix C develops a set of Monte Carlo simulations to illustrate the procedure in practice and the role of instrument relevance. We now turn to implement the procedure in the market for US automobiles.

5 Implementing Our Procedure in the Market for Cars

Step 0. Construct Menu of Models We consider a menu of seven different models of firm conduct. The first five are motivated by [Feenstra and Levinsohn \(1995\)](#) and [Berry et al. \(1999\)](#), and the last two by differences in production flexibility across manufacturers.

1. *Bertrand pricing*: All car manufacturers choose prices to maximize profits, taking competitors’ prices as given.
2. *Cournot quantity setting*: All car manufacturers choose quantities of each car model to sell, prices adjust to clear the market.
- 3-7. *Mixed Models*: Subset of manufacturers $\mathcal{B}$ set prices, remaining subset $\mathcal{C}$ set quantities.
 3. $\mathcal{B}$: Asian firms, $\mathcal{C}$: US and European firms.
 4. $\mathcal{B}$: US firms, $\mathcal{C}$: Asian and European firms.
 5. $\mathcal{B}$: Asian and US firms, $\mathcal{C}$: European firms.
 6. $\mathcal{B}$: all firms except the Big Three, $\mathcal{C}$: the Big Three (GM, Ford, Fiat Chrysler).
 7. $\mathcal{B}$: Tesla, $\mathcal{C}$: all other firms.

While Bertrand pricing is a standard assumption in the empirical IO literature, in the market for cars [Feenstra and Levinsohn \(1995\)](#) and [Berry et al. \(1999\)](#) have also considered

Cournot conduct and “mixed” models where some manufacturers compete in prices while others compete in quantities. This specification reflects potential asymmetries between manufacturers. For instance, firms with more flexible production systems may be better positioned to compete on price, while those with rigid capacity may effectively compete in quantities. Following [Feenstra and Levinsohn \(1995\)](#), we consider three mixed models based on brand nationality (American, Asian, and European): (i) only Asian manufacturers play Bertrand, (ii) only American manufacturers play Bertrand, and (iii) only European manufacturers play Cournot.¹⁴

Models 6 and 7 are motivated by differences in production flexibility across manufacturers. Firms with unionized workforces may face greater rigidity in adjusting output due to collective bargaining agreements and legacy supplier contracts, while newer entrants, unconstrained by such arrangements, may have more flexibility to adjust production rapidly. Accordingly, Model 6 lets only the Big Three US manufacturers (Ford, GM, and Fiat Chrysler, the only firms with UAW-unionized workforces during our sample period) play Cournot, while Model 7 lets only Tesla (a non-legacy entrant with a non-unionized workforce) play Bertrand.

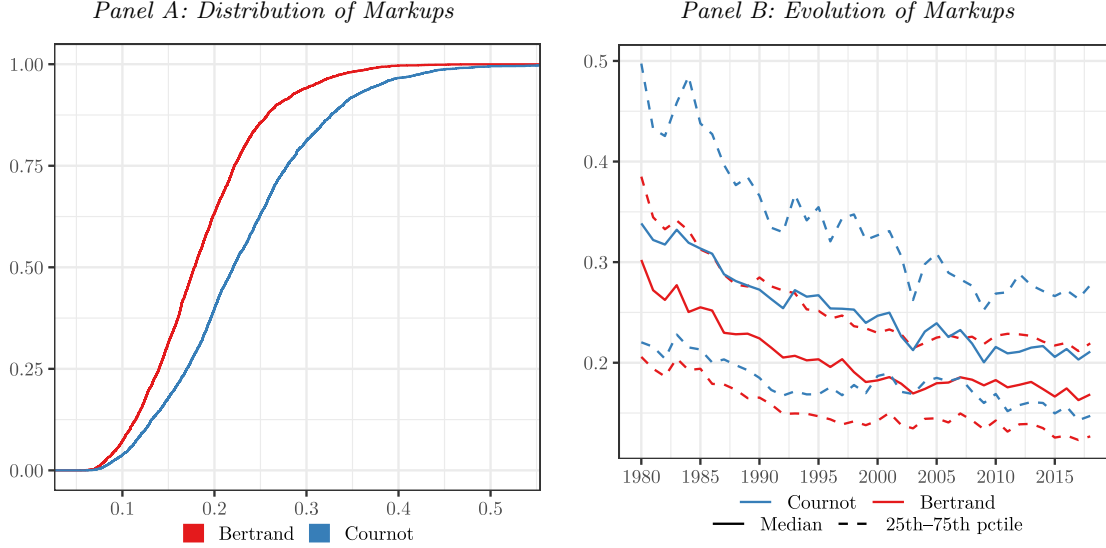
Step 1. Obtain Implied Markups and Marginal Costs Under Each Model For each model m we consider, Δ_{mt} can be computed in each market from the data on equilibrium outcomes and the demand estimates in [Grieco et al. \(2024\)](#). For the Bertrand and Cournot models, implied markups are respectively given by Equations (3) and (4). We derive the implied markups for the mixed models of conduct Δ_{Mt} in Appendix D.2. The implied costs for all models in our menu can then be recovered as $c_{mt} = p_t - \Delta_{mt}$.

In Figure 1, Panel A, we plot the distribution of implied markups (as Lerner indices $\frac{\Delta_{mjt}}{p_{jt}}$) under Bertrand and Cournot, the two extremes of our conduct menu. In line with theory ([Magnolfi, Quint, Sullivan, and Waldfogel, 2022](#)), Cournot implies higher markups than Bertrand; all mixed models lie between these two extremes. Markups remain below price under both extremes, so implied marginal costs are positive throughout. While the different conduct assumptions yield different levels of markups, Figure 1, Panel B, shows that all models of conduct imply declining trends from 1980 to 2018, in line with the findings of [Grieco et al. \(2024\)](#).

We further explore the implications of conduct by examining in Table 2 the Bertrand and Cournot markups for the four best-selling models in each of the sedan, SUV, truck, and van categories. Across these car models, Bertrand markups are 4-14 percentage points below Cournot markups, which is substantial in this industry.

¹⁴Using data for 1987, [Feenstra and Levinsohn \(1995\)](#) find evidence for a Nash equilibrium where American firms choose prices, European firms choose quantities, and Japanese firms choose either prices or quantities.

FIGURE 1: Markup Implications of Models of Conduct



The figure plots the distribution of implied markups under Bertrand and Cournot, the extremes of the conduct menu (Panel A), and the evolution of the median, 25th, and 75th percentile markups over the full sample in [Grieco et al. \(2024\)](#) (Panel B).

TABLE 2: Model-Implied Markups for Top Car Models by Segment

Sedans			SUVs		
Car Model	Bertrand	Cournot	Car Model	Bertrand	Cournot
Toyota Camry	23.1	28.7	Toyota RAV4	21.6	28.5
Honda Civic	28.7	33.7	Nissan Rogue	21.9	26.8
Honda Accord	20.7	24.3	Honda CR-V	20.1	24.8
Toyota Corolla	30.2	37.4	Chevrolet Equinox	23.8	32.8
Trucks			Vans		
Car Model	Bertrand	Cournot	Car Model	Bertrand	Cournot
Ford F Series	18.8	31.6	Dodge Caravan	26.4	37.7
Chevrolet Silverado	17.8	31.5	Ford Transit	21.2	32.8
Ram Pickup	15.1	24.6	Chrysler Pacifica	25.8	36.4
Toyota Tacoma	23.4	29.7	Honda Odyssey	17.1	21.6

The table reports estimated markups (as Lerner indices, Δ_{mjt}/p_{jt}) in percent units under Bertrand and Cournot conduct assumptions for the top-selling car models in each vehicle segment during 2018.

Step 2. Estimate Model-Implied Scale Economies and Cost Shocks To estimate the model-implied economies of scale, we need to specify a marginal cost function and construct instruments.

Instrument Choice: We construct instruments that satisfy the requirements for economic distinctness developed in [Section 4.2](#): variables that move equilibrium quantities and markup

differences through separate economic channels, generating the non-proportional responses that let one model of conduct be falsified against another.

The most transparent source of such variation combines a variable that moves demand with one that moves cost. The former moves quantities through consumers’ choices; the latter moves them through firms’ marginal costs and pricing. Because the two enter the equilibrium at different points, they produce non-proportional responses across models of conduct. For the cost channel we use the rival exchange rate: following [Grieco et al. \(2024\)](#), country-of-assembly exchange rates are an effective source of cost variation, and aggregating over rivals yields variation that is excluded from a model’s own cost equation and operates through rivals’ pricing rather than its own demand.

Our demand-side instruments are, in turn, economically distinct not only from this cost variation but from one another. As [Section 4.2](#) discusses and [Appendix A.5](#) formalizes, under random-coefficient demand, asymmetric substitution patterns break the proportionality that would otherwise render demand-side variation redundant. Following [Gandhi and Houde \(2020\)](#), we construct local differentiation instruments from the proximity of rival products to each model in the space of the demand model’s continuous characteristics (footprint, horsepower, and fuel economy), which summarize precisely these substitution patterns. We pair these with three further demand channels: market structure (the number of rival models within a category $\times$ luxury $\times$ sport $\times$ EV group, capturing competitive intensity); consumer demographics (average income and age, which affect price sensitivity); and an SUV indicator interacted with a time trend, which predicts demand-driven variation in production volumes and is particularly valuable for identifying economies of scale.

Our main test ([Table 4](#)) is conducted with this instrument set. [Appendix F.1](#) reports the test under alternative instrument formulations: the cost, structure, and demographic instruments on their own and the BLP functional form.

Economies of Scale Estimates: Following [Grieco et al. \(2024\)](#), we specify the vector of cost shifters w_{jt} to include the exchange rate, a quadratic time trend, years since design, the log of continuous characteristics (height, footprint, horsepower, miles per gallon, curb weight, number of trims), and indicators for release year, segment, electric, sport, and brand fixed effects. [Table 3](#) presents 2SLS estimates of the economies of scale parameter γ_m under different models of conduct, while [Table 12](#) in [Appendix D.3](#) reports full regression results. [Table 20](#) in [Appendix F.2](#) explores the robustness of these estimates to alternative cost specifications, including platform-level economies of scale and a log-linear and quadratic functional form.

The estimates suggest meaningful economies of scale across all specifications, with scale elasticities ranging from -0.125 to -0.135 . The magnitude of the economies of scale es-

TABLE 3: Economies of Scale Estimates

Model m	γ_m Estimate	Standard Error
1. Bertrand	-0.135	0.030
2. Cournot	-0.125	0.031
3. Mixed (Asian firms Bertrand)	-0.129	0.031
4. Mixed (US firms Bertrand)	-0.128	0.030
5. Mixed (European firms Cournot)	-0.134	0.030
6. Mixed (Big Three Cournot)	-0.132	0.031
7. Mixed (Tesla Bertrand)	-0.125	0.031

The table reports estimates of parameter γ_m obtained by estimating Equation (5) via 2SLS under different models of conduct m (reported in different rows). To account for potential delayed effects and unexpected shocks, the variable that interacts with γ_m is the average of the two previous years’ actual model production. All specifications include cost shifters and fixed effects (see Table 12 for full regression results). Standard errors are clustered by car model.

timates is stable across different conduct assumptions, suggesting this finding is robust to how we model competitive interaction. These estimates sit within the range of comparable estimates in the IO literature: [Verboven \(1996\)](#) finds -0.11 , [Fuss and Waverman \(1990\)](#) find a coefficient of -0.07 , while [Goldberg and Verboven \(2001\)](#) find -0.006 to -0.03 , although the authors point out that these estimates may be attenuated due to the presence of a quota. Our Cournot estimate of -0.125 is closest to [Verboven \(1996\)](#) and smaller in magnitude than the external scale elasticity of -0.16 for motor vehicle production reported by [Bartelme et al. \(2025\)](#). The finding of economies of scale is also consistent with industry evidence of economies of scale in car assembly and parts manufacturing.¹⁵

Steps 3-5. Conduct Testing We implement the testing procedure to evaluate our seven models of conduct. Table 4, Panel A presents the results.

The test results favor the model where all firms engage in Cournot competition. This model is included in the model confidence set with a p -value of 1.00, while all other specifications are rejected. The effective F -statistics for all pairwise comparisons clear the [Duarte et al. \(2024\)](#) critical values for both worst-case size and best-case power (reported in Appendix D.4), so the instruments are sufficiently strong to deliver reliable inference, yielding a model confidence set containing only one model.

The finding that the Cournot model best fits the data in this market is consistent with several features of this industry. First, it can approximate yearly production targets, with prices that adjust via dealers’ incentives (e.g., [Tremblay, Tremblay, and Isariyawongse, 2013](#)). Moreover, choosing quantities facilitates parts sourcing. Finally, capacity constraints are salient for automobile manufacturing, and Cournot can approximate a two-stage capac-

¹⁵For example, [Doner, Noble, and Ravenhill \(2021\)](#) state: “By 2023, profitable economies of scale [require] annual sales of 1 million vehicles using a given platform and 300,000 units for engines.”

TABLE 4: Conduct Test Results

	T^{RV}						
Models	2	3	4	5	6	7	MCS p -values
<i>Panel A: Test Results Allowing for Economies of Scale</i>							
1. Bertrand	4.37	2.16	3.30	−0.76	1.52	4.25	0.000
2. Cournot		−5.47	−3.79	−4.63	−4.94	−2.37	1.000
3. B: Asia, C: US, EU			0.52	−2.49	−2.24	5.31	0.000
4. B: US, C: Asia, EU				−4.05	−1.34	3.59	0.000
5. B: Asia, US, C: EU					1.79	4.54	0.000
6. B: non-Big 3, C: Big 3						4.86	0.000
7. B: Tesla, C: all others							0.018
<i>Panel B: Test Results Under Constant Marginal Cost</i>							
1. Bertrand	7.48	4.86	6.87	1.09	3.04	7.15	0.000
2. Cournot		−7.23	−4.94	−7.58	−7.47	−3.38	1.000
3. B: Asia, C: US, EU			0.89	−4.85	−4.86	6.64	0.000
4. B: US, C: Asia, EU				−7.49	−2.43	4.43	0.000
5. B: Asia, US, C: EU					2.71	7.25	0.000
6. B: non-Big 3, C: Big 3						7.17	0.000
7. B: Tesla, C: all others							0.001

The table reports the RV test statistics T^{RV} and the MCS p -values for the seven models of conduct. Panel A contains results allowing for economies of scale, corresponding to the procedure in Section 4. Panel B contains results under constant marginal cost. A negative RV test statistic suggests a better fit of the row model. With MCS p -values below 0.05 a row model is rejected from the model confidence set. Appendix D.4 reports the corresponding effective F -statistics. Test statistics in the table do not account for demand uncertainty; Appendix F.6 presents robustness results.

ity and price competition game (Kreps and Scheinkman, 1983; Hendel, 1994).

For comparison with existing methods in the literature, Table 4, Panel B shows results for the conduct test under constant marginal cost. While allowing for scale economies will make a meaningful difference for our counterfactuals, the test results are similar under this more restrictive cost specification, with a model confidence set that only contains Cournot.

In the pairwise comparison, Cournot decisively beats Bertrand with economies of scale ($T^{\text{RV}} = 4.37$). Given this, we would also expect Cournot to outperform the standard specification in the literature, Bertrand with constant marginal cost, which further restricts the scale elasticity to zero. The estimated scale elasticity under Bertrand conduct is itself statistically different from zero (Table 3), suggesting that the constant marginal cost restriction does not hold in this industry. Indeed, the pairwise test also rejects the standard specification in favor of Cournot with economies of scale.

A natural concern is whether our preferred model's superior in-sample fit translates to reliable out-of-sample predictions. To assess this, we perform a leave-one-out cross-validation exercise: for each year, we estimate the marginal cost regression using data from all other years and predict prices for the held-out year's market structure. Cournot produces

lower prediction errors than Bertrand in every one of the 17 years in our sample (Figure 2 in Appendix D.5). We interpret this as evidence that Cournot predicts held-out markets more accurately than Bertrand within the range of year-to-year variation observed in the data.

6 Evaluating Tariffs

In May 2018, the Trump administration opened a Section 232 investigation into whether automotive imports threatened national security, raising the prospect of tariffs of up to 25% on imported cars and parts. Broad auto tariffs were not ultimately imposed in that episode, but the threat has since recurred: the second Trump administration has again threatened, imposed, and paused stacked tariffs on cars and car parts (we discuss this episode further in Section 6.5).

We now evaluate the effects of imposing stacked and reciprocal tariffs in the 2018 US automobile market. To do so, we use the model chosen by our model selection procedure: Cournot quantity setting with economies of scale. To disentangle the effect of tariffs on finished cars and car parts, we perform three counterfactuals. In the first counterfactual, denoted C1, we impose a 25% tariff on all foreign-manufactured cars (including those assembled in Mexico and Canada) imported to the US.¹⁶ Second, in C2, we impose a 25% tariff on both foreign-manufactured cars and foreign-manufactured car parts imported to the US. Third, in C3, we impose 25% tariffs on all foreign imports and 25% reciprocal tariffs on US-manufactured parts exported for foreign assembly. The chosen 25% tariff level is in line with previous proposals by the Trump administration in 2018.¹⁷

6.1 Implementation of Counterfactuals

Under the Cournot model with economies of scale, the first-order conditions in each market are given by Equation (2), where $\Delta_{mt} = \Delta_{Ct}$ takes the form specified in Equation (4) and the implied costs $c_{mt} = c_{Ct}$ are obtained by taking the exponential of Equation (5). To simulate counterfactual outcomes, we hold the demand and marginal cost parameters, as well as the cost shock ω_{Ct} , fixed at their estimated levels. Across the counterfactuals, we need to turn on and off specific tariffs depending on the car model’s country of assembly and parts production. To simplify the exposition, we define an indicator variable for foreign assembly of model j , ϕ_j .

¹⁶In 2018, a 2.5% MFN tariff applied to passenger vehicles from countries without a US free trade agreement (primarily Japan and the EU). As we do not account for these tariffs when estimating marginal cost, our counterfactuals can be interpreted as imposing an effective tariff of 28.125% rather than 25% on these vehicles.

¹⁷E.g., <https://www.piie.com/blogs/trade-and-investment-policy-watch/2018/trumps-propose-d-auto-tariffs-would-throw-us-automakers>.

To perform C1, we need to modify the first-order conditions to account for a 25% tariff on foreign assembled cars. Tariffs apply to the port cost, essentially an unobserved wholesale price between manufacturer and dealer. Since port costs are unobserved, we must model how they are set. We consider two approaches: (i) modeling the port price as a fraction λ of the observed retail price p_{jt} which the firm internalizes when making profit maximizing choices, or (ii) modeling the port cost as a fraction ν of the implied marginal cost c_{mjt} which the firm takes as given with respect to its action. In the market for cars, [Goldberg \(1995\)](#) and [Goldberg and Verboven \(2001\)](#) model unobserved wholesale prices by adopting the first approach. Meanwhile, [Coşar et al. \(2018\)](#) adopt the second approach.

To account for the tariff, we need to appropriately modify the first-order conditions for foreign-assembled car models. Following the first approach, we could impose the 25% tariff as a $(25 \times \lambda)\%$ ad valorem tax levied on the producers of foreign-assembled cars; in the second approach, we could scale νc_{mjt} by 25%. Notice that for $\nu = \lambda(1 - \phi_j 0.25\lambda)^{-1}$, these two approaches are equivalent and the modified first-order conditions with tariffs become¹⁸

$$p_{jt} = \Delta_{mjt} + (1 + \nu 0.25)^{\phi_j} c_{mjt}(\cdot). \quad (8)$$

As our main specification for C1, we set $\nu = 1$ (or equivalently $\lambda = 0.8$), which is broadly in line with [Goldberg \(1995\)](#). As a robustness check, we set $\nu = 0.68$ (equivalent to $\lambda = 0.58$, corresponding to the estimate from [Coşar et al. 2018](#))—results are in Appendix F.3.

To perform C2 and C3, we need to further modify the first-order conditions to account for tariffs on car parts. To perform these counterfactuals, we maintain several assumptions. First, we assume for both foreign- and domestically assembled cars that car assembly is 29% of marginal cost while parts represent the remaining 71% ([Menk, Chen, and Cregger, 2012](#)). We also assume a fixed value for the pass-through of the tariff to the price of parts, denoted Λ for all cars. From the AALA data, we know for each model of car j the fraction of the total value of parts that were produced outside the US, which we denote μ_j , and the fraction produced in the US, denoted $(1 - \mu_j)$. Thus, across both C2 and C3, we can augment the first-order conditions to account for parts tariffs by scaling the appropriate part of the marginal cost. Specifically, in C2, we add tariffs on the foreign parts used in US-assembled vehicles so that the first-order conditions for each product j become

$$p_{jt} = \Delta_{mjt} + \left(1 + \nu 0.25\right)^{\phi_j} \left(1 + \mu_j \Lambda 0.71 \cdot 0.25\right)^{(1-\phi_j)} c_{mjt}(\cdot).$$

In C3, we now include reciprocal tariffs on US parts used in foreign production, so that the

¹⁸Dividing both sides of Equation (8) by $(1 + 0.25\phi_j\nu)$ and plugging in $\nu = \lambda(1 - \phi_j 0.25\lambda)^{-1}$ yields $(1 - \phi_j\lambda \times 0.25)p_{jt} = (1 - \phi_j\lambda \times 0.25)\Delta_{mjt} + c_{mjt}(\cdot)$ which can be interpreted as levying an ad valorem tax on the firms.

first-order conditions for each product j become

$$p_{jt} = \Delta_{mjt} + \left((1 + \nu 0.25) (1 + (1 - \mu_j) \Lambda 0.71 \cdot 0.25) \right)^{\phi_j} \left(1 + \mu_j \Lambda 0.71 \cdot 0.25 \right)^{(1 - \phi_j)} c_{mjt}(\cdot).$$

In our main specifications, we set $\Lambda = 1$, which is akin to assuming in-house production of all parts. This is consistent with [Ganapati and Hottman \(2026\)](#), who decompose tariff pass-through within supplier relationships into strategic and scale components and find that total pass-through to buyer prices remains near one. In [Appendix F.3](#), we consider alternative values of Λ .

With each counterfactual’s modified first-order conditions, we solve for equilibrium prices using the generalization of the [Morrow and Skerlos \(2011\)](#) procedure developed in [Duarte \(2025\)](#). We emphasize that our counterfactuals characterize the medium-term equilibrium of the US market alone, not a global-market equilibrium: foreign sales of foreign-assembled vehicles are held fixed, and we trace tariffs’ effects only through US-market outcomes. This matters because we model scale economies arising from total production, so that changes in foreign sales in reaction to US tariffs could impact firms’ marginal costs and US market outcomes. However, we have only estimated demand for cars in the US market. Thus, we keep foreign sales constant in our counterfactuals and recompute global production figures based on changes in US sales predicted by our model. However, it turns out that holding foreign sales constant is conservative with respect to our counterfactual. Under economies of scale, each model’s marginal cost depends on its total production across markets, so the market-by-market separability that holds under constant cost ([Bulow, Geanakoplos, and Klemperer, 1985](#)) breaks down. A US tariff that contracts a model’s US sales raises its marginal cost and leads it to contract foreign output as well, amplifying the scale channel rather than offsetting it. A two-market simulation (US vs a composite foreign market) in [Appendix F.4](#) is consistent with this logic, with the tariff’s price effects rising relative to the fixed-foreign baseline. A further maintained assumption concerns parts sourcing. In computing the parts-tariff counterfactuals (C2 and C3), we hold each model’s foreign parts share μ_j fixed, so manufacturers do not re-source toward domestic suppliers in response to the parts tariff. This reflects the medium-run horizon of our counterfactuals, over which sourcing relationships are difficult to reconfigure.

Finally, we note that our counterfactual analysis extrapolates the estimated cost function to production levels that differ substantially from the baseline, with decreases in import volumes but also meaningful increases in domestic production for certain car models (with the median model-level production increase reaching 12.5% in C3). A natural concern is whether such production increases are feasible while achieving the same returns to scale we estimate in-sample. We maintain that this assumption is plausible on two grounds. First, according to Federal Reserve data, capacity utilization in the US automotive sector

was approximately 75% in 2018,¹⁹ suggesting that substantial spare capacity existed in the industry, making it plausible for domestic producers to expand output in the short term while preserving similar cost structures. Second, as we show in Figure 3 in Appendix E.2, the counterfactual production increases are well within the range of year-to-year production changes observed in our sample, suggesting that our cost function is not being evaluated outside the support of the data.

6.2 Counterfactual Results

The counterfactual analysis reveals substantial effects from stacked tariffs.²⁰ Table 5 shows average price increases of 9.73% across all models in C1. This average masks heterogeneity by assembly location: foreign-assembled cars experience 26.20% price increases, while the prices of US-assembled cars decrease by 1.03%.

TABLE 5: Counterfactual Results – Prices

	# Models	Fraction US		% Change Prices		
		Assembly	Parts	C1	C2	C3
All	314	0.35	0.23	+9.73 (0.29)	+13.56 (0.38)	+15.32 (0.43)
US	111	1.00	0.50	−1.03 (0.12)	+6.55 (0.24)	+6.40 (0.25)
Non-US	203	0.00	0.09	+26.20 (0.75)	+24.29 (0.69)	+28.99 (0.81)

The table reports the sales-weighted average of the percentage change in prices corresponding to tariff counterfactuals C1-C3. Standard errors are in parentheses.

The price decrease for domestic manufacturers reflects strategic substitutability under Cournot competition with economies of scale: as tariffs raise foreign car prices, domestic firms gain market share, benefit from lower marginal costs due to increased production volumes, and respond by expanding output rather than raising prices. This echoes the finding in Berry et al. (1999) that European car prices fell in response to voluntary export restraints on Japanese vehicles. In Section 6.3 below, we define pass-through formally, compare rates across supply models, and decompose the sources of the gap.

Table 5 also shows the effect of stacked and reciprocal tariffs on car prices in the US market. When tariffs are also levied on car parts, the marginal cost of assembling cars in the US increases. For cars assembled in the US, stacking tariffs on parts causes prices

¹⁹See <https://fred.stlouisfed.org/series/CAPUTLG33611SQ>.

²⁰Standard errors in the counterfactual tables are from a parametric bootstrap that draws the demand parameters from the sampling distribution reported by Grieco et al. (2024) under a diagonal covariance, re-estimates the economies-of-scale cost function, and re-solves each counterfactual; we report the standard deviation across 196 feasible draws (of 200).

to rise on average by 6.55%, as opposed to falling by 1.03%. As this increase generates some substitution back to foreign firms, their overall price change under stacked tariffs is slightly lower (24.29% vs 26.20%), due to economies of scale and strategic substitutability via Cournot competition. Finally, comparing C2 and C3 shows the limited effect that reciprocal tariffs on car parts have on car prices to US consumers. Overall, the response of foreign manufacturers is to increase prices by an additional 4.70 percentage points, enabling US-assembled cars to lower prices by a modest 0.15 percentage points. Accounting for all three types of tariffs, the average price of cars is increased by 15.32% in the US market.

TABLE 6: Counterfactual Results – Consumer Surplus, Profits, and Revenues

	Change in Billions \$		
	C1	C2	C3
<i>Consumer surplus</i>	−25.3 (1.1)	−57.4 (1.5)	−59.6 (1.6)
Buyers of US-assembled cars	+4.6 (0.5)	−22.1 (0.7)	−21.6 (0.7)
Buyers of imported cars	−29.8 (0.7)	−35.1 (0.9)	−37.9 (0.9)
<i>Profit</i>	−6.3 (0.7)	−14.5 (1.2)	−15.2 (1.3)
Profit, US-assembled models	+20.4 (0.8)	+7.0 (1.1)	+9.3 (1.2)
Profit, imported models	−26.7 (0.3)	−21.5 (0.2)	−24.4 (0.3)
<i>Tariff revenue</i>	+12.9 (0.5)	+43.5 (0.3)	+43.2 (0.4)
Cars	+12.9 (0.5)	+17.8 (0.5)	+15.6 (0.5)
Parts	—	+25.6 (0.2)	+27.6 (0.1)
Total welfare	−18.6 (0.5)	−28.4 (1.0)	−31.6 (1.0)

The table reports overall welfare changes and tax revenue corresponding to tariff counterfactuals C1-C3. Consumer surplus is disaggregated by pre-tariff purchase type. Profit is disaggregated by model assembly location, summed across all manufacturers. Total welfare excludes parts-manufacturing profit, which we do not observe. Unindented rows report group totals, indented rows their decomposition; total welfare is the sum of the three group totals. Standard errors in parentheses.

Tariffs on cars and car parts have sizable effects on consumer welfare. In Table 6, we report changes in consumer welfare, measured in US dollars, by the purchase decision made by the household. In C1, tariffs on imported cars hurt consumers overall. However, there are small welfare gains for buyers of US-assembled cars, as these cars are sold at lower prices when the prices of foreign cars increase due to economies of scale and Cournot strategic substitution, while the losses fall on buyers of imported cars. Comparing C1 to C2 shows the important effect that stacked tariffs have in this market. The additional losses to households are large; overall, the loss in consumer surplus more than doubles. Now, since the costs and prices of US-assembled cars increase, the additional losses compared to C1 disproportionately fall on purchasers of US-assembled cars. Finally, as the price effects of reciprocal tariffs are minimal, these responses by foreign governments in C3 have limited

effects on US car consumers. Welfare losses are also larger for high-income consumers, as they are more likely to stay in the market when prices increase; Appendix E.1 reports household-level detail by demographic group and purchase decision.

Levying stacked and reciprocal tariffs on cars and car parts has differential effects on profits, depending on choices of assembly location and parts sourcing. Table 6 reports the effect on profits split by model assembly location; as we do not observe data on parts prices or costs, the table only includes profits from assembled vehicles. When tariffs are levied only on imported cars (C1), profit on US-assembled models rises while profit on imported models falls by more, as production shifts toward domestic assembly. Stacking tariffs on parts imported to the US (C2) erodes the gain to US-assembled models, which rely on imported parts, while reciprocal tariffs on US parts (C3) have a more limited effect. Because this split is by where models are assembled rather than by firm nationality, the gains and losses do not line up with firm origin; in Appendix E.1 we report the effect on profits for individual manufacturers, and in Appendix E.4 on a model-specific basis.

In each scenario, the welfare losses to consumers and firms easily outweigh government revenues. Levying tariffs only on imported cars in C1 results in a total welfare loss to consumers and firms of \$31.6 billion, of which only \$12.9 billion is offset by tax revenue. Comparing C2 to C1, stacked tariffs magnify the welfare losses: the losses to consumers and firms more than double, while the total net surplus lost in the US market for cars increases to \$28.4 billion. Interestingly, the revenue from tariffs on parts exceeds that from tariffs on car imports. Finally, comparing C2 to C3, we find additional losses of \$3.2 billion from reciprocal tariffs. These welfare effects depend importantly on the model of conduct and cost: as we show in Appendix F.5, the standard specification in the literature, Bertrand competition with constant marginal cost, understates the total net surplus loss in C3 by nearly \$3.5 billion.

6.3 Understanding Price Responses and Pass-Through

The counterfactual results above show that tariffs raise foreign car prices substantially while lowering domestic prices under our preferred model. To connect these findings with the empirical trade literature, we now explore the pass-through of tariffs to consumers. In our counterfactuals, assembled vehicles and parts are hit at a common 25% rate. Thus, we define pass-through as the ratio of the percentage change in retail prices to 25%, capturing the tariff elasticity of retail price when a policymaker sets a common rate on both upstream and downstream goods. This retail-price concept differs from the border-price pass-through measured in the trade literature (Amiti et al., 2019; Fajgelbaum et al., 2020; Cavallo et al., 2021), but it is close in spirit to the consumer-price concept in Flaaen et al. (2020).

Panel A of Table 7 reports the resulting pass-through rates under our preferred model

for both US- and non-US-assembled vehicles across all three counterfactuals. For non-US-assembled vehicles, pass-through exceeds one in C1 (1.05) and rises further in C3 (1.16), broadly consistent with the empirical literature on the 2018 Trump administration tariffs (Amiti et al., 2019; Fajgelbaum et al., 2020; Flaaen et al., 2020; Cavallo et al., 2021).²¹ For US-assembled vehicles, pass-through is essentially zero under car-only tariffs but turns positive (0.26) once parts tariffs are added, reflecting the reliance of domestic assembly on imported components.

TABLE 7: Pass-Through Rates and Decomposition

<i>Panel A: Pass-Through Rates</i>						
	C1		C2		C3	
	US	Non-US	US	Non-US	US	Non-US
Pass-through	-0.04 (0.00)	1.05 (0.03)	0.26 (0.01)	0.97 (0.03)	0.26 (0.01)	1.16 (0.03)
<i>Panel B: Pass-Through Decomposition – C1, Non-US-Assembled</i>						
	Bertrand Const. MC	Conduct Adj.	Scale Adj.	Eq. Adj.	Cournot Econ. of Scale	
Model-specific recovered costs	0.78	-0.06	+0.33	—	1.05	
Common recovered costs	0.73	+0.20	+0.33	-0.21	1.05	

The table reports, in Panel A, sales-weighted average pass-through under Cournot with economies of scale, computed as the ratio of the percentage price change from Table 5 to 25%. Standard errors are in parentheses. Panel B decomposes the C1 non-US-assembled pass-through into a conduct adjustment and a scale adjustment, reading left to right. “Model-specific recovered costs” uses each fitted supply model to both recover costs and run the counterfactual. “Common recovered costs” evaluates both models at the Cournot-implied cost vector; the equilibrium adjustment (Eq. Adj.) appears because Bertrand evaluated at Cournot-implied costs does not reproduce the observed baseline equilibrium.

To understand what drives the above-one pass-through for foreign vehicles, we decompose the 26.20% price increase for foreign-assembled vehicles in C1. Under our preferred model, this increase is almost entirely a marginal cost effect: it decomposes into a marginal-cost increase equal to 26.37% of the baseline price and a markup decline equal to 0.16% of the baseline price. The tariff directly raises the cost of imported vehicles and, because domestic firms cut prices under Cournot, demand shifts away from foreign cars more strongly, foreign output falls more, and the associated loss of scale raises marginal cost further.

Panel B of Table 7 decomposes the gap between Cournot and Bertrand pass-through for non-US-assembled vehicles in C1 into a conduct adjustment and a scale adjustment. In the model-specific recovered-cost row, each fitted supply model is used both to recover

²¹The fact that different models of supply imply different pass-through rates suggests that, when variation in tariffs or tax rates is available in the data, pass-through may be used to distinguish models of supply. This is an old idea in IO (e.g., Sumner, 1981); Dearing et al. (2024) show that instrument-based methods, such as the ones we use, essentially generalize this intuition and make it more broadly applicable.

marginal costs and to run the tariff counterfactual; this is the relevant comparison for policy prediction across estimated models. Starting from Bertrand pass-through of 0.78, the conduct adjustment is small (-0.06) and the scale adjustment is large ($+0.33$), giving Cournot pass-through of 1.05. The conduct adjustment is small, not because conduct is unimportant, but because moving from Cournot to Bertrand also changes the recovered cost vector, which offsets part of the strategic difference.

The common recovered-cost row evaluates both counterfactuals at the Cournot-implied cost vector, isolating the role of conduct. Starting from Bertrand pass-through of 0.73, the conduct adjustment rises to $+0.20$, revealing a larger difference between quantity-setting and price-setting behavior. An equilibrium-adjustment term of -0.21 appears because Bertrand evaluated at Cournot-implied costs does not reproduce the observed baseline equilibrium.

Appendix E.3 examines alternative demand specifications. Removing the random coefficient on price leads firms to cut markups much more in response to the tariff: for foreign-assembled vehicles in C1, the markup change goes from -0.16% under our standard demand system to -1.28% . Under logit, the markup change remains close to baseline (-0.18%), but the induced decline in foreign sales is larger, so the loss of scale is larger and marginal cost rises from 26.37% to 30.15% of the baseline price. This is why logit produces pass-through above one under Cournot with economies of scale.²²

6.4 Beyond Welfare: Effects on Employment

While our counterfactual analysis reveals substantial welfare losses from tariffs, these policies may be motivated by objectives beyond consumer welfare, particularly employment creation in domestic manufacturing and the reshoring of production. Our model does not directly capture non-price and non-quantity adjustment margins for auto manufacturers. However, we can use our counterfactual predictions to perform a back-of-the-envelope quantification of the employment effects of each tariff scenario to provide a more comprehensive assessment of trade-offs. We develop this analysis in this subsection and provide in Appendix E.5 some results on how tariffs provide incentives for the reshoring of parts production. Because these calculations rest on additional assumptions that we cannot test with our data, we treat the employment results as more exploratory than the price and welfare results.²³

To estimate employment effects in both vehicle assembly and parts manufacturing for the US in 2018, we assume that the annual ratio of labor expenditure to total revenue remains

²²This does not contradict the standard sub-unitary pass-through result for logit demand (Birchall et al., 2024; Miravete et al., 2025), which applies under Bertrand with constant marginal cost; under that benchmark, we indeed find pass-through ≈ 0.78 .

²³The parts-employment figures are particularly dependent on strong assumptions: the revenue-to-jobs proportionality must continue to hold, and assemblers must not re-source toward domestic suppliers over the horizon we model.

constant after tariff imposition.²⁴ For each of the two manufacturing sectors, indexed by k , this ratio is defined as:

$$\kappa_k = \frac{\bar{w}_k L_k}{R_k},$$

where L_k represents total US employment in sector k , $\bar{w}_k$ indicates the sector’s average wage, and R_k denotes total sector revenue. Further assuming that wages are unaffected in the medium run, employment in counterfactuals C1-C3 (L'_k) can then be computed from counterfactual revenue R'_k as:

$$L'_k = \frac{\kappa_k}{\bar{w}_k} R'_k.$$

This formula implies that if wages were to increase by a factor $(1 + \delta)$ following tariff imposition, employment would scale by $1/(1 + \delta)$.²⁵

To implement this approach, we calculate κ_k for both assembly and parts sectors using data from the Bureau of Labor Statistics’ Current Employment Statistics (for employment and wages) and the Census Bureau’s Annual Survey of Manufactures (for total revenue).²⁶ In 2018, κ_k is 0.042 for assembly and 0.096 for parts.²⁷

In each counterfactual C1-C3, our model predicts counterfactual revenues for cars assembled in the US as $R'_{\text{assembly}} = \sum_j \text{assembled in US } q'_j p'_j$, where q'_j is the counterfactual quantity of model j in 2018 and p'_j is the counterfactual price. To compute counterfactual revenues in parts manufacturing, we scale the appropriate fraction of the counterfactual marginal cost (c'_j) corresponding to US-produced parts by the counterfactual quantity. R'_{parts} sums this model-level revenue over all models sold in the US, regardless of country of assembly. We calculate it as $R'_{\text{parts}} = \sum_j q'_j \times c'_j \times 0.71 \times (1 - \mu_j)$.

Table 8 presents our employment impact estimates across the three tariff scenarios. The results reveal that tariffs would indeed create domestic jobs in both assembly and parts manufacturing, though with varying effectiveness across scenarios.

Under the car-only tariff scenario (C1), employment increases substantially in both sectors, with approximately 73,800 new assembly jobs (a 31.6% increase) and 34,810 new parts manufacturing jobs (a 5.8% increase). When tariffs are extended to imported parts (C2), the employment gains in assembly and parts are substantially reduced to 35,110 jobs (a 15.0% increase) and 2,230 jobs (0.4%), respectively.²⁸ This stark reduction in parts

²⁴This assumption is in line with the Cobb-Douglas production function that underlies our cost function specification.

²⁵For example, a 10% wage increase would reduce the employment gains by approximately 9%, from 44,060 total jobs in C3 to 40,050. The welfare cost per job created would correspondingly increase from \$717,300 to \$789,000, reinforcing our conclusion that the stacked tariffs are an inefficient tool for job creation.

²⁶The vehicle assembly industry corresponds to NAICS 3361, while parts manufacturing is represented by NAICS 3363.

²⁷Both values of κ_k have remained stable over time in the data.

²⁸While these are large employment changes, they are in line with other findings in the literature: [Flaen](#)

TABLE 8: Employment Effects by Tariff Scenario

	C1	C2	C3
Assembly Employment:			
Added jobs (thousands)	73.80 (2.71)	35.11 (2.07)	42.76 (2.12)
Percentage change (%)	31.59 (1.16)	15.03 (0.89)	18.30 (0.91)
Parts Employment:			
Added jobs (thousands)	34.81 (1.43)	2.23 (1.75)	1.30 (1.80)
Percentage change (%)	5.80 (0.24)	0.37 (0.29)	0.22 (0.30)
Total Jobs Created (thousands)	108.61	37.34	44.06
Annual Welfare Loss per Job (thousands \$)	171.7	761.1	717.3

Parts-employment changes use $\sum_j (q'_j c'_j - q_j c_j) 0.71(1 - \mu_j)$, with resource marginal cost evaluated at each scenario's production. US parts used in foreign-assembled vehicles are included. Labor-expenditure-to-revenue ratios and wages retain their baseline values. Standard errors in parentheses use 196 feasible seeded parametric-bootstrap draws and the same revenue formula. Annual welfare loss per job divides total welfare loss by total jobs created.

employment reflects how tariffs on intermediate inputs can undermine the employment benefits of final goods protection. The parts tariff raises the production cost of domestically assembled cars, causing their prices to swing from a decrease of 1.03% in C1 to an increase of 6.55% in C2 (Table 5). Higher prices for domestic cars reduce their sales, which in turn reduces demand for US-manufactured parts, the very industry the parts tariff is intended to protect. One countervailing margin lies outside our medium-run analysis: facing the parts tariff, some assemblers would likely substitute from foreign toward domestic parts suppliers, partially offsetting the decline in US parts employment that holding parts sourcing fixed implies. Adding reciprocal tariffs (C3) raises the assembly-employment gain to 42,760 jobs (an 18.3% increase), while the parts-employment gain falls to 1,300 jobs (0.2%).

To assess the efficiency of tariffs as an employment-creation tool, we calculate the welfare cost per job created, shown in the bottom row of Table 8. Each job created under the car-only tariff scenario (C1) costs \$171,700 in lost welfare per year. When tariffs extend to parts (C2), this cost rises sharply to \$761,100 per job. Under the reciprocal tariff scenario (C3), the cost remains high at \$717,300 per job.

The car-only tariff (C1) creates jobs at a welfare cost (\$171,700 each) of roughly twice the sector's average annual compensation of approximately \$75,000 in 2018. The stacked-tariff scenarios are costlier still, at \$717,300–\$761,100 per job. Across all scenarios, tariffs

et al. (2020) find that the 2018 tariffs create around 1,800 jobs in the washing machine industry. This is comparable to our findings since the auto manufacturing industry is 40-60 times larger than the washing machine industry.

are a highly inefficient mechanism for job creation; the high costs reflect how increased input costs severely undermine the employment benefits while magnifying welfare losses. Our stacked-tariff estimates are broadly in line with previous studies that quantify the cost of creating jobs in US manufacturing using tariffs as a policy instrument: [Flaaen et al. \(2020\)](#) quantify the cost as \$817,000 in the washing machine industry for the 2018 tariffs, and [Hufbauer and Lowry \(2012\)](#) find a cost per job from tariffs on Chinese tires in 2009 of \$900,000. Beyond tariffs, [Montag \(2026\)](#) finds that the Whirlpool-Maytag acquisition preserved domestic jobs at a cost of \$345,000 of consumer harm per worker. These employment effects depend importantly on the model of conduct and cost: as we show in Appendix F.5, a model of Bertrand competition with constant marginal cost yields negative point estimates for parts employment under C2 and C3, whereas the point estimates under our preferred model are positive.

6.5 An Initial Look at the 2025–2026 Tariffs

Our counterfactuals ask what would have happened in the 2018 market had the tariffs threatened that year been imposed. In 2025–2026 the United States enacted a broadly similar policy of duties on imported vehicles and parts, which lets us ask whether the mechanisms that drive our 2018 analysis appear in realized data. The comparison is imperfect in two ways: the realized policy differs from our threatened-tariff experiment, and it struck a later market with a different product mix and demand. We therefore read the comparison as a check on the model’s mechanisms rather than an exact quantitative validation of its predictions. All details on data construction, the 2025 tariff schedule, and implementation can be found in Appendix G.

To obtain a first, real-time measure of market participants’ reactions, we examine the responses of automobile manufacturers’ stock prices. These responses were large. Around the March 26, 2025 Section 232 proclamation, foreign manufacturers’ stock prices declined 8.6% on average over a seven-trading-day window and suppliers’ stock prices declined 3.3%, while the stock prices of US manufacturers (a mixed group that includes less import-exposed and electric-vehicle producers) moved little. The same group pattern held at the April 3 effective date (foreign -9.5% , suppliers -5.0% , US roughly flat), with substantial firm-level heterogeneity by exposure: Tesla rose 14.6% while Volkswagen, Ford, and GM fell 11.6, 3.0, and 2.3%. The market thus moved in the direction implied by the model, with the most import-exposed firms suffering the largest stock-price losses.

In the year following the tariff announcement, realized prices and quantities responded very differently from one another. Import quantities declined sharply. Vehicle import shipments from April through December 2025 were 24% below the same months of 2024, with the declines broadly reflecting the tariff exposure of the country of assembly; shipments

remained depressed in early 2026. Despite the sharp stock-price reaction at the announcement and the reduction in imports, retail car prices over the following year responded far more weakly (see Panel A in Table 9). The BLS new-vehicle CPI (which reflects shifts in the composition of vehicles purchased and tracks transaction prices) rose 0.1% from February 2025 to May 2026. To build a more appropriate yardstick for our model’s predictions, we construct a panel of MSRPs for 100 car models spanning model years 2025 and 2026, and match these observations to our 2018 data.²⁹ Table 9 reports observed and predicted price changes on this common sample, using either equal weights across car models or weights proportional to their 2018 sales. Observed MSRP increases average 2.4% under equal weights and 1.6% under sales weights, while our C2 counterfactual would have predicted 14.6% and 10.5% increases, respectively.

TABLE 9: From the Stylized Experiment to the Enacted Policy

	Price change (%)		Import units (%)
	Equal weights	2018 sales weights	Total change
<i>Panel A: Observed</i>			
Import shipments, Apr.–Dec. 2025 vs. 2024			–24
BLS new-vehicle CPI, Feb. 2025 – May 2026	+0.1 (official index)		
Matched-trim MSRP, MY2025 → MY2026	+2.4	+1.6	
<i>Panel B: Model (C2, Cournot with economies of scale)</i>			
(a) Stylized experiment: uniform 25%, no offset	+14.6	+10.5	–51
(b) Enacted rates on cars and parts	+7.7	+5.4	–31
(c) Adding the assembler offset	+5.6	+2.9	–40
(d) Adjusting the customs-value base	+5.3	+2.8	–38

In Panel A, observed imports are HTS 8703 border shipments. Observed MSRP changes and model price predictions use the same 100 matched car models. Equal weights assign the same weight to each car model; sales weights use 2018 sales. Observed changes compare matched car models between 2025 and 2026. In Panel B, model-predicted imports are the change in total US sales of all foreign-assembled models. The duty-base weight is 0.80 in rows (b)-(c) and 0.77 in row (d).

Another key difference between our counterfactual environment and the 2025 tariffs is that the enacted policy was milder than a uniform 25%, in three ways; the rows of Panel B in Table 9 take these up in turn. First, in row (b) we adjust the rates: countries negotiated vehicle duties down from the 25% baseline (15% for the European Union, Japan, and Korea, and 10% up to a negotiated quota for the United Kingdom), duties on USMCA-compliant vehicles were applied only on their non-US content, and tariffs on USMCA-compliant parts, which account for much of the imported content in US assembly, were never levied at all. Second, in row (c), we give US-assembled models an offset against parts duties fixed at

²⁹The matched observations account for 80.6% of 2018 sales of US-assembled vehicles and 39.5% of sales of foreign-assembled vehicles.

3.75% of 2018 MSRP.³⁰ Third, in row (d) we allow the duty base itself to adjust, as declared customs values of imported vehicles fell about 4% once the duty took effect, consistent with related-party transfer prices adjusting to the tariff. Under equal weights, these adjustments lower the predicted price increase from 14.6% in row (a) to 7.7% in row (b), 5.6% in row (c), and 5.3% in row (d). Under 2018 sales weights, the corresponding predictions are 10.5%, 5.4%, 2.9%, and 2.8%. Reflecting the enacted policy thus closes a substantial portion of the price gap; the quantity column shows the same pattern: after adjusting for the exact implementation, import shipments fell 24% against a predicted 38%.

The remaining gap may reflect forces outside our model. First, the 2025 market reacted in ways our 2018 counterfactual does not consider. The offered product set has become markedly more US-assembled as low-volume import variants exited, and demand growth for electric vehicles is concentrated in US-assembled models. The fleet hit by the 2025 tariff therefore increasingly differs from the fleet on which our predictions are computed. Second, firms had reasons to delay repricing in the short run. Tariff rates fell repeatedly within the year as agreements were negotiated, so manufacturers hesitated to adjust MSRPs in response to a shock that might shrink further,³¹ and pre-tariff inventories together with the model-year pricing cycle meant tariff-related costs reached posted prices only with a lag. Consumers likewise anticipated purchases ahead of the duties, a within-year timing shift that let much of 2025's sales clear at pre-tariff prices.³² Manufacturers also appear to have held prices down under political pressure: the Trump administration explicitly warned automakers against raising prices in response to the tariffs,³³ a constraint on pricing absent from any equilibrium model. Consistent with firms absorbing the shock rather than passing it on, the 2025 Form 10-K filings of Ford show profits falling by \$2 billion due to the tariffs. With costs mounting, firms in 2026 have begun to signal that they can no longer shield buyers, and some have started to raise prices.³⁴ The transition from the short run to the medium run appears to be underway.

³⁰Because the offset accrues only to US assembly, it operates as a subsidy to domestic production. Under economies of scale, US-assembled models see costs fall and expand production while imports lose scale and see costs rise.

³¹In spring 2025, several manufacturers publicly committed to hold list prices temporarily; e.g., Hyundai's Customer Assurance program, launched "in response to dynamic market conditions and the potential impact of tariffs," guaranteed current MSRPs from April 4 through June 2, 2025 (Hyundai Motor America press release, April 4, 2025, <https://www.hyundainews.com/en-us/releases/4418>).

³²See "Automakers are eating the cost of tariffs — for now," NPR, July 30, 2025, <https://www.npr.org/2025/07/30/nx-s1-5482857/automakers-eating-tariff-costs>, documenting both the pre-tariff inventories on dealer lots and the pull-forward of purchases.

³³See "Trump Warns Auto Makers Not to Raise Prices," *Wall Street Journal*, 2025, <https://www.wsj.com/business/autos/trump-tariffs-automaker-prices-warning-928bc7a9>.

³⁴See "Experts: Tariffs to Hit Auto Prices," *Detroit Free Press*, March 27, 2026, p. A1.

7 Conclusion

This paper makes two key contributions. First, we develop a new method for testing firm conduct under economies of scale, revealing that Cournot competition with substantial scale effects best characterizes the US automobile market. Our approach provides a framework applicable to other manufacturing sectors where economies of scale matter, and distinguishing the correct model of supply (conduct and cost) from data may help shape credible policy analysis. Second, we demonstrate how stacked tariffs propagate through global value chains, producing effects that would be hard to predict without a granular evaluation of the foreign parts content of domestic vehicles. Our counterfactual analysis shows how tariffs on imported vehicles alone induce price decreases for domestic cars and welfare gains for their buyers, while additional tariffs on parts reverse these effects, causing domestic prices to rise by 6.6% and sharply increasing total welfare losses to \$28.4 billion. The employment gains from such policies come at a substantial welfare cost, rising from \$171,700 per job under car-only tariffs to \$761,100 per job when parts tariffs are included. These findings highlight the importance of accounting for both conduct and global value chains when evaluating trade policy, especially in industries with complex international production networks.

Our analysis is subject to several caveats that echo concerns raised in [Berry et al. \(1999\)](#). First, we employ a static model on both demand and supply, abstracting from potentially important dynamic considerations. On the demand side, consumers may anticipate or delay purchases in expectation of future price changes. On the supply side, important dynamic aspects of manufacturers' decision-making include capacity adjustments and product development. Second, we do not model the used car market, which may serve as an important substitute for new vehicles. For both limitations, our focus on medium-run effects provides some justification: production relocation and product development face substantial frictions in this time frame, while the used vehicle stock remains relatively fixed. Third, our counterfactuals involve substantial price changes, requiring extrapolation beyond observed variation. While such extrapolation always requires caution, we take some comfort in the fact that the demand system from [Grieco et al. \(2024\)](#) was estimated using long panel variation capturing substantial price movements, and our assumption of stable preferences is standard. These limitations suggest our results should be interpreted as informative but incomplete estimates of automotive trade policy effects.

Finally, while our counterfactuals describe the 2018 market, we take an initial look at the broadly similar tariffs enacted in 2025–2026. A growing literature evaluates the accuracy of structural predictions against ex-post realizations, both for trade models (e.g., [Adão, Costinot, and Donaldson, 2025](#)) and for merger simulations (e.g., [Bhattacharya, Illanes, Kreps, Salas, and Stillerman, 2025](#)); a key finding of the latter is that supply-side

misspecification, particularly an incorrect model of firm conduct, is an important source of prediction error. In our setting, realized retail price changes in the short run did not match the medium-run responses our counterfactuals predict: firms muted their price increases and absorbed much of the tariff, a margin that lies outside our model. Even so, the price gap shrinks considerably once the counterfactual reflects the policy as enacted. Once the market has adjusted further toward the medium run, our method could be used for a retrospective exercise, speaking to the importance of data-driven selection of the conduct model and cost structure for accurate policy predictions.

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Online Appendix

Appendix A General Falsification Problem: Non-constant Marginal Cost with Differentiated Products

In Section 4.2 of the paper, we focus on the problem of instrument relevance for testing models of conduct in the context of our application: distinguishing between particular conduct models (Bertrand, Cournot, or a hybrid) under a functional form assumption for marginal costs.

In this appendix, we develop a more formal and general analysis of what instruments are needed to distinguish models of conduct under scale economies. We adopt here the falsification lens of [Berry and Haile \(2014\)](#), who show that positing a model of conduct yields falsifiable restrictions in differentiated products markets, even under non-constant marginal costs. This lens is useful because [Duarte et al. \(2024\)](#) establishes that instruments are irrelevant for testing a pair of models with the RV test when neither model is falsified by the instruments, so that $E[\omega_{mjt} \mid \mathbf{w}_{jt}, \mathbf{z}_{jt}] = 0$ for $m = 1, 2$. Building on the framework in [Dearing et al. \(2024\)](#), we characterize the economic determinants of instrument relevance for our testing for a broader range of models under general economies of scale.

A.1 Preliminaries

As in the text, we assume data in each market are generated by equilibrium play in a true model of firm behavior $p_t = \Delta_{0t} + c_{0t}$. We normalize market size to 1, so that we may use s_{jt} and q_{jt} interchangeably. Each candidate model m yields its own set of implied markups Δ_{mt} as a function of observables and demand primitives; the marginal costs implied by that model, c_{mt} , are then calculated as $p_t - \Delta_{mt}$. We require the following for any model m to have well-defined first-order conditions:

Assumption 1. (*Equilibrium Uniqueness*) For any model m , including the true model, either: (i) A unique equilibrium exists, or (ii) The equilibrium selection rule consistently yields the same p_t for any given (c_{mt}, x_t, ξ_t) .

This assumption is analogous to Assumption 13 in [Berry and Haile \(2014\)](#).

Demand and cost may include random shocks, varying across markets, that are unobserved by the researcher. We now assume that marginal costs follow a general functional form separable in the unobserved shock and a function of the observable cost shifters $\mathbf{w}_{jt}$ and own quantities, or $c_{0jt} = \bar{c}_{0j}(q_{jt}, \mathbf{w}_{jt}) + \omega_{0jt}$.³⁵ While this cost function departs from

³⁵In principle, the function $\bar{c}_{0j}$ can depend on the full vector q_t , or lags. Restricting $\bar{c}_{0j}$ to depend on

the functional form used in the body of the paper, which was chosen for its applicability to the car market, it allows us to maintain the falsifiable restriction in [Berry and Haile \(2014\)](#) which assumes that marginal cost is additively separable in ω_{0jt} . The researcher can observe w_{jt} , but does not know the function $\bar{c}_{0j}$ or observe ω_{0jt} . To falsify incorrect models, instruments z_{jt} are constructed, which are assumed to be mean independent of unobserved cost shocks under the true model:

Assumption 2. (*Instrument Exogeneity*) For each j , $c_{0jt} = \bar{c}_{0j}(q_{jt}, w_{jt}) + \omega_{0jt}$, and z_{jt} is a vector of d_z excluded instruments such that $E[\omega_{0jt} \mid w_{jt}, z_{jt}] = 0$.

For the true model, $E[\omega_{0jt} \mid w_{jt}, z_{jt}] = 0$ by Assumption 2, or equivalently, $E[p_{jt} - \Delta_{0jt} - \bar{c}_{0j}(q_{jt}, w_{jt}) \mid w_{jt}, z_{jt}] = 0$ for the true cost function $\bar{c}_{0j}(\cdot, \cdot)$. For a candidate model m and candidate cost function $\bar{c}_{mj}$, then, defining $\omega_{mjt} = p_{jt} - \Delta_{mjt} - \bar{c}_{mj}(q_{jt}, w_{jt})$, the condition $E[\omega_{mjt} \mid w_{jt}, z_{jt}] = 0$, which adapts the falsifiable restriction in Theorem 9 of [Berry and Haile \(2014\)](#) to our parametric setting, serves as a falsifiable restriction. If any collection of cost functions $\bar{c}_{mj}(\cdot, \cdot)$ allows this restriction to be satisfied, we say that model m is *not falsified by the instruments* z_{jt} . If no cost functions $\{\bar{c}_{mj}(q_{jt}, w_{jt})\}_{j=1}^J$ satisfy this restriction almost surely over w_{jt} and z_{jt} , then model m is falsified.

Since prices in the data are generated by the true model, we can rewrite the implied cost shock as $\omega_{mjt} = \Delta_{0jt} - \Delta_{mjt} + \bar{c}_{0j}(q_{jt}, w_{jt}) - \bar{c}_{mj}(q_{jt}, w_{jt}) + \omega_{0jt}$. Thus, the condition for falsification is as follows:³⁶

Lemma 1. *Under Assumptions 1 and 2, model m is falsified by instruments z_{jt} if and only if for some j there exists no function $\bar{c}_{mj}$ such that*

$$E[\Delta_{0jt} - \Delta_{mjt} \mid w_{jt}, z_{jt}] = E[\bar{c}_{mj}(q_{jt}, w_{jt}) - \bar{c}_{0j}(q_{jt}, w_{jt}) \mid w_{jt}, z_{jt}] \quad a.s. \quad (9)$$

Here, as in the rest of the section, the expectation is over unobservables, and *a.s.* (*almost surely*) is over realizations of the observables (w_{jt}, z_{jt}) .

A.2 Intuition: Constant vs Non-Constant Marginal Costs

The expectations in Equation (9) are taken conditional on the realization of w_{jt} , so a product's own cost shifters cannot offer any additional variation with which to form instruments. Since $\bar{c}_{mj}$ and $\bar{c}_{0j}$ are functions only of w_{jt} and q_{jt} , any instrument can only move $\bar{c}_{mj}$ and $\bar{c}_{0j}$ through its effect on own quantity q_{jt} .

To build intuition, we will first consider the case, explored in [Dearing et al. \(2024\)](#), where marginal costs do not depend on quantity produced. In that case, $\bar{c}_{mj}$ and $\bar{c}_{0j}$ can't

own quantity simplifies the exposition, and is in line with our application. Extensions, including economies of scope, are straightforward and generate no further qualitative insight.

³⁶Proofs of all lemmas and propositions are in Appendix A.6.

depend on the instruments at all, so the right-hand side of Equation (9) is constant in z_{jt} ; falsification therefore depends on whether an instrument moves Δ_{0jt} and Δ_{mjt} differentially. The following example illustrates how this would work.

Example 1: In this and the subsequent examples, we consider a simple environment with two single-product firms and logit demand, so that market shares $j \in \{1, 2\}$ are given by

$$s_{jt} = \frac{\exp(x_{jt}\beta - \alpha p_{jt})}{1 + \exp(x_{1t}\beta - \alpha p_{1t}) + \exp(x_{2t}\beta - \alpha p_{2t})},$$

where x_{jt} are characteristics of product j in market t and α and β are coefficients. We suppose that the true conduct model is Cournot, and we wish to falsify Bertrand competition.

For this example, consider the case where true marginal cost is constant in quantity and linear in a scalar w_{jt} , or $\bar{c}_{0j} = w_{jt}\tau$, and hold unobservables fixed for ease of exposition. The researcher forms an instrument using variation in the cost shifter of firm 2, $z_{1t} = w_{2t}$. Because cost shifters do not enter the markup function of either model, z_{jt} affects Δ_{0jt} and Δ_{mjt} only through its effect on observed prices p_t .

Under the true (Cournot) model, the cost pass-through matrix can be calculated to be

$$P_{Ct} = \begin{bmatrix} \frac{s_{0t}}{1-s_{2t}} & 0 \\ 0 & \frac{s_{0t}}{1-s_{1t}} \end{bmatrix}$$

and therefore a change in $z_{1t} = w_{2t}$ only changes p_{2t} , not p_{1t} . Firm 1's markup under the two models can be calculated to be $\Delta_{01t} = \Delta_{C1t} = \frac{1}{\alpha} \left(1 + \frac{s_{1t}}{s_{0t}}\right)$ and $\Delta_{m1t} = \Delta_{B1t} = \frac{1}{\alpha} \left(1 + \frac{s_{1t}}{1-s_{1t}}\right)$, where $s_{0t} = 1 - s_{1t} - s_{2t}$. It's useful to rewrite these as

$$\Delta_{01t} = \frac{1}{\alpha} (1 + \exp(x_{1t}\beta - \alpha p_{1t})) \quad \text{and} \quad \Delta_{m1t} = \frac{1}{\alpha} \left(1 + \frac{\exp(x_{1t}\beta - \alpha p_{1t})}{1 + \exp(x_{2t}\beta - \alpha p_{2t})}\right)$$

which makes it clear that an increase in p_{2t} (following an increase in c_{2t}) will increase Δ_{m1t} but not Δ_{01t} . Thus, the left-hand side of Equation (9) changes with z_{1t} while the right-hand side does not, so equality cannot be maintained for different values of z_{1t} , falsifying the wrong model. •

Falsification becomes more difficult when $\bar{c}_{0j}$ and $\bar{c}_{mj}$ are allowed to depend on the quantity produced: since q_{1t} changes in response to a change in the instrument, the change in the left-hand side of Equation (9) can potentially be matched by a change in the right-hand side due to the differential dependence of $\bar{c}_{mj}$ and $\bar{c}_{0j}$ on q_{jt} . In fact, with a single instrument, falsification may be impossible, as a cost function $\bar{c}_{mj}$ for a misspecified model $\Delta_m \neq \Delta_0$ can potentially be constructed to satisfy Equation (9), as stated in Remark 1 for

the log marginal cost case. This is further illustrated in the next example:

Example 2: Suppose that true marginal cost does not depend on the quantity produced, $\bar{c}_{0j} = w_{jt}\tau$, but the researcher does not know this, and must allow for the possibility that $\bar{c}_{mj}$ can depend on q_{jt} as well as w_{jt} . Consider again using a scalar rival cost shifter instrument $z_{1t} = w_{2t}$ to falsify Bertrand competition when the true model is Cournot. The lack of falsification is easiest to see in the case where w_{2t} is the only variation across markets, i.e., unobservables and w_{1t} are held fixed. With only w_{2t} varying, there is a one-to-one mapping from the realization of w_{2t} to the realizations of both prices p_t and market shares q_t ,³⁷ so whatever variation in c_{1t} is needed to rationalize the incorrect model m can be attributed to the dependence of $\bar{c}_{m1}$ on q_{1t} ; letting $w_2(s)$ denote the value of w_{2t} which would lead to market share $q_{1t} = s$, and suppressing the dependence on w_{1t} since it is fixed, the cost function

$$\bar{c}_{m1}(q_{1t}) = \bar{c}_{01}(q_{1t}) + E(\Delta_{01t} - \Delta_{m1t} | w_{2t} = w_2(q_{1t}))$$

along with

$$\bar{c}_{m2}(w_{2t}, q_{2t}) = \bar{c}_{02}(w_{2t}, q_{2t}) + E(\Delta_{02t} - \Delta_{m2t} | w_{2t})$$

mechanically satisfies Equation (9), and therefore falsification fails (Lemma 1). (In the absence of variation in unobservables or w_{1t} , the difference in markups $\Delta_{0jt} - \Delta_{mjt}$ is a deterministic function of w_{2t} , and the expectations on the right-hand side are degenerate, but are written this way for consistency. Also note that if true marginal cost $\bar{c}_{0j}$ does depend on q_{jt} , falsification still fails, it's just less transparent because even with diagonal model-implied passthrough P_{0t} , the cost shifter w_{2t} moves both prices by changing both firms' marginal costs via their market shares.)

With variation in both firms' cost shifters and unobservables, things get a little less transparent, as $\bar{c}_{mj}$ can only depend on w_{jt} and q_{jt} , and must match its "target" in expectation over unobservables and for each realization of other observables; however, with just one instrument, a rationalizing cost function can sometimes exist, and is guaranteed to exist under certain conditions,³⁸ so falsification may fail. •

A.3 Recasting in terms of Marginal Effects

As in Dearing et al. (2024), to better understand the economic content of the falsifiable restriction and the determinants of falsification, it's useful to restate Lemma 1 in terms of

³⁷This is particularly straightforward since passthrough under the true model (Cournot) is diagonal, and we can calculate $\frac{dp_t}{dz_{1t}} = \begin{bmatrix} 0 \\ \tau \frac{s_{0t}}{1-s_{1t}} \end{bmatrix}$, and $\frac{dq_t}{dz_{1t}} = \tau \frac{s_{0t}}{1-s_{1t}} \begin{bmatrix} \alpha s_{1t} s_{2t} \\ -\alpha s_{2t} (1-s_{2t}) \end{bmatrix}$ since $\frac{\partial \bar{c}_{02}}{\partial w_{2t}} = \tau$.

³⁸Specifically, when $E[\Delta_{0jt} - \Delta_{mjt} | w_{jt}, q_{jt}, z_{jt}] = E[\Delta_{0jt} - \Delta_{mjt} | w_{jt}, q_{jt}]$ – that is, if the instrument moves $\Delta_{0jt} - \Delta_{mjt}$ only through a product's own cost shifter and quantity – the cost functions $\bar{c}_{mj}(w_{jt}, q_{jt}) = \bar{c}_{0j}(w_{jt}, q_{jt}) + E(\Delta_{0jt} - \Delta_{mjt} | w_{jt}, q_{jt})$ will satisfy the condition in Lemma 1 for falsification to fail.

the marginal impacts of the instruments on markups. We first ensure that both sides of Equation (9) can be differentiated with respect to the continuous-valued instruments:³⁹

Assumption 3. (*Smoothness*) For any model m under consideration, including the true one $m = 0$, $E[\Delta_{mjt} \mid \mathbf{w}_{jt}, z_{jt}]$ is differentiable in each continuously-varying instrument z_{jt} ; any cost function under consideration including the true one is differentiable in quantity; and all these derivatives have integrable bounds.

For Equation (9) to hold at different values of z_{jt} , a change in z_{jt} must have the same marginal impact on the two sides of the equation, leading to the following:

Proposition 1. *Suppose that Assumptions 1, 2, and 3 hold. Then model m is falsified by the instruments z_{jt} if for some j there exists no function $\bar{c}_{mj}$ such that*

$$E \left[\frac{d\Delta_{0jt}}{dz_{jt}^{(k)}} - \frac{d\Delta_{mjt}}{dz_{jt}^{(k)}} \mid \mathbf{w}_{jt}, z_{jt} \right] = E \left[\left(\frac{\partial \bar{c}_{mj}(q_{jt}, \mathbf{w}_{jt})}{\partial q_{jt}} - \frac{\partial \bar{c}_{0j}(q_{jt}, \mathbf{w}_{jt})}{\partial q_{jt}} \right) \frac{dq_{jt}}{dz_{jt}^{(k)}} \mid \mathbf{w}_{jt}, z_{jt} \right] \quad (10)$$

a.s. for all k .

Conversely, if all instruments are continuous-valued and have connected support and cost functions $\bar{c}_{mj}$ exist satisfying Equation (10) for all k , then model m is not falsified, so under these added assumptions Proposition 1 can be thought of as if-and-only-if. (If there are some discrete-valued instruments, the cost functions $\bar{c}_{mj}$ must also satisfy Equation (9) for each realization of these instruments for model m to not be falsified.)

As noted above, since the expectations in Equation (9) are conditional on a realization of $\mathbf{w}_{jt}$, the instruments can only change the right-hand side through q_{jt} , hence the right-hand side of Equation (10). In the case of constant marginal costs, the right-hand side of Equation (10) is zero, and falsification depends on whether the left-hand side is zero. As for the left-hand side, instruments can affect Δ_{0jt} and Δ_{mjt} in two ways. Markups are typically expressed as functions of prices, market shares, and price elasticities; depending on the instrument, it may have a direct effect on markups (a product characteristic affecting market share even at fixed prices), and an additional effect through its effect on prices. Defining $P_{mt} = \left(I - \frac{\partial \Delta_m}{\partial p} \right)^{-1}$ as the conduct-implied cost pass-through matrix for model m (and likewise for the true model $m = 0$), we can explicitly decompose the left-hand side of Equation (10) into these two effects, as

$$E \left[\frac{\partial \Delta_{0jt}}{\partial z_{jt}^{(k)}} - \frac{\partial \Delta_{mjt}}{\partial z_{jt}^{(k)}} \mid \mathbf{w}_{jt}, z_{jt} \right] + E \left[\left(P_{mt}^{-1} - P_{0t}^{-1} \right)_j \frac{dp_0}{dz_{jt}^{(k)}} \mid \mathbf{w}_{jt}, z_{jt} \right],$$

³⁹ Assumption 3 holds for all models in our examples and application.

where the first term denotes the difference in the direct effects of instruments on markups, while the second denotes the difference in indirect effects through prices, which depends on the inverse pass-through matrices of the two models and the effect of instruments on prices under the true model.⁴⁰ Except in knife-edge cases where the two effects cancel out, falsification with constant marginal costs can be achieved if either the indirect or direct effect is non-zero. (In Example 1 above, there was no direct effect since costs do not enter directly into markups, but the indirect effect was nonzero, allowing falsification.) On the other hand, falsification with non-constant marginal costs will typically fail with one instrument, as in Example 2 above, since $\frac{\partial \bar{c}_{mj}}{\partial q_{jt}}$ can be chosen judiciously to satisfy Equation (10).

A.4 Non-Constant Costs and Multiple Instruments

A possible solution when marginal costs depend on quantity is to use additional instruments. The next example shows why this can work:

Example 3: Suppose again that demand is logit, $\bar{c}_{0j} = w_{jt}\tau$, and the researcher does not know there are no economies of scale and must allow for $\bar{c}_{mj}$ to depend on q_{jt} . The researcher again wants to falsify the Bertrand model when the truth is Cournot. This time, the researcher chooses two instruments: a rival cost shifter, $z_{1t}^{(1)} = w_{2t}^{(1)}$, and an own product characteristic which is excluded from cost, $z_{1t}^{(2)} = x_{1t}^{(1)}$. Let $\tau^{(1)}$ be the coefficient with which $w_{2t}^{(1)}$ enters $\bar{c}_{02}$, and $\beta^{(1)}$ the coefficient with which $x_{1t}^{(1)}$ enters into logit demand, and assume for exposition that $\tau^{(1)}, \beta^{(1)} > 0$. From Proposition 1, we can falsify model m unless a cost function $\bar{c}_{m1}$ exists satisfying

$$E \left[\frac{d\Delta_{01t}}{dz_{1t}^{(k)}} - \frac{d\Delta_{m1t}}{dz_{1t}^{(k)}} \mid w_{1t}, z_{1t} \right] = E \left[\left(\frac{\partial \bar{c}_{m1}}{\partial q_{1t}} - \frac{\partial \bar{c}_{01}}{\partial q_{1t}} \right) \frac{dq_{1t}}{dz_{1t}^{(k)}} \mid w_{1t}, z_{1t} \right] \quad (11)$$

almost surely for both instruments. We'll show why no such cost function can exist, and therefore why the model is falsified, again focusing on the case where $\frac{\partial \bar{c}_{01}}{\partial q_{1t}} = 0$ for clearest intuition. In that case, note first that $dq_{1t}/dz_{1t}^{(k)} > 0$ for both instruments—under the true model (Cournot), an increase in a rival's cost, and an increase in one's own product quality, both result in a higher own market share in equilibrium. The first occurs because, under Cournot, passthrough is diagonal, so an increase in $w_{2t}^{(1)}$ results in an increase in p_{2t} and no change in p_{1t} , increasing q_{1t} . The latter is more subtle, because an increase in own quality

⁴⁰Note that while the matrices P_{mt} and P_{0t} give the "initial" pass-through of costs to equilibrium prices, they exclude the additional feedback loop where these changes in prices would lead to changes in market shares, therefore changes in marginal costs, which would yield an additional change in equilibrium prices. The term $\frac{dp_0}{dz_{jt}^{(k)}}$, on the other hand, is the full effect of a change in instrument values on equilibrium prices, including this additional effect which is absent when marginal costs don't depend on quantity.

results in an increase in own price; but the equilibrium price adjustment is small enough that $x_{1t}\beta - \alpha p_{1t}$ increases with an increase in $x_{1t}^{(1)}$. With no change in p_{2t} , this then leads to an increase in q_{1t} as well.⁴¹

Again, rewriting firm 1's markup under the two models as

$$\Delta_{01t} = \frac{1}{\alpha} (1 + \exp(x_{1t}\beta - \alpha p_{1t})) \quad \text{and} \quad \Delta_{m1t} = \frac{1}{\alpha} \left(1 + \frac{\exp(x_{1t}\beta - \alpha p_{1t})}{1 + \exp(x_{2t}\beta - \alpha p_{2t})} \right)$$

allows us to see the effect of each instrument on $\Delta_{01t} - \Delta_{m1t}$:

- An increase in $w_{2t}^{(1)}$, by increasing p_{2t} without changing p_{1t} , does not change Δ_{01t} , but increases Δ_{m1t} . Thus, for the first instrument, the left-hand side of Equation (11) is negative.
- An increase in $x_{1t}^{(1)}$, by increasing $x_{1t}\beta - \alpha p_{1t}$ without changing $x_{2t}\beta - \alpha p_{2t}$, increases both Δ_{01t} and Δ_{m1t} , but it increases Δ_{01t} *by more*; so for the second instrument, the left-hand side of equation (11) is *positive*.

This makes it clear why the two instruments together suffice to falsify the wrong model: since $\frac{dq_{1t}}{dz_{1t}^{(k)}}$ is positive for both instruments and (by assumption) $\frac{\partial \bar{c}_{01}}{\partial q_{1t}} = 0$, for each realization of observables, the expected value of $\frac{\partial \bar{c}_{m1}}{\partial q_{1t}}$ would need to be strictly negative to satisfy the exclusion restriction for the first instrument, but strictly positive to satisfy the exclusion restriction for the second instrument.

It's useful to return to the economics of the situation for further intuition. An increase in firm 2's cost under the true model leads to an increase in p_{2t} but no change in p_{1t} ; because rival's cost passthrough is positive under Bertrand, however, the Bertrand model "sees" the lack of a change in p_{1t} as evidence of a partly offsetting decrease in c_{1t} , expressed as an increase in firm 1's markup. Since firm 1's cost shifter did not change, this decrease in marginal costs must be attributed to the increased level of output, requiring $\frac{\partial \bar{c}_{m1}}{\partial q_{1t}}$ to be smaller than its true value of 0. On the other hand, when firm 1's quality increases and price adjusts, this leads to a smaller change in the measured markup under Bertrand than under Cournot, or an *increase* in imputed marginal cost under Bertrand relative to Cournot; since (again) w_{1t} did not change, this greater marginal cost increase must now be attributed to the increase in output, requiring $\frac{\partial \bar{c}_{m1}}{\partial q_{1t}}$ to this time be *larger* than its true value of 0. Since $\frac{\partial \bar{c}_{m1}}{\partial q_{1t}}$ can't simultaneously be less than and greater than 0, the model is falsified.

When $\frac{\partial \bar{c}_{01}}{\partial q_{1t}} \neq 0$, the logic is more complicated (since each instrument has an additional effect operating through the change in both firms' marginal costs), but one can again show that no single cost function $\bar{c}_{m1}$ can satisfy the exclusion restrictions for both instruments,

⁴¹To be more thorough, under Cournot, $dp_t/dw_{2t}^{(1)} = \tau^{(1)} [s_{0t}/(1-s_{1t})]$ and $dp_t/dx_{1t}^{(1)} = \frac{\beta^{(1)}}{\alpha} [s_{1t}/(1-s_{2t})]$. From the latter, $d(x_{1t}\beta - \alpha p_{1t})/dx_{1t}^{(1)} = \beta^{(1)} - \alpha \frac{\beta^{(1)}}{\alpha} s_{1t}/(s_{0t} + s_{1t}) > 0$.

and the wrong model is therefore falsified. •

This example makes falsification particularly transparent, because the left-hand side of Equation (11) is positive for one instrument and negative for the other, while $\frac{dq_{1t}}{dz_{1t}^{(k)}}$ has the same sign for both instruments. Things won't always be this clean. Still, this illustrates the more general point that when two instruments affect the environment through economically distinct channels, there is no reason they should change the difference in measured markups between two models identically; and if they don't (relative to how they shift own market share), then the requirements for a cost function to fit the observed changes will be different for the two instruments, and no single cost function will be able to "satisfy" both, allowing the model to be falsified.

Generalization to richer cost functions The condition in Equation (7) ensures that the two instruments separately imply different values of the economies-of-scale coefficient, so no single value can satisfy both moment restrictions. It provides a parametric analogue to the argument in Section 6, Case 2 of [Berry and Haile \(2014\)](#): a candidate model can be falsified when two distinct subsets of instruments, each sufficient to identify the marginal-cost function under that model, imply different cost coefficients and hence different implied cost shocks. This argument, which is not tied to the log-log specification in Equation (5), extends naturally to the paper's parametric IV estimator when the marginal-cost equation contains K endogenous regressors.

Let $\mathbf{c}_{mjt} = f(p_{jt} - \Delta_{mjt})$ denote a known transformation of the marginal cost implied by model m . Let $\mathbf{q}_{jt}$ be a K -vector of known functions of the relevant production quantities. These K endogenous regressors may arise either from distinct underlying endogenous production objects or from multiple transformations of the same object. For example, a specification allowing for both scale and scope economies might include own-model production and firm- or platform-level production, whereas a quadratic-cost specification includes q_{jt} and q_{jt}^2 , two transformations of the same underlying quantity.

The cost equation then becomes $\mathbf{c}_{mjt} = \mathbf{q}_{jt}'\gamma_m + w_{jt}'\tau_m + \omega_{mjt}$. Let tildes denote residuals from population projections on w_{jt} . Given a set of $L > K$ excluded instruments, consider all K -instrument subsets indexed by S for which $E[\tilde{Z}_{S,jt}\tilde{\mathbf{q}}_{jt}']$ is nonsingular, and suppose that every instrument belongs to at least one such subset. Each subset S implies

$$\gamma_m^{(S)} = \left(E[\tilde{Z}_{S,jt}\tilde{\mathbf{q}}_{jt}']\right)^{-1} E[\tilde{Z}_{S,jt}\tilde{\mathbf{c}}_{mjt}].$$

If all such subsets imply a common coefficient vector γ_m , then all L moment restrictions are consistent with the same cost specification. The instruments are therefore not economically distinct and are irrelevant for testing model m . Equivalently, the instruments are

economically distinct if and only if

$$\text{rank} \left[E[\tilde{Z}_{jt} \tilde{\mathbf{q}}'_{jt}], E[\tilde{Z}_{jt} \tilde{\mathbf{c}}_{mjt}] \right] = K + 1.$$

Thus, at least $K + 1$ excluded instrument directions are necessary: K linearly independent directions to identify the cost coefficients and an additional direction outside their span to test conduct. Instruments operating through different economic channels will generically satisfy these conditions in differentiated-products settings. We illustrate with two examples.

Example: Economies of scale and scope. Suppose marginal cost takes the form

$$\log(c_{0jt}) = \gamma_1 \log(q_{jt}) + \gamma_2 \log(Q_{jt}^-) + w'_{jt} \tau_0 + \omega_{0jt},$$

where q_{jt} is own-model production and Q_{jt}^- is the total production of other models by the same firm on the same platform. In the notation above, $f(c) = \log(c)$,

$$\mathbf{q}_{jt} = \begin{pmatrix} \log(q_{jt}) \\ \log(Q_{jt}^-) \end{pmatrix}, \quad \mathbf{c}_{mjt} = \log(p_{jt} - \Delta_{mjt}).$$

Both q_{jt} and Q_{jt}^- are endogenous, so estimating γ_1 and γ_2 absorbs two directions of variation from the instrument set. At least three economically distinct instrument directions are therefore needed. The ratio condition in Equation (7) becomes the rank condition

$$\text{rank} \left[E[\tilde{Z}_{jt} \tilde{\mathbf{q}}'_{jt}], E[\tilde{Z}_{jt} \tilde{\mathbf{c}}_{mjt}] \right] = 3.$$

As with the single-endogenous-regressor case, instruments operating through different economic channels will generically satisfy this condition in differentiated-products settings. For instance, a rival cost shifter, an own-product characteristic, and a measure of the number of rival models on competing platforms provide three sources of variation entering the equilibrium system at different points. These instruments can generate linearly independent variation in own-model production, platform production, and model-implied marginal cost, satisfying the rank condition.

Example: Quadratic cost. Suppose marginal cost instead takes the form

$$c_{0jt} = \gamma_1 q_{jt} + \gamma_2 q_{jt}^2 + w'_{jt} \tau_0 + \omega_{0jt}.$$

In this case, $f(c) = c$,

$$\mathbf{q}_{jt} = \begin{pmatrix} q_{jt} \\ q_{jt}^2 \end{pmatrix}, \quad \mathbf{c}_{mjt} = p_{jt} - \Delta_{mjt}.$$

Both q_{jt} and q_{jt}^2 are endogenous regressors. Again, estimating γ_1 and γ_2 absorbs two directions of variation from the instrument set, so at least three economically distinct instrument directions are needed to retain variation for testing conduct. The rank condition is

$$\text{rank} \left[E[\tilde{Z}_{jt} \tilde{q}_{jt}'], E[\tilde{Z}_{jt} \tilde{c}_{mjt}] \right] = 3.$$

Equivalently, because $\mathbf{c}_{mjt} - \mathbf{c}_{0jt} = \Delta_{0jt} - \Delta_{mjt}$, the residualized instrument correlations with q_{jt} , q_{jt}^2 , and $\Delta_{0jt} - \Delta_{mjt}$ must span three dimensions. Although q_{jt}^2 is a deterministic function of q_{jt} , the relevant instrument correlations need not be proportional. An instrument that has an approximately linear relationship with q_{jt} across markets may have a different relationship with q_{jt}^2 than an instrument whose relationship with q_{jt} is nonlinear, for example because it shifts demand in a different part of the distribution of market sizes. In practice, an own-product characteristic, a rival cost shifter, and a demographic variable such as average market income will typically generate sufficient independent variation to satisfy the rank condition.

A.5 Redundant vs “Economically Distinct” Instruments

For falsification with non-constant costs to work, the two chosen instruments must have economically distinct impacts on market outcomes. Multiple instruments will not allow for falsification when they provide essentially the same information.

For example, suppose that w_{jt} includes two separate cost shifters, so that $\bar{c}_{0j} = w_{jt}^{(1)} \tau^{(1)} + w_{jt}^{(2)} \tau^{(2)} + q_{jt} \gamma_0$. As before, the researcher wants to falsify the Bertrand model when the truth is Cournot. The researcher constructs two instruments: $z_{1t}^{(1)} = w_{2t}^{(1)}$ and $z_{1t}^{(2)} = w_{2t}^{(2)}$. However, the two instruments affect Δ_{01t} , Δ_{m1t} , and q_{1t} identically, just scaled in proportion to the corresponding coefficients $\tau^{(1)}$ and $\tau^{(2)}$. This implies that the same misspecified cost function $\bar{c}_{m1}$ that satisfies Equation (10) for $z_{1t}^{(1)}$, will also satisfy it for $z_{1t}^{(2)}$, making falsification impossible.⁴²

We can formalize this insight somewhat by noting that in any setting where falsification is impossible with a single instrument (the typical case with economies of scale), falsification remains impossible with $K > 1$ instruments if there exist a set of constants $\{\zeta_{jk}\}_{k>1}$ such that $\frac{dq_{jt}}{dz_{jt}^{(k)}} = \zeta_{jk} \frac{dq_{jt}}{dz_{jt}^{(1)}}$ and $\left(\frac{d\Delta_{0jt}}{dz_{jt}^{(k)}} - \frac{d\Delta_{mjt}}{dz_{jt}^{(k)}} \right) = \zeta_{jk} \left(\frac{d\Delta_{0jt}}{dz_{jt}^{(1)}} - \frac{d\Delta_{mjt}}{dz_{jt}^{(1)}} \right)$ for each $k > 1$. That is, additional instruments bring no additional power to falsify a model if their economic effects are simply rescalings of those of existing instruments. Notice that this is analogous to the condition (7) in Remark 2 in the main body of the paper.

⁴²That is, since both instruments affect outcomes only through c_{2t} , we will have $\frac{d(\Delta_{01t} - \Delta_{m1t})}{dz_{1t}^{(k)}} = \tau^{(k)} \frac{d(\Delta_{01t} - \Delta_{m1t})}{dc_{2t}}$ and $\frac{dq_{1t}}{dz_{1t}^{(k)}} = \tau^{(k)} \frac{dq_{1t}}{dc_{2t}}$ for $k = 1, 2$, so if (10) holds for $k = 1$ it will hold for $k = 2$.

This may sound so obvious that it doesn't bear discussion, but redundancy of instruments in this way can occasionally crop up in unexpected places. In our well-trodden example environment of simple logit demand, for example, instruments formed from a rival's cost shifter and a product characteristic of the same rival turn out to be redundant. That is, when we allow for economies of scale, falsifying the Bertrand model when true firm conduct is Cournot is impossible using the two instruments $z_{1t}^{(1)} = w_{2t}^{(1)}$ and $z_{1t}^{(2)} = x_{2t}^{(1)}$. The failure of falsification, however, is due to a rather knife-edge fact about the particular demand system. In simple logit, demand depends on product characteristics and prices only through a single index $\delta_{jt} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$ for each product. Given that fact, for most standard conduct models, equilibrium markups depend on product characteristics and marginal costs only through the terms $\{x_{jt}\beta - \alpha c_{jt}\}_{j=1,\dots,J}$.⁴³ This means a change in a product characteristic has the same equilibrium impacts on market shares and implied markups as a change in its marginal cost, scaled by the appropriate coefficients. However, without the single index restriction, this would not be the case; in a richer demand system with random coefficients, for example, the effects of a product characteristic and a cost shifter of the same product would not simply be rescalings of each other, and we would expect to be able to falsify the incorrect model of conduct.

This discussion is simply meant to highlight the fact that if multiple instruments are to allow falsification in the presence of economies of scale, they must be *economically distinct*, which we define informally as having equilibrium effects on market shares and implied markups which do not both vary (across instruments) by the same scalar multipliers. In particular, two economically distinct instruments either differentially move quantities under the true model or differentially move the difference in markups. As stated in Remark 3, it should be easy to select economically distinct instruments, meaning falsification under non-constant marginal costs is typically possible in a standard differentiated product environment. Example 3 illustrated this, pairing a characteristic of one product with a cost shifter of a rival product. We just noted that under certain demand systems such as simple logit, it is not sufficient to pair a cost shifter and product characteristic of the same rival product, but outside of the "single index" demand systems discussed above, we would expect cost side instruments and demand side instruments to typically be economically distinct. In addition, with more than two firms, instruments that shift the marginal costs of different rival products may in fact be economically distinct, particularly in demand models sufficiently rich for some products to be "closer substitutes" than others.

⁴³ Appendix C of [Dearing et al. \(2024\)](#) explores the set of conduct models where this occurs; in brief, if each firm is maximizing profits under some assumption about how rivals will respond to their actions, their problem is $\max \sum_{j \in F} (p_{jt} - c_{jt}) s_{jt}(p_t)$, we can think instead of the firms choosing markups to solve $\max \sum_{j \in F} \Delta_{jt} s_{jt}(\delta_t)$, and write $\delta_{jt} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$ instead as $\delta_{jt} = (x_{jt}\beta - \alpha c_{jt} + \xi_{jt}) - \alpha \Delta_{jt}$, to make clear that the Δ_{jt} that form equilibrium can depend on x_t and c_t only through $x_t\beta - \alpha c_t + \xi_t$.

A.6 Proofs

Proof of Lemma 1. As we note in the text, in our parametric framework, the falsifiable restriction in Equation (28) of [Berry and Haile \(2014\)](#) is⁴⁴

$$E[\omega_{mjt} \mid \mathbf{w}_{jt}, z_{jt}] = E[p_{jt} - \Delta_{mjt} - \bar{c}_{mj}(q_{jt}, \mathbf{w}_{jt}) \mid \mathbf{w}_{jt}, z_{jt}] = 0 \quad a.s.$$

Since observed prices are generated under the true model as

$$p_{jt} = \Delta_{0jt} + c_{0jt} = \Delta_{0jt} + \bar{c}_{0j}(q_{jt}, \mathbf{w}_{jt}) + \omega_{0jt}$$

and $E[\omega_{0jt} \mid \mathbf{w}_{jt}, z_{jt}] = 0$ under Assumption 2, the falsifiable restriction is equivalent to

$$E[\Delta_{0jt} + \bar{c}_{0j}(q_{jt}, \mathbf{w}_{jt}) + \omega_{0jt} - \Delta_{mjt} - \bar{c}_{mj}(q_{jt}, \mathbf{w}_{jt}) \mid \mathbf{w}_{jt}, z_{jt}] = 0 \quad a.s.$$

or equivalently

$$E[\Delta_{0jt} - \Delta_{mjt} \mid \mathbf{w}_{jt}, z_{jt}] = E[\bar{c}_{mj}(q_{jt}, \mathbf{w}_{jt}) - \bar{c}_{0j}(q_{jt}, \mathbf{w}_{jt}) \mid \mathbf{w}_{jt}, z_{jt}] \quad a.s.$$

giving the result. $\square$

Proof of Proposition 1. In our parametric framework, the falsifiable restriction in Equation (28) of [Berry and Haile \(2014\)](#) is that for all j there exists a cost function $\bar{c}_{mj}$ such that:

$$E[p_{jt} - \Delta_{mjt} \mid \mathbf{w}_{jt}, z_{jt}] = E[\bar{c}_{mj}(q_{jt}, \mathbf{w}_{jt}) \mid \mathbf{w}_{jt}, z_{jt}] \quad a.s.$$

By plugging in for p_t as in the proof of Lemma 1, a model m is not falsified if for all j there exists a cost function $\bar{c}_{mj}$ such that

$$E[\Delta_{0jt} - \Delta_{mjt} \mid \mathbf{w}_{jt}, z_{jt}] = E[\bar{c}_{mj}(q_{jt}, \mathbf{w}_{jt}) - \bar{c}_{0j}(q_{jt}, \mathbf{w}_{jt}) \mid \mathbf{w}_{jt}, z_{jt}] \quad a.s.$$

The result thus follows by extending Lemma 2 in [Dearing et al. \(2024\)](#) to this restriction, so that a model is falsified if for some j there exists no cost function $\bar{c}_{mj}$ such that for all k

$$E \left[\frac{d\Delta_{0jt}}{dz_{jt}^{(k)}} - \frac{d\Delta_{mjt}}{dz_{jt}^{(k)}} \mid \mathbf{w}_{jt}, z_{jt} \right] = E \left[\frac{d\bar{c}_{mj}(q_{jt}, \mathbf{w}_{jt})}{dz_{jt}^{(k)}} - \frac{d\bar{c}_{0j}(q_{jt}, \mathbf{w}_{jt})}{dz_{jt}^{(k)}} \mid \mathbf{w}_{jt}, z_{jt} \right] \quad a.s.$$

$\square$

⁴⁴See Section 6, Case 2 in [Berry and Haile \(2014\)](#) for a discussion of their non-parametric environment.

Appendix B Distributions of the RV- and F -statistics

The derivations in this appendix are not tied to the log-log specification of Equation (5) and apply to arbitrary known transformations of model-implied costs and quantities. Throughout this appendix, let $\mathbf{c}_{mjt}$ denote a known scalar transformation of $p_{jt} - \Delta_{mjt}$, and let $\mathbf{q}_{jt} \in \mathbb{R}^{d_q}$ denote a known vector-valued transformation of q_{jt}^p , where $d_q < d_z$. The specification in Section 4 is the special case in which both transformations are logarithmic and $d_q = 1$. Let $\tilde{q}_{jt} \in \mathbb{R}^{d_q}$ be the best linear predictor of $\mathbf{q}_{jt}$ using z_{jt} and $\mathbf{w}_{jt}$, and let $z_{jt}^{e,0}$ be the d_z -dimensional residual vector from projecting z_{jt} on $\tilde{q}_{jt}$ and $\mathbf{w}_{jt}$. Set $r = d_z - d_q$, and let S_e be an $r \times d_z$ selection matrix such that $E[S_e z_{jt}^{e,0} z_{jt}^{e,0'} S_e']$ is nonsingular. We write $z_{jt}^e = S_e z_{jt}^{e,0}$ and use the same convention for the sample counterparts.

Test statistics We first give explicit definitions of the test statistics T^{RV} and F utilized in the paper. With data on n observations, the in-sample analogue of our measure of lack-of-fit is given as $\hat{Q}_m = \hat{g}_m' \hat{W} \hat{g}_m$ where $\hat{W} = (\frac{1}{n} \sum_{j,t} \hat{z}_{jt}^e \hat{z}_{jt}^{e'})^{-1}$, $\hat{g}_m = \frac{1}{n} \sum_{j,t} \hat{z}_{jt}^e \hat{\omega}_{mjt}$, $\hat{z}_{jt}^{e,0}$ is the residual in a projection on $\hat{q}$ and $\mathbf{w}$,

$$\hat{z}_{jt}^{e,0} = z_{jt} - \hat{\Lambda}_q \hat{q}_{jt} - \hat{\Lambda}_w \mathbf{w}_{jt},$$

where

$$\begin{bmatrix} \hat{\Lambda}_q' \\ \hat{\Lambda}_w' \end{bmatrix} = \left[\begin{pmatrix} \hat{q} \\ \mathbf{w} \end{pmatrix}' \begin{pmatrix} \hat{q} \\ \mathbf{w} \end{pmatrix} \right]^{-1} \begin{pmatrix} \hat{q} \\ \mathbf{w} \end{pmatrix}' z.$$

We set $\hat{z}_{jt}^e = S_e \hat{z}_{jt}^{e,0}$. The estimated cost error $\hat{\omega}_{mjt}$ is the residual in a sample 2SLS projection,

$$\hat{\omega}_{mjt} = \mathbf{c}_{mjt} - \mathbf{q}_{jt}' \hat{\gamma}_m - \mathbf{w}_{jt}' \hat{\tau}_m$$

with

$$\begin{pmatrix} \hat{\gamma}_m \\ \hat{\tau}_m \end{pmatrix} = \left[\begin{pmatrix} \hat{q} \\ \mathbf{w} \end{pmatrix}' (\mathbf{q}, \mathbf{w}) \right]^{-1} \begin{pmatrix} \hat{q} \\ \mathbf{w} \end{pmatrix}' \mathbf{c}_m,$$

Here $\hat{q}_{jt} = \hat{\zeta}_z' z_{jt} + \hat{\zeta}_w' \mathbf{w}_{jt}$ is the in-sample prediction obtained from

$$\begin{bmatrix} \hat{\zeta}_z \\ \hat{\zeta}_w \end{bmatrix} = [(z, \mathbf{w})' (z, \mathbf{w})]^{-1} (z, \mathbf{w})' \mathbf{q}.$$

The influence functions below also use the following sample moments of the unselected

residual vector:

$$\hat{\Sigma}_e^0 = n^{-1} \sum_{j,t} \hat{z}_{jt}^{e,0} \hat{z}_{jt}^{e,0'}, \quad \hat{g}_m^0 = n^{-1} \sum_{j,t} \hat{z}_{jt}^{e,0} \hat{\omega}_{mjt}.$$

For the derivative of the inverse square root of the full-rank covariance matrix, define

$$A_W = - \left(I_r \otimes W^{-1/2} + W^{-1/2} \otimes I_r \right)^{-1} \left(W^{1/2} \otimes W^{1/2} \right). \quad (12)$$

Its sample analogue $A_{\hat{W}}$ is obtained by replacing W with $\hat{W}$ in (12).

The variance estimator used in the RV test statistic is

$$\hat{\sigma}_{\text{RV}}^2 = 4 \left[\hat{g}_1' \hat{W}^{1/2} \hat{V}_{11}^{\text{RV}} \hat{W}^{1/2} \hat{g}_1 + \hat{g}_2' \hat{W}^{1/2} \hat{V}_{22}^{\text{RV}} \hat{W}^{1/2} \hat{g}_2 - 2 \hat{g}_1' \hat{W}^{1/2} \hat{V}_{12}^{\text{RV}} \hat{W}^{1/2} \hat{g}_2 \right]$$

where $\hat{V}_{\ell k}^{\text{RV}}$ is an estimator of the covariance between $\sqrt{n} \hat{W}^{1/2} \hat{g}_\ell$ and $\sqrt{n} \hat{W}^{1/2} \hat{g}_k$. Our proposed $\hat{V}_{\ell k}^{\text{RV}}$ is given by $\hat{V}_{\ell k}^{\text{RV}} = n^{-1} \sum_{j,t} \hat{\psi}_{\ell jt} \hat{\psi}_{k jt}'$ where $\text{vec}(\cdot)$ stacks the columns of a matrix and

$$\begin{aligned} \hat{\psi}_{mjt} &= \hat{W}^{1/2} \left(\hat{z}_{jt}^e \hat{\omega}_{mjt} - \hat{g}_m - S_e \hat{\Lambda}_q \hat{q}_{jt}^e \hat{z}_{jt}^{r'} \hat{Z} \hat{g}_m^0 \right) \\ &\quad + (\hat{g}_m' \otimes I_r) A_{\hat{W}} \text{vec} \begin{pmatrix} \hat{z}_{jt}^e \hat{z}_{jt}^{e'} - \hat{W}^{-1} - S_e \hat{\Sigma}_e^0 \hat{Z} \hat{z}_{jt}^r \hat{q}_{jt}^{e'} \hat{\Lambda}_q' S_e' \\ - S_e \hat{\Lambda}_q \hat{q}_{jt}^e \hat{z}_{jt}^{r'} \hat{Z} \hat{\Sigma}_e^0 S_e' \end{pmatrix}. \end{aligned}$$

The terms involving $\hat{q}_{jt}^e$ account for the additional sampling variability arising from estimation of the best linear predictor $\tilde{q}$. The construction uses $\hat{q}_{jt}^e = \mathbf{q}_{jt} - \hat{\tilde{q}}_{jt}$, $\hat{z}_{jt}^r = z_{jt} - z'w(w'w)^{-1}w_{jt}$, and $\hat{Z} = (\frac{1}{n} \sum_{j,t} \hat{z}_{jt}^r \hat{z}_{jt}^{r'})^{-1}$. The vector influence-function representation above is convenient for the asymptotic derivations below and can be augmented to account for demand estimation and clustering. For computation, it reduces to the simpler scalar representation in the following remark.

Remark 4. For implementation, the estimated vector influence function above can be reduced to an equivalent scalar contribution. Define $\hat{\chi}_{mjt} = 2 \hat{g}_m' \hat{W}^{1/2} \hat{\psi}_{mjt}$. The variance estimator above can then be written as $\hat{\sigma}_{\text{RV}}^2 = n^{-1} \sum_{j,t} (\hat{\chi}_{1jt} - \hat{\chi}_{2jt})^2$. To simplify $\hat{\chi}_{mjt}$, first consider the component of $\hat{\psi}_{mjt}$ that captures the first-stage sampling error from $\hat{z}^e$ in $\hat{W}$:

$$\begin{aligned} [2 \hat{g}_m' \hat{W}^{1/2}] (-\hat{g}_m' \otimes I_r) A_{\hat{W}} \text{vec} \begin{pmatrix} S_e \hat{\Sigma}_e^0 \hat{Z} \hat{z}_{jt}^r \hat{q}_{jt}^{e'} \hat{\Lambda}_q' S_e' + S_e \hat{\Lambda}_q \hat{q}_{jt}^e \hat{z}_{jt}^{r'} \hat{Z} \hat{\Sigma}_e^0 S_e' \end{pmatrix} \\ = 2 \left(\hat{g}_m' \hat{W} S_e \hat{\Lambda}_q \hat{q}_{jt}^e \right) \hat{z}_{jt}^{r'} \hat{Z} \hat{\Sigma}_e^0 S_e' \hat{W} \hat{g}_m = 2 \left(\hat{g}_m' \hat{W} S_e \hat{\Lambda}_q \hat{q}_{jt}^e \right) \hat{z}_{jt}^{r'} \hat{Z} \hat{g}_m^0 \end{aligned}$$

where, in the first equality, the quadratic form follows because, for an arbitrary symmetric $r \times r$ direction H , $2 \hat{g}_m' \hat{W}^{1/2} (\hat{g}_m' \otimes I_r) A_{\hat{W}} \text{vec}(H)$ is the derivative of $\hat{g}_m' \hat{W} \hat{g}_m$ with respect to $\hat{W}^{-1}$ in direction H , which has the sandwich form $-\hat{g}_m' \hat{W} H \hat{W} \hat{g}_m$; the factor 2 arises

because the two terms inside vec are transposes of each other. The second equality follows because $\hat{\Sigma}_e^0 S_e' \hat{W} S_e$ is a projector onto the column space of $\hat{\Sigma}_e^0$, which contains $\hat{g}_m^0$, and hence $\hat{\Sigma}_e^0 S_e' \hat{W} \hat{g}_m = \hat{\Sigma}_e^0 S_e' \hat{W} S_e \hat{g}_m^0 = \hat{g}_m^0$. After contraction by $2\hat{g}_m' \hat{W}^{1/2}$, the contribution of the first-stage sampling error through $\hat{W}$ is therefore identical in magnitude and opposite in sign to its contribution through $\hat{g}_m$, so these two terms cancel in $\hat{\chi}_{mjt}$. Finally, applying the same quadratic-form identity to the remaining part of the sampling error in $\hat{W}$, namely $\hat{z}_{jt}^e \hat{z}_{jt}^{e'} - \hat{W}^{-1}$, the scalar contribution simplifies to

$$\hat{\chi}_{mjt} = 2\hat{v}_{mjt}\hat{\omega}_{mjt} - \hat{v}_{mjt}^2 - \hat{Q}_m, \quad \text{where } \hat{v}_{mjt} = \hat{z}_{jt}^{e'} \hat{W} \hat{g}_m.$$

We use this scalar representation, aggregated at the cluster level, to construct $\hat{\sigma}_{\text{RV}}^2$ with the `pyRVtest` package in our empirical analysis.

The F -statistic in the current context is a joint test statistic for the two hypotheses: $H_{0,m} : \pi_m = 0$ in the equations $\omega_{mjt} = z_{jt}^{e'} \pi_m + e_{mjt}$ for $m = 1, 2$. Formulaically, it is

$$F = (1 - \hat{\rho}^2) \frac{n}{2r} \frac{\hat{\sigma}_2^2 \hat{g}_1' \hat{W} \hat{g}_1 + \hat{\sigma}_1^2 \hat{g}_2' \hat{W} \hat{g}_2 - 2\hat{\sigma}_{12} \hat{g}_1' \hat{W} \hat{g}_2}{\hat{\sigma}_1^2 \hat{\sigma}_2^2 - \hat{\sigma}_{12}^2},$$

where

$$\hat{\rho}^2 = \frac{(\hat{\sigma}_1^2 - \hat{\sigma}_2^2)^2}{(\hat{\sigma}_1^2 + \hat{\sigma}_2^2)^2 - 4\hat{\sigma}_{12}^2}, \quad \hat{\sigma}_m^2 = \frac{\text{trace}(\hat{V}_{mm}^{\text{AR}} \hat{W}^{-1})}{r}, \quad \hat{\sigma}_{12} = \frac{\text{trace}(\hat{V}_{12}^{\text{AR}} \hat{W}^{-1})}{r},$$

$\hat{V}_{\ell k}^{\text{AR}} = n^{-1} \sum_{j,t} \hat{\phi}_{\ell jt} \hat{\phi}_{k jt}'$ where $\hat{\phi}_{mjt} = \hat{W} \hat{z}_{jt}^e (\hat{\omega}_{mjt} - \hat{z}_{jt}^{e'} \hat{\pi}_m)$, and $\hat{\pi}_m = \hat{W} \hat{g}_m$ is the OLS estimator in the r -dimensional full-rank representation of π_m .

Assumptions The following assumptions list direct analogues of Assumptions 1–4 in [Duarte et al. \(2024\)](#). The only key new requirement is in Assumption 5, part (ii), which ensures that the instruments are strong for pinning down the economies of scale. We refer to [Duarte et al. \(2024\)](#) for further discussion.

Assumption 4. z_{jt} is a vector of d_z excluded instruments, so that $E[z_{jt} \omega_{0jt}] = 0$.

Assumption 5. (i) $\{p_{jt}, \Delta_{0jt}, \Delta_{1jt}, \Delta_{2jt}, z_{jt}, w_{jt}, q_{jt}^p, \omega_{0jt}\}_{j,t}$ are jointly iid with the observables being $p_{jt}, \Delta_{1jt}, \Delta_{2jt}, z_{jt}, w_{jt}, q_{jt}^p$; (ii) $E[(\mathbf{c}_{1jt} - \mathbf{c}_{2jt})^2] > 0$, $E[(z_{jt}', w_{jt}')'(z_{jt}', w_{jt}')]$ is positive definite, and $E[(z_{jt}', w_{jt}')'(\mathbf{q}_{jt}', w_{jt}')]$ has full column rank; (iii) the entries of $\mathbf{c}_{1jt}$, $\mathbf{c}_{2jt}$, z_{jt} , w_{jt} , $\mathbf{q}_{jt}$, ω_{1jt} , and ω_{2jt} have finite fourth moments.

Assumption 6. The error term in Equation (6), e_{mjt} , is homoskedastic, i.e., $E[e_{mjt}^2 z_{jt}^e z_{jt}^{e'}] = \sigma_m^2 E[z_{jt}^e z_{jt}^{e'}]$ with $\sigma_m^2 > 0$ for $m \in \{1, 2\}$ and $E[e_{1jt} e_{2jt} z_{jt}^e z_{jt}^{e'}] = \sigma_{12} E[z_{jt}^e z_{jt}^{e'}]$ with $\sigma_{12}^2 < \sigma_1^2 \sigma_2^2$.

Assumption 7. Define $\pi_m = Wg_m$. For both $m = 1$ and $m = 2$,

$$\pi_m = h_m / \sqrt{n} \quad \text{for some finite vector } h_m \in \mathbb{R}^r.$$

Establishing Critical Values To establish the critical values for our test statistic and F -statistic, we prove the following result, which is a direct analog of Proposition 4 in Duarte et al. (2024) with the only difference being that r replaces d_z :

To state the result, let $\mathbf{e}_1 = (1, 0, \dots, 0)'$ denote the first basis vector in $\mathbb{R}^r$ and define

$$b_{\pm} = 2(\sigma_1^2 + \sigma_2^2 \pm 2\sigma_{12})^{1/2},$$

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \left(\frac{1}{b_-} + \frac{1}{b_+} \right) \begin{pmatrix} \sqrt{n} W^{1/2} g_1 \\ -\sqrt{n} W^{1/2} g_2 \end{pmatrix} + \left(\frac{1}{b_-} - \frac{1}{b_+} \right) \begin{pmatrix} -\sqrt{n} W^{1/2} g_2 \\ \sqrt{n} W^{1/2} g_1 \end{pmatrix}.$$

Set $\iota = 1 + \mathbf{1}\{\|a_1\| \leq \|a_2\|\}$ and

$$\mu_- = \|a_{\iota}\| - \|a_{3-\iota}\|, \quad \mu_+ = \|a_1\| + \|a_2\|.$$

Finally, let $\rho = \text{Corr}(e_{\iota jt} - e_{3-\iota, jt}, e_{1jt} + e_{2jt})$. Under Assumption 7, $\sqrt{n} W^{1/2} g_m = W^{-1/2} h_m$, so these nuisance parameters do not depend on n .

Proposition 2. Suppose Assumptions 4–7 hold. Then

$$(i) \quad \begin{pmatrix} |T^{\text{RV}}| \\ F \end{pmatrix} \xrightarrow{d} \begin{pmatrix} |\Psi'_- \Psi_+| / \left(\|\Psi_-\|^2 + \|\Psi_+\|^2 + 2\rho \Psi'_- \Psi_+ \right)^{1/2} \\ \left(\|\Psi_-\|^2 + \|\Psi_+\|^2 - 2\rho \Psi'_- \Psi_+ \right) / (2r) \end{pmatrix}$$

where $\hat{\rho}^2 \xrightarrow{p} \rho^2$ and

$$\begin{pmatrix} \Psi_- \\ \Psi_+ \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_- \mathbf{e}_1 \\ \mu_+ \mathbf{e}_1 \end{pmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \otimes I_r \right),$$

$$(ii) \quad H_0^{\text{RV}} \text{ holds if and only if } \mu_- = 0,$$

$$(iii) \quad H_{0,1} \text{ and } H_{0,2} \text{ hold if and only if } \mu_+ = 0,$$

$$(iv) \quad 0 \leq \mu_- \leq \mu_+.$$

To establish Proposition 2, we establish the following lemma, which generalizes Lemma A.1 in Duarte et al. (2024) to the setting of this paper. Given this lemma, Proposition 2 follows by the use of the same proof as for Proposition 4 in Duarte et al. (2024).

To state the following lemma and give a formulation of σ_{RV}^2 , we introduce population versions of $\hat{\psi}_{mjt}$ and $\hat{\phi}_{mjt}$ along with notation for their variances. Let $\Sigma_e^0 = E[z_{jt}^{e,0} z_{jt}^{e,0'}]$ and $g_m^0 = E[z_{jt}^{e,0} \omega_{mjt}]$ denote the population moments of the unselected residualized instruments.

Let

$$\begin{aligned}\psi_{mjt} &= W^{1/2} \left(z_{jt}^e \omega_{mjt} - g_m - S_e \Lambda_q q_{jt}^e z_{jt}^{r'} Z g_m^0 \right) \\ &\quad + (g'_m \otimes I_r) A_W \text{vec} \begin{pmatrix} z_{jt}^e z_{jt}^{e'} - W^{-1} - S_e \Sigma_e^0 Z z_{jt}^r q_{jt}^{e'} \Lambda_q' S_e' \\ - S_e \Lambda_q q_{jt}^e z_{jt}^{r'} Z \Sigma_e^0 S_e' \end{pmatrix}.\end{aligned}$$

Also let $\phi_{mjt} = W z_{jt}^e e_{mjt}$, where $q_{jt}^e = \mathbf{q}_{jt} - \tilde{q}_{jt}$,

$$z_{jt}^r = z_{jt} - E[z'w]E[w'w]^{-1}w_{jt},$$

$Z = E[z_{jt}^r z_{jt}^{r'}]^{-1}$, and

$$\begin{bmatrix} \Lambda_q' \\ \Lambda_w' \end{bmatrix} = E[(\tilde{q}, w)'(\tilde{q}, w)]^{-1} E[(\tilde{q}, w)'z].$$

Then define

$$\begin{aligned}V_{\ell k}^{\text{RV}} &= E[\psi_{\ell jt} \psi'_{kjt}], & V_{\ell k}^{\text{AR}} &= E[\phi_{\ell jt} \phi'_{kjt}], \\ V^{\text{RV}} &= E[(\psi'_{1jt}, \psi'_{2jt})'(\psi'_{1jt}, \psi'_{2jt})].\end{aligned}$$

Here, V^{RV} is a matrix with V_{11}^{RV} , V_{12}^{RV} , and V_{22}^{RV} as its entries. Finally,

$$\sigma_{\text{RV}}^2 = 4 \left[g_1' W^{1/2} V_{11}^{\text{RV}} W^{1/2} g_1 + g_2' W^{1/2} V_{22}^{\text{RV}} W^{1/2} g_2 - 2g_1' W^{1/2} V_{12}^{\text{RV}} W^{1/2} g_2 \right].$$

The sample covariance matrices $\hat{V}_{\ell k}^{\text{RV}}$ and their population counterparts $V_{\ell k}^{\text{RV}}$ are formed from these influence functions; $\hat{\sigma}_{\text{RV}}^2$ and σ_{RV}^2 are the corresponding sample and population delta-method variances.

Lemma 2. *Suppose Assumptions 4 and 5 hold. Parts (i) and (ii) hold for $\ell, k \in \{1, 2\}$. If Assumption 7 also holds, parts (iii) and (iv) hold for $\ell, k, m \in \{1, 2\}$.*

$$\begin{aligned}(i) \quad & \sqrt{n} \begin{pmatrix} \hat{W}^{1/2} \hat{g}_1 - W^{1/2} g_1 \\ \hat{W}^{1/2} \hat{g}_2 - W^{1/2} g_2 \end{pmatrix} \xrightarrow{d} N(0, V^{\text{RV}}), & (ii) \quad & \hat{V}_{\ell k}^{\text{RV}} \xrightarrow{p} V_{\ell k}^{\text{RV}}, \\ (iii) \quad & \sqrt{n} (\hat{\pi}_m - \pi_m) \xrightarrow{d} N(0, V_{mm}^{\text{AR}}), & (iv) \quad & \hat{V}_{\ell k}^{\text{AR}} \xrightarrow{p} V_{\ell k}^{\text{AR}}.\end{aligned}$$

Remark 5. When the RV test is not degenerate, i.e., when $\sigma_{\text{RV}}^2 > 0$, it follows from parts (i), (ii), and a first-order Taylor approximation that $T^{\text{RV}} \xrightarrow{d} N(0, 1)$ under H_0^{RV} so that the RV test is asymptotically valid. Details of this step can be found in [Rivers and Vuong \(2002\)](#); [Hall and Pelletier \(2011\)](#) and are omitted. When non-degeneracy fails to hold, a

first-order Taylor approximation does not capture the behavior of T^{RV} .

Proof of Lemma 2. To prove their analog of this lemma, [Duarte et al. \(2024\)](#) proceeds in three steps to establish (i) and (ii) before commenting on the minor modifications in the argument that establish (iii) and (iv). Here we comment on the modifications needed due to the introduction of economies of scale. Step (1) shows that $\frac{1}{\sqrt{n}} \sum_{j,t} (\psi'_{1jt}, \psi'_{2jt})' \xrightarrow{d} N(0, V^{\text{RV}})$ and $\check{V}_{\ell k}^{\text{RV}} := \frac{1}{n} \sum_{j,t} \psi_{\ell jt} \psi'_{kjt} \xrightarrow{p} V_{\ell k}^{\text{RV}}$ for $\ell, k \in \{1, 2\}$, and this step requires no new arguments. Step (2) establishes that $\sqrt{n}(\hat{W}^{1/2} \hat{g}_m - W^{1/2} g_m) - \frac{1}{\sqrt{n}} \sum_{j,t} \psi_{mjt} = o_p(1)$ for $m \in \{1, 2\}$, and step (3) proves that $\text{trace}((\hat{V}_{\ell k}^{\text{RV}} - \check{V}_{\ell k}^{\text{RV}})'(\hat{V}_{\ell k}^{\text{RV}} - \check{V}_{\ell k}^{\text{RV}})) = o_p(1)$ for $\ell, k \in \{1, 2\}$. In the approximations of the last two steps, the only new complication to handle is the estimation of the best linear predictor $\tilde{q}$ and the economies of scale parameter γ_m .

In contrast to Remark 2 of [Duarte et al. \(2024\)](#), estimating $\tilde{q}$ changes the d_z -dimensional residualized instruments $z^{e,0}$ in our setting. Consequently, when $g_m \neq 0$, the corresponding equality used there does not hold to first order. Applying S_e to the resulting expansion gives

$$\hat{g}_m - g_m = \frac{1}{n} \sum_{j,t} \left\{ z_{jt}^e \omega_{mjt} - g_m - S_e \Lambda_q q_{jt}^e z_{jt}^{r'} Z g_m^0 \right\} + O_p(n^{-1}).$$

The term involving q_{jt}^e is the first-order effect of estimating $\tilde{q}$ on the residualized instruments. For the covariance matrix, first consider the d_z -dimensional residuals:

$$\begin{aligned} n^{-1}(\hat{z}^{e,0'} \hat{z}^{e,0} - z^{e,0'} z^{e,0}) &= n^{-1} z^{e,0'} (\hat{z}^{e,0} - z^{e,0}) \\ &\quad + n^{-1} (\hat{z}^{e,0} - z^{e,0})' z^{e,0} \\ &\quad + n^{-1} (\hat{z}^{e,0} - z^{e,0})' (\hat{z}^{e,0} - z^{e,0}). \end{aligned}$$

Here we can write $\hat{z}^{e,0} - z^{e,0}$ as

$$\begin{aligned} \hat{z}^{e,0} - z^{e,0} &= (\tilde{q}, w) \begin{pmatrix} \Lambda'_q \\ \Lambda'_w \end{pmatrix} - (\hat{q}, w) \begin{pmatrix} \hat{\Lambda}'_q \\ \hat{\Lambda}'_w \end{pmatrix} \\ &= (\tilde{q}, w) \begin{pmatrix} \Lambda'_q - \hat{\Lambda}'_q \\ \Lambda'_w - \hat{\Lambda}'_w \end{pmatrix} - (\hat{q} - \tilde{q}) \hat{\Lambda}'_q. \end{aligned}$$

where in turn $\hat{q} - \tilde{q} = z(\hat{\zeta}_z - \zeta_z) + w(\hat{\zeta}_w - \zeta_w)$. Since $n^{-1} z^{e,0'} (\tilde{q}, w) = O_p(n^{-1/2})$ and $n^{-1} z^{e,0'} z = \Sigma_e^0 + O_p(n^{-1/2})$, standard arguments (among which a key one is $\hat{\zeta}_z - \zeta_z =$

$n^{-1}Zz^{r'}q^e + O_p(n^{-1})$ imply that

$$\begin{aligned}\hat{\Sigma}_e^0 &= n^{-1}z^{e,0'}z^{e,0} - n^{-1}\Sigma_e^0 Zz^{r'}q^e \Lambda_q' \\ &\quad - n^{-1}\Lambda_q q^{e'} z^r Z \Sigma_e^0 + O_p(n^{-1}).\end{aligned}$$

Premultiplying and postmultiplying this expansion by S_e and S_e' gives the first-order perturbation of the full-rank covariance matrix:

$$\hat{W}^{-1} - W^{-1} = \frac{1}{n} \sum_{j,t} \begin{pmatrix} z_{jt}^e z_{jt}^{e'} - W^{-1} - S_e \Sigma_e^0 Z z_{jt}^r q_{jt}^{e'} \Lambda_q' S_e' \\ - S_e \Lambda_q q_{jt}^{e'} z_{jt}^r Z \Sigma_e^0 S_e' \end{pmatrix} + O_p(n^{-1}).$$

Applying the derivative A_W in (12) to this covariance perturbation and combining it with the preceding expansion of $\hat{g}_m - g_m$ gives

$$\hat{W}^{1/2} \hat{g}_m - W^{1/2} g_m = \frac{1}{n} \sum_{j,t} \psi_{mjt} + O_p(n^{-1}).$$

The first component of ψ_{mjt} therefore comes from the expansion of $\hat{g}_m - g_m$, while its second component applies the inverse-square-root derivative A_W in (12) to the covariance perturbation.

It remains to establish that $n^{-1} \sum_{j,t} \|\hat{\psi}_{mjt} - \psi_{mjt}\|^2 = o_p(1)$. This follows from the preceding stochastic expansions and the consistency of the sample analogues entering $\hat{\psi}_{mjt}$, after which the remainder of the proof follows the corresponding argument in [Duarte et al. \(2024\)](#). $\square$

Appendix C Monte Carlo Simulations

We now illustrate the performance of the procedure developed in Section 4 of the paper through Monte Carlo simulations.

Setup We simulate data for 150,000 markets using PyBLP ([Conlon and Gortmaker, 2020](#)). In each market, two single-product firms compete. Consumer utility follows a logit specification with a price coefficient $\alpha = -0.9$ and two product characteristics (including a constant), with coefficients $\beta = [1, 5]$. For each product-market pair, we draw one characteristic independently from a uniform distribution $U[0, 1]$.

On the supply side, marginal costs include two observed shifters (also drawn from $U[0, 1]$) and a constant, with coefficients $\tau = [3, 3, 3]$. When present, economies of scale enter through a parameter $\gamma = -0.5$ that multiplies quantity. The unobserved demand (ξ_{jt}) and cost (ω_{jt}) shocks follow a bivariate normal distribution with unit variances and correlation

0.9, which is the default in PyBLP. We select demand and cost parameters to generate reasonable elasticities and outside good shares. In all simulations, the true model is Bertrand competition, while we attempt to falsify Cournot competition.

Results Table 10 reports results from six simulation experiments that illustrate our theoretical findings on instrument relevance. For each experiment, we report both the RV test statistic T^{RV} and the F -statistic that diagnoses instrument strength, computed with the package `pyRVtest` (Duarte et al., 2022).

TABLE 10: Illustrating Instrument Relevance

Experiments	(1)	(2)	(3)	(4)	(5)	(6)
DGP and Instruments:						
Econ. of Scale (γ)	0	-0.5	-0.5	-0.5	-0.5	-0.5
Instrument	$w_{-j}^{(1)}$	$w_{-j}^{(1)}$	$w_{-j}^{(1)}, w_{-j}^{(2)}$	$w_{-j}^{(1)}, x_j^{(1)}$	$w_{-j}^{(1)}, x_{-j}^{(1)}$	$x_j^{(1)}, x_{-j}^{(1)}$
<i>Panel A: Bertrand vs. Cournot</i>						
T^{RV}	-2.5	0.0	-0.4	-3.2	-0.2	-5.0
	**			***		***
F	2,894.2	0.0	0.2	1,443.6	0.0	4,189.4
	††† ^^^			††† ^^^		††† ^^^
<i>Panel B: Model-implied Estimated Economies of Scale ($\hat{\gamma}_B, \hat{\gamma}_C$)</i>						
$\hat{\gamma}_B$	—	-0.6	-0.8	-0.5	-0.7	-0.5
	—	(0.46)	(0.32)	(0.02)	(0.23)	(0.02)
$\hat{\gamma}_C$	—	2.3	2.0	-0.7	2.2	-0.7
	—	(0.49)	(0.33)	(0.02)	(0.24)	(0.02)

The table reports, for each experiment 1-6, the RV test statistics T^{RV} and effective F -statistic in Panel A, and estimated economies of scale parameters in Panel B. For Panel A, a negative RV test statistic suggests a better fit of the true Bertrand model. The symbols ** and *** indicate rejection of the null of equal fit at the 5% and 1% significance levels, respectively. The symbols ††† and ^^^ indicate that F is above the appropriate critical values for worst-case size below 0.075, and best-case power above 0.95, respectively.

In Simulation 1, we consider constant marginal costs ($\gamma = 0$) and use a single rival cost shifter as an instrument. With a large F -statistic and significant RV test statistic ($T^{\text{RV}} = -2.5$), the results confirm that a researcher can falsify a wrong model of conduct with a single relevant instrument when marginal costs are constant (Dearing et al., 2024). However, once we introduce economies of scale ($\gamma = -0.5$) in Simulation 2 while maintaining the single rival cost shifter instrument, both statistics drop to near zero, indicating a complete failure of falsification as predicted by Remark 1.

In line with Remark 2, adding a second rival cost shifter in Simulation 3 fails to resolve this problem. These instruments are not economically distinct, and yield an F -statistic of only 0.2, indicating they remain weak for testing. The breakthrough comes in Simulation 4, where we pair a rival cost shifter with an own product characteristic - these are economically

distinct instruments and available in standard datasets, in line with Remark 3. These instruments successfully falsify the wrong model even with non-constant costs, yielding both a significant test statistic ($T^{\text{RV}} = -3.2$) and strong instruments ($F = 1,443.6$). This finding illustrates how instruments that affect the environment through different economic channels can overcome the challenges posed by non-constant marginal costs.

The final two specifications further validate our theoretical framework by exploring alternative instrument pairs, further illustrating Remarks 2 and 3. Simulation 5 combines a rival cost shifter with a rival product characteristic, but these instruments fail to be economically distinct and provide no power for testing. In contrast, Simulation 6 pairs own and rival product characteristics, achieving strong power as diagnosed by the effective F -statistic ($F = 4,189.4$) and clear rejection of the wrong model ($T^{\text{RV}} = -5.0$).

Panel B reports the estimated economies of scale parameters (γ) under both Bertrand and Cournot specifications across our experiments. These estimates highlight how instrument choice affects our ability to separately identify cost structure and conduct. With weak instruments (columns 2, 3, and 5), the estimates vary substantially between models. In contrast, when using economically distinct instruments (columns 4 and 6), we obtain more stable and similar estimates across specifications.

Appendix D Additional Empirical Details

D.1 Reproduced Demand Estimates from [Grieco et al. \(2024\)](#)

TABLE 11: Demand Estimates Reproduced from [Grieco et al. \(2024\)](#)

	β	σ	Demographic interactions						
			Income	Inc. sq.	Age	Rural	Fam. size 2	FS 3–4	FS 5+
Price	−3.112	—	0.094	−0.462	2.065	—	—	—	—
Van	−7.614	5.538	—	—	—	—	1.737	3.681	5.840
SUV	−0.079	3.617	—	—	—	—	—	—	—
Truck	−7.463	6.309	—	—	—	3.007	—	—	—
Footprint	0.534	1.873	—	—	—	—	0.481	0.459	0.636
Horsepower	1.018	1.246	—	—	—	—	—	—	—
Miles/Gal.	−0.965	1.645	—	—	—	—	—	—	—
Luxury	—	2.624	—	—	—	—	—	—	—
Sport	−3.046	2.617	—	—	—	—	—	—	—
EV	−5.549	3.798	—	—	—	—	—	—	—
Euro. brand	—	1.921	—	—	—	—	—	—	—
U.S. brand	—	2.141	—	—	—	—	—	—	—
Constant	—	—	0.362	—	—	—	—	—	—

The table reports the point estimates from the main demand specification in [Grieco et al. \(2024\)](#).

D.2 Mixed Models of Conduct

To define the implied markups for the mixed models, we first establish some notation. For any matrix A_{Mt} , we order Bertrand and Cournot players in market t , and partition

$$A_{Mt} = \begin{bmatrix} A_{BBt} & A_{BCt} \\ A_{CBt} & A_{CCt} \end{bmatrix},$$

where subscripts B and C denote firms competing in prices and quantities, respectively. In a mixed conduct equilibrium, Bertrand firms choose prices taking rivals' prices and Cournot firms' quantities as given, while Cournot firms choose quantities taking rivals' quantities and Bertrand firms' prices as given. Following the derivation in Proposition 3 of [Feenstra and Levinsohn \(1995\)](#), equilibrium markups for Cournot and Bertrand players can be expressed in terms of the partitioned demand derivative matrix D_{Mt} and ownership matrix Ω_{Mt} .

Cournot players The first-order conditions for quantity-setting firms yield markups analogous to the pure Cournot case in Equation (4), but restricted to the Cournot block:

$$\Delta_{Ct} = -(\Omega_{CCt} \odot (D_{CCt}^{-1})') s_{Ct}.$$

This follows because quantity-setting firms internalize only the price effects of their own output changes on their products, taking the Bertrand firms' prices—and hence Bertrand firms' residual demand—as given. The relevant demand derivatives are therefore only those within the Cournot block, D_{CCt} .

Bertrand players When a Bertrand firm changes its price, the Cournot firms hold their quantities fixed, so their prices adjust: setting $dq_{Ct} = D_{CBt} dp_{Bt} + D_{CCt} dp_{Ct} = 0$ gives $dp_{Ct} = -D_{CCt}^{-1} D_{CBt} dp_{Bt}$. The Bertrand firms therefore face the residual demand derivative $D_{BBt} - D_{BCt} D_{CCt}^{-1} D_{CBt}$, and their first-order conditions yield:

$$\Delta_{Bt} = - \left[\Omega_{BBt} \odot (D_{BBt} - D_{BCt} D_{CCt}^{-1} D_{CBt})' \right]^{-1} s_{Bt}.$$

This expression differs from the pure Bertrand markup in Equation (3) because a Bertrand firm's price change shifts demand for the Cournot products (captured by D_{CBt}), the Cournot firms' prices move to keep their quantities unchanged (captured by D_{CCt}^{-1}), and those price responses feed back into the demand faced by the Bertrand firms (captured by D_{BCt}). When there are no cross-group demand effects ($D_{BCt} = D_{CBt} = 0$), this reduces to the pure Bertrand markup formula applied to the Bertrand block.

Stacking these, the full vector of mixed-model markups is:

$$\Delta_{Mt} = \begin{bmatrix} - \left[\Omega_{BBt} \odot (D_{BBt} - D_{BCt} D_{CCt}^{-1} D_{CBt})' \right]^{-1} s_{Bt} \\ - (\Omega_{CCt} \odot (D_{CCt}^{-1})') s_{Ct} \end{bmatrix}.$$

The derivation above follows Proposition 3 in [Feenstra and Levinsohn \(1995\)](#), adapted to our notation. Specifically, their subscripts 1 and 2 correspond to B (Bertrand) and C (Cournot) here, and their matrix J is our ownership matrix Ω . [Feenstra and Levinsohn \(1995\)](#) write the residual demand derivative on their matrix of cross-elasticities E as $-(I_1 - E_{11}) + E_{12}(I_2 - E_{22})^{-1}E_{21}$ (their equation (A22)); because their demand derivative is proportional to $-(I - E)$, the block $(I_2 - E_{22})^{-1}$ corresponds to $-D_{CCt}^{-1}$ in our notation, which is why the same expression carries a plus sign in their formulation and a minus sign in ours.

D.3 Marginal Cost Regression Results

Table 3 reports the economies of scale parameters obtained from the regression in Equation (5) for the seven models of conduct we consider in the car industry. In Table 12, we report all parameter estimates from these regressions.

TABLE 12: Full Implied Marginal Cost Regression Results

	Mixed Models						
	Bertrand	Cournot	Asian Bert.	US Bert.	Europe Cour.	Big Three Cour.	Tesla Bert.
Constant	−2.809 (1.248)	−3.681 (1.292)	−3.128 (1.259)	−3.157 (1.250)	−2.906 (1.249)	−2.955 (1.258)	−3.664 (1.293)
RXR	0.070 (0.038)	0.073 (0.038)	0.069 (0.038)	0.074 (0.038)	0.071 (0.038)	0.068 (0.038)	0.073 (0.038)
t	−0.140 (0.016)	−0.125 (0.016)	−0.130 (0.016)	−0.141 (0.016)	−0.141 (0.016)	−0.132 (0.016)	−0.125 (0.016)
t^2	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)	0.002 (0.000)
log(height)	−0.837 (0.166)	−0.913 (0.172)	−0.870 (0.168)	−0.881 (0.166)	−0.847 (0.166)	−0.857 (0.168)	−0.914 (0.172)
log(footprint)	0.075 (0.169)	0.055 (0.170)	0.057 (0.168)	0.078 (0.167)	0.084 (0.169)	0.052 (0.168)	0.054 (0.170)
log(horsepower)	0.365 (0.055)	0.403 (0.056)	0.381 (0.055)	0.380 (0.054)	0.367 (0.055)	0.375 (0.056)	0.402 (0.056)
log(MPG)	−0.032 (0.069)	−0.010 (0.070)	−0.016 (0.069)	−0.027 (0.068)	−0.030 (0.069)	−0.022 (0.069)	−0.009 (0.070)
log(curbweight)	1.009 (0.169)	1.098 (0.172)	1.041 (0.170)	1.048 (0.169)	1.012 (0.169)	1.033 (0.170)	1.100 (0.172)
log(# trims)	0.078 (0.026)	0.067 (0.026)	0.071 (0.026)	0.072 (0.026)	0.077 (0.026)	0.073 (0.026)	0.068 (0.026)
Release Year	−0.118 (0.026)	−0.115 (0.026)	−0.117 (0.026)	−0.112 (0.025)	−0.117 (0.026)	−0.119 (0.026)	−0.116 (0.026)
SUV	0.054 (0.027)	0.051 (0.027)	0.056 (0.027)	0.056 (0.026)	0.056 (0.026)	0.054 (0.027)	0.051 (0.027)
Truck	−0.238 (0.050)	−0.277 (0.051)	−0.260 (0.050)	−0.230 (0.049)	−0.236 (0.050)	−0.265 (0.051)	−0.278 (0.051)
Van	−0.118 (0.047)	−0.140 (0.047)	−0.126 (0.047)	−0.115 (0.046)	−0.117 (0.047)	−0.130 (0.047)	−0.141 (0.047)
PHEV/EV	0.153 (0.058)	0.172 (0.057)	0.160 (0.057)	0.168 (0.057)	0.156 (0.058)	0.157 (0.058)	0.177 (0.058)
Sport	0.021 (0.037)	0.032 (0.037)	0.026 (0.037)	0.027 (0.036)	0.022 (0.037)	0.023 (0.037)	0.031 (0.037)
Years Since Design	−0.004 (0.003)	−0.004 (0.003)	−0.004 (0.003)	−0.004 (0.003)	−0.004 (0.003)	−0.005 (0.003)	−0.004 (0.003)
log(q_{jt}^p)	−0.135 (0.030)	−0.125 (0.031)	−0.129 (0.031)	−0.128 (0.030)	−0.134 (0.030)	−0.132 (0.031)	−0.125 (0.031)

The table reports full results of estimating Equation (5) via 2SLS under different models of conduct. Table 3 reports γ from these regressions. Specifications include brand fixed effects. Standard errors are clustered by car model.

D.4 Effective F -Statistics for the Conduct Test

Table 4 reports the Rivers–Vuong test statistics and model confidence set p -values for the seven models of conduct. Table 13 reports the corresponding effective F -statistics for all pairwise comparisons, under both the economies-of-scale specification (Panel A) and constant marginal cost (Panel B). Every F -statistic exceeds the critical values for both a worst-case size of 0.075 and a best-case power of 0.95, so the test is correctly sized and well-powered throughout. Cournot is the only model included in the model confidence set under economies of scale (Panel A) as the strong instruments are able to separate Cournot from every alternative. The same conclusion holds under constant marginal cost (Panel B), where the 95% set is again the singleton {Cournot}.

TABLE 13: Effective F -Statistics for the Conduct Test

Models	Effective F					
	2	3	4	5	6	7
<i>Panel A: Economies of Scale</i>						
1. Bertrand	13.7	13.3	13.4	10.7	15.2	13.6
2. Cournot		16.2	15.3	14.8	13.9	11.4
3. B: Asia, C: US, EU			16.9	14.6	10.5	15.6
4. B: US, C: Asia, EU				14.9	15.7	14.7
5. B: Asia, US, C: EU					13.6	14.5
6. B: non-Big 3, C: Big 3						13.6
<i>Panel B: Constant Marginal Cost</i>						
1. Bertrand	24.3	22.9	29.0	18.0	23.5	23.6
2. Cournot		25.5	21.6	25.7	23.4	17.4
3. B: Asia, C: US, EU			22.0	24.4	17.8	23.7
4. B: US, C: Asia, EU				40.4	22.2	20.7
5. B: Asia, US, C: EU					20.8	24.8
6. B: non-Big 3, C: Big 3						22.5

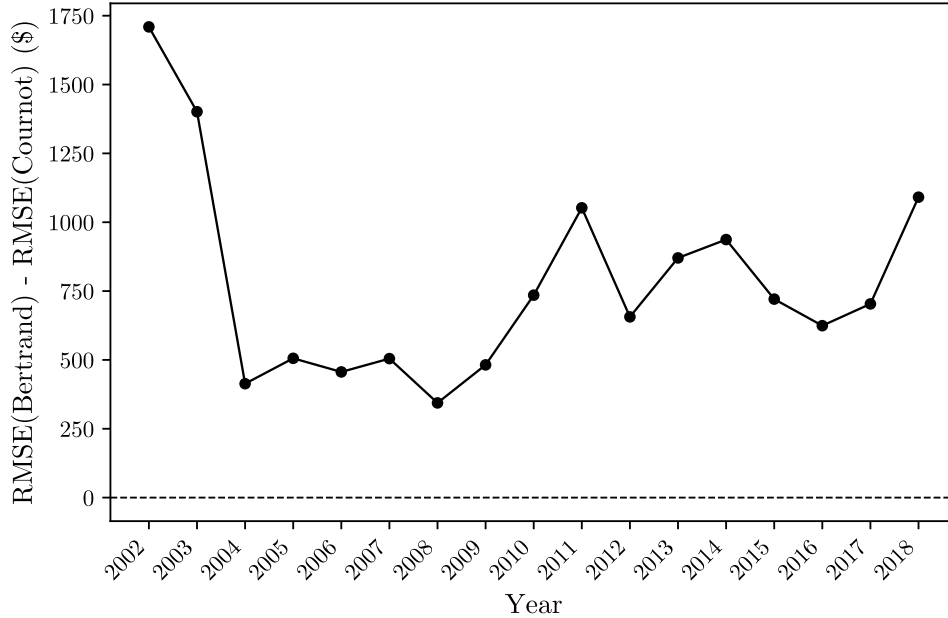
The table reports the effective F -statistics for all pairwise comparisons in the seven-model conduct test of Table 4, under economies of scale (Panel A) and constant marginal cost (Panel B). Every F -statistic exceeds the critical values for both a worst-case size of 0.075 and a best-case power of 0.95.

D.5 Cross-Validation of Preferred Model

To assess the out-of-sample predictive performance of our preferred model, we perform a leave-one-out cross-validation exercise. For each year in our sample, we estimate the marginal cost regression in Equation (5) using data from all other years and then predict prices using the held-out year’s market structure. Figure 2 plots the difference in root mean squared error (RMSE) of price predictions under Bertrand versus Cournot. In every one of the 17 years in our sample, the Cournot model produces lower prediction errors. This

indicates that Cournot predicts held-out prices more accurately than Bertrand within the support of observed year-to-year variation in the data.

FIGURE 2: Cross-Validation: RMSE Difference (Bertrand – Cournot)



The figure plots differences in root mean squared prediction error from a leave-one-out cross-validation exercise. For each year, the marginal cost regression is estimated using all other years, and price predictions are computed using the held-out year's market structure. Positive values indicate that Cournot produces more accurate out-of-sample price predictions. Prices are in dollars.

Appendix E Additional Counterfactuals: Details and Results

E.1 Disaggregating Consumer Welfare and Firm Profits

This appendix disaggregates the welfare effects summarized in Table 6: consumer surplus at the household level, by demographic group and by purchase decision, and profits at the level of individual manufacturers.

Table 14 explores changes in household-level welfare, measured in US dollars, both by demographic type and by the purchase decision made by the household. In C1, averaging over all households, tariffs on imported cars cost \$496 per household; conditional on buying a car, the loss is \$1,416. The welfare losses are larger for high-income consumers, as they are more likely to stay in the market when prices increase: the average loss rises from \$114 per household in the bottom income quartile to \$1,157 in the top. By purchase decision, buyers of imported cars lose \$3,589 on average, while buyers of US-assembled cars gain \$385, as

these cars are sold at lower prices when the prices of foreign cars increase due to economies of scale and Cournot strategic substitution. These gains extend to low-income (\$311) and rural low-income (\$322) buyers of American cars, groups often described as the intended beneficiaries of tariff protection. Comparing C1 to C2 shows the effect of stacked tariffs at the household level: losses roughly double across demographic groups, and, since the cost and prices of US-assembled cars now increase, purchasers of US-assembled cars swing from a \$385 gain to a \$1,597 loss. As the price effects of reciprocal tariffs are minimal, C3 changes these figures little. Table 6 reports the aggregate counterparts of the purchase-type split.

TABLE 14: Counterfactual Results – Household-Level Consumer Surplus

	Change in \$				Change in \$		
	C1	C2	C3		C1	C2	C3
All	−496 (20.6)	−1,125 (29.4)	−1,169 (30.8)	<u>Purchase Type</u>			
				Buy any car	−1,416 (34.4)	−2,629 (41.7)	−2,710 (42.7)
<u>Demog. Type</u>							
Income							
1st Inc. Q	−114 (13.2)	−203 (25.2)	−207 (25.9)	Buy American	385 (43.9)	−1,597 (51.1)	−1,557 (54.1)
2nd Inc. Q	−196 (21.6)	−526 (47.5)	−540 (49.0)	1st Inc. Q	311 (46.3)	−1,332 (51.1)	−1,300 (54.0)
3rd Inc. Q	−553 (35.1)	−1,236 (52.8)	−1,283 (56.7)	1st Inc. Q × Rural	322 (47.7)	−1,090 (56.2)	−1,066 (58.8)
4th Inc. Q	−1,157 (108.9)	−2,621 (205.8)	−2,735 (218.3)	Buy Import	−3,589 (53.8)	−3,984 (42.6)	−4,229 (44.9)
HH Size							
1	−236 (17.2)	−601 (32.2)	−615 (32.7)				
3-4	−672 (30.0)	−1,386 (56.4)	−1,455 (60.4)				
Rural	−420 (22.8)	−1,002 (35.4)	−1,033 (36.5)				
Urban	−507 (20.7)	−1,143 (28.7)	−1,189 (30.2)				

The table reports the average compensating variation across households within a group for counterfactuals C1-C3. The last three columns report changes based on pre-tariff purchase decisions. Standard errors are in parentheses.

Table 15 reports the effect on profits for individual manufacturers. When the tariffs are levied only on imported cars (C1), large importers like VW see disproportionate losses,

while firms with mostly domestic production (e.g., Tesla and Honda) see large gains. The Big 3 automakers (GM, Ford, Fiat Chrysler), with intermediate levels of US production, see only modest losses. Stacking tariffs on parts imported to the US (C2) changes the relative effect on profits across firms: relative to C1, losses accrue for firms with large US assembly and low US parts, including GM, Ford, and Tesla. While reciprocal tariffs on US parts (C3) have a limited impact for most manufacturers compared to C2, Fiat Chrysler, a firm with disproportionate foreign assembly and US parts, sees its losses rise sharply, from 2.56% under C2 to 6.07% under C3.

TABLE 15: Counterfactual Results – Profits by Manufacturer

	# Models	Fraction US		% Change Variable Profits		
		Assembly	Parts	C1	C2	C3
All	314	0.35	0.23	−5.32 (0.57)	−12.28 (1.04)	−12.88 (1.09)
GM	42	0.76	0.41	−1.81 (0.64)	−14.91 (0.91)	−15.39 (0.89)
Toyota	39	0.58	0.44	−10.36 (0.95)	−11.95 (1.86)	−13.96 (1.89)
Ford	26	0.74	0.47	−2.66 (0.51)	−7.60 (1.01)	−7.85 (1.04)
Fiat-Chrysler	27	0.65	0.57	−4.32 (0.68)	−2.56 (0.89)	−6.07 (0.97)
Honda	17	0.91	0.49	+32.31 (0.69)	+18.09 (1.98)	+20.63 (2.07)
Volkswagen	29	0.16	0.11	−56.16 (1.59)	−53.31 (2.06)	−54.93 (2.14)
Tesla	3	1.00	0.51	+14.48 (1.56)	−2.33 (3.21)	−2.01 (3.17)

The table reports, for a subset of manufacturers, the percentage change in total variable profits corresponding to tariff counterfactuals C1-C3. Standard errors are in parentheses.

E.2 Support for Counterfactual Production Changes

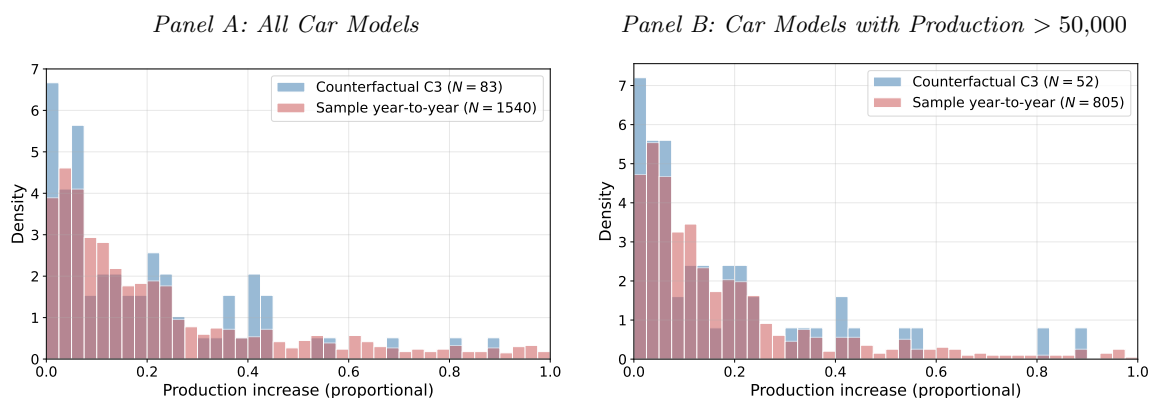
As discussed in Section 6.1, our counterfactual analysis changes production levels for affected car models. To assess whether these changes require extrapolation beyond the support of the data, we compare counterfactual production changes to the year-to-year production changes observed in our sample.

The relevant metric is the proportional change in q_{jt}^p , since this is what enters the economies-of-scale component of the cost function. In counterfactual C3, 83 of 314 models

experience a production increase, with a median increase of 14.7% and a 95th percentile of 108.3%. By comparison, model-level year-to-year production increases in our sample ($N = 1,540$) have a median of 18.6% and a 95th percentile of 245.5%. The typical counterfactual increase is thus smaller than historical year-to-year variation, and even the upper tail remains well within the historical range.

Figure 3 presents the full distribution of production increases for the counterfactual and the sample. Panel A includes all models; Panel B restricts attention to models with worldwide production above 50,000 units, for which capacity constraints are most relevant. In both cases, the counterfactual production increases are well within the range of year-to-year variation observed in the data.

FIGURE 3: Distribution of Production Increases: Counterfactual C3 vs. Sample Year-to-Year



The figure plots counterfactual C3 production increases (blue) against year-to-year production increases in the estimation sample (red). Production changes are proportional changes in q_{jt}^p . Panel B restricts to models with production above 50,000 units.

E.3 Marginal Cost and Markup Decomposition Under Alternative Demand Specifications

Table 16 reports the decomposition of counterfactual price changes into marginal cost and markup components under three demand specifications: the baseline GMY random-coefficients model, GMY without the random coefficient on price (GMY-no price RC), and logit. As discussed in Section 6.3, removing the random coefficient on price leads firms to cut markups more aggressively in response to C1 tariffs (-1.28% vs. -0.16% for foreign-assembled vehicles under Cournot). Under logit, the markup change is similar to GMY, but the induced decline in foreign sales is larger, producing a larger marginal cost increase (30.15% vs. 26.37% of the baseline price).

TABLE 16: Counterfactual Tariff Effect – Marginal Cost and Markup Decomposition

Demand	Assembly	Cournot, Econ. of Scale			Bertrand, Constant MC		
Model		Price	MC	Markup	Price	MC	Markup
C1:							
GMY	US	−1.03	−1.65	0.62	0.47	0.00	0.47
GMY	Non-US	26.20	26.37	−0.16	19.59	19.55	0.04
GMY-no price RC	US	−1.06	−1.47	0.41	0.37	0.00	0.37
GMY-no price RC	Non-US	25.58	26.86	−1.28	19.02	19.63	−0.60
Logit	US	−0.94	−0.86	−0.08	−0.09	0.00	−0.09
Logit	Non-US	29.98	30.15	−0.18	19.44	19.57	−0.13
C2:							
GMY	US	6.55	5.89	0.66	7.40	6.85	0.55
GMY	Non-US	24.29	24.36	−0.07	19.79	19.55	0.24
GMY-no price RC	US	6.22	6.28	−0.06	7.08	6.90	0.18
GMY-no price RC	Non-US	23.69	24.91	−1.22	19.18	19.63	−0.44
Logit	US	10.31	10.90	−0.59	6.70	6.95	−0.25
Logit	Non-US	28.37	28.94	−0.57	19.32	19.57	−0.25
C3:							
GMY	US	6.40	5.70	0.70	7.45	6.85	0.61
GMY	Non-US	28.99	29.04	−0.05	22.72	22.50	0.22
GMY-no price RC	US	6.06	6.10	−0.04	7.12	6.90	0.22
GMY-no price RC	Non-US	27.94	29.41	−1.48	21.99	22.59	−0.60
Logit	US	10.20	10.81	−0.61	6.68	6.95	−0.26
Logit	Non-US	32.85	33.43	−0.58	22.30	22.57	−0.26

The table reports counterfactual results under alternative demand specifications. Price columns show the sales-weighted average effect in percentage terms. MC and Markup columns refer to the change in the respective variable divided by the pre-tariff average price.

E.4 Which Car Models Are at Risk of Exit?

While our main analysis examines medium-run effects holding the product set fixed, manufacturers facing substantial profit declines might eventually discontinue certain models or relocate production. Table 17 examines this possibility by identifying models at risk of discontinuation under tariff scenario C3.

To assess how car manufacturers’ product portfolios could be affected, we focus on two popular segments (SUVs and sedans), and identify segment-specific profit thresholds based on the lowest variable profit levels observed in the pre-tariff market: \$15.1 million for sedans (represented by the Fiat 500) and \$15.9 million for SUVs (represented by the Nissan Rogue Hybrid).⁴⁵ These figures are consistent with other estimates in the literature: Sabal (2025) quantifies median market entry costs for automobiles at \$8-15 million. Models falling

⁴⁵We exclude from this analysis vehicles with less than 2,500 sales in 2018—these may not have been on sale for the full year (e.g., because discontinued), or correspond to very small niches.

TABLE 17: Models at Risk Under Counterfactual C3

Car Model	Profit (millions \$)	Assembly	US Content (%)	Profit Change (%)
<i>Panel A: Sedan Segment</i>				
Honda Clarity PHEV	-0.7	Japan	0	-102
Kia Stinger	0.2	Korea	0	-99
Ford Fusion PHEV	5.6	Mexico	30	-86
Kia Optima Hybrid	6.1	Korea	1	-75
Nissan 370z	6.5	Japan	0	-68
Kia Cadenza	11.5	Korea	0	-62
Subaru BRZ	9.4	Japan	0	-54
Fiat 500	8.2	Mexico	19	-46
Fiat 124 Spider	11.3	Japan	0	-45
Mitsubishi Lancer	9.4	Japan	1	-38
<i>Panel B: SUV Segment</i>				
Nissan Armada	15.2	Japan	5	-90
Toyota Land Cruiser	3.1	Japan	5	-90
Mitsubishi Outlander PHEV	5.4	Japan	0	-71
Kia Niro PHEV	13.5	Korea	2	-39

The table reports models with the largest profit declines under the C3 tariff scenario, showing post-tariff variable profits under C3, assembly location, US parts content percentage, and percentage change in profits relative to baseline.

significantly below these thresholds post-tariff would be candidates for exit or production relocation.

Ten sedan models and four SUV models fall at or below these thresholds under C3. Among sedans, variable profit declines by 102% for the Honda Clarity PHEV, 99% for the Kia Stinger, 86% for the Ford Fusion PHEV, and 75% for the Kia Optima Hybrid. Among SUVs, it declines by 90% for both the Nissan Armada and Toyota Land Cruiser, and by 71% for the Mitsubishi Outlander PHEV. These reductions, and especially the negative profit for the Honda Clarity PHEV,⁴⁶ would likely trigger exit decisions, rather than just price adjustments.

The table reveals two further patterns. First, the most severely affected models are assembled in Asia (primarily Japan and Korea) with minimal US parts content. This group includes niche performance vehicles (the 370z, Stinger, BRZ, and 124 Spider), large SUVs (the Armada and Land Cruiser), and electrified models (the Clarity PHEV, Optima Hybrid, Outlander PHEV, and Niro PHEV). Second, the affected set also contains two Mexican-assembled models whose US-sourced components are exposed to reciprocal tariffs under C3: the Ford Fusion PHEV and Fiat 500, with US content shares of 30% and 19%, respectively.

⁴⁶For the Clarity, the tariff-induced decline in sales raises average variable cost sufficiently that its counterfactual price falls below average variable cost plus tariff payments.

E.5 Reshoring Incentives Under Tariffs

The stacked and reciprocal tariffs in C3 create complex incentives for manufacturers to adjust their supply chains. To explore these dynamics, we simulate a counterfactual scenario where each of the top-selling models increases its US/Canadian parts content by 10 percentage points after the implementation of C3 tariffs, holding all other factors constant. This allows us to isolate the strategic incentives for reshoring parts production on a model-by-model basis.

It is important to note that this exercise should be interpreted as an enhanced back-of-the-envelope calculation rather than a full evaluation of parts sourcing decisions—this is not a margin of decision that is directly captured by our model. Thus, we quantify only the potential benefit of reshoring in terms of decreasing the tariff bill, while abstracting from the costs and constraints that might make domestic sourcing more expensive or even infeasible. In reality, manufacturers source parts globally for numerous reasons beyond cost, including access to specialized expertise, quality considerations, capacity constraints, and technological advantages. Our analysis thus quantifies only one side of the trade-offs involved in the decision to reshore parts production, as it does not account for the potentially substantial costs that led manufacturers to establish global supply chains in the first place.

TABLE 18: Effects of Increasing US Content by 10 Percentage Points Under C3

Car Model	Price Effect(%)		Share Effect(%)		Profit Effect(%)		US Content (%)	Assembly
	C3	Change	C3	Change	C3	Change		
Ford F-Series	6.6	−1.1	−2.9	5.7	−1.9	7.9	53	US
Chevrolet Silverado	5.8	−1.5	4.6	11.3	0.1	13.1	46	US
Ram Pickup	6.0	−1.3	1.3	8.6	0.6	11.0	56	US
Toyota RAV4	31.7	2.5	−67.4	−3.9	−66.1	−3.7	35	J
Nissan Rogue	10.0	−1.8	10.0	9.4	10.3	11.5	25	US
Honda CR-V	13.3	−2.0	−13.9	9.2	−7.6	10.2	23	US
Toyota Camry	3.3	−1.8	20.9	10.5	19.7	11.2	65	US
Chevrolet Equinox	45.4	1.9	−83.6	−1.5	−88.7	−1.2	45	CN
Honda Civic	7.4	−1.4	7.8	5.6	12.4	6.1	48	US
Honda Accord	5.4	−2.1	6.3	12.1	10.5	13.0	65	US

The table reports effects for the top 10 models by 2018 sales volume. For each model, we show the C3 tariff effect on price, market share, and profits (in percentage terms), followed by the change in percentage points when that specific model increases its US parts content by 10 percentage points. Assembly locations are US except for Toyota RAV4 (Japan) and Chevrolet Equinox (Canada).

Table 18 presents the results of this counterfactual for the top-selling models in 2018. Several patterns emerge that highlight the complex incentives created by stacked and reciprocal tariffs. First, increasing US parts content generally leads to lower price increases across most models, as indicated by the negative values in the price change column. For example, the Honda CR-V would see a 2.0 percentage point reduction in its price increase

under C3, while the Toyota Camry would experience a 1.8 percentage point smaller price increase.

However, the Toyota RAV4 and Chevrolet Equinox are exceptions. Despite being among the most impacted models under C3 (with a 31.7% and 45.4% price increase, and a 66.1% and 88.7% profit decline, respectively), increasing their US content would counterintuitively worsen their situation, with price effects increasing by an additional 2.5 and 1.9 percentage points. This stems from the models' foreign assembly location, where the reciprocal tariffs create perverse incentives that penalize the reshoring of parts production.

For market shares, the reshoring effects are generally positive for US-assembled vehicles, with the Honda Accord showing the largest improvement (12.1 percentage points). These share gains translate into profit gains, with all US-assembled models seeing profit enhancements ranging from 6.1 to 13.1 percentage points.

These findings highlight two insights about reshoring incentives under a tariff regime like C3. First, the benefits of increasing domestic content are not monotonic in initial US content levels. For instance, the Honda Accord (65% US content) receives a larger profit benefit (13.0 percentage points) from increasing its domestic content than the Honda Civic (48% US content) with only a 6.1 percentage point improvement. Second, assembly location alters the reshoring calculus. For the Toyota RAV4 assembled in Japan and the Chevrolet Equinox assembled in Canada, increasing US content actually amplifies their competitive disadvantage in the C3 scenario.

In summary, while reshoring parts production could mitigate some negative tariff impacts for certain models, the benefits are unevenly distributed and sometimes counterintuitive. A more complete analysis would need to weigh these tariff avoidance benefits against the fundamental economic reasons that led to global sourcing in the first place. These findings show how the complexity of global value chains can produce unexpected outcomes when disrupted by stacked tariff policies, highlighting the importance of considering these nuanced effects when evaluating trade interventions.

Appendix F Robustness

F.1 Testing with Alternative Instruments

Our main conduct test combines cost, market structure, and demographic instruments with the Gandhi–Houde local differentiation instruments. This appendix isolates what each component contributes. Table 19 repeats the test under three formulations: the main specification in Panel A; the cost, structure, and demographic instruments on their own in Panel B; and those instruments combined with the BLP instruments (computed as within-market averages of rival characteristics) in Panel C. The two product-characteristic sets are

built from the same three demand characteristics (footprint, horsepower, and fuel economy), so the comparison between Panels A and C holds the underlying variation fixed and isolates the functional form of the instruments.

TABLE 19: Conduct Test under Alternative Instrument Formulations

	T^{RV}						
Models	2	3	4	5	6	7	MCS p -values
<i>Panel A: Cost, Structure, and Demographic IVs + Gandhi-Houde IVs – Main Specification</i>							
1. Bertrand	4.37	2.16	3.30	−0.76	1.52	4.25	0.000
2. Cournot		−5.47	−3.79	−4.63	−4.94	−2.37	1.000
3. B: Asia, C: US, EU			0.52	−2.49	−2.24	5.31	0.000
4. B: US, C: Asia, EU				−4.05	−1.34	3.59	0.000
5. B: Asia, US, C: EU					1.79	4.54	0.000
6. B: non-Big 3, C: Big 3						4.86	0.000
7. B: Tesla, C: all others							0.018
<i>Panel B: Cost, Structure, and Demographic IVs</i>							
1. Bertrand	3.34	2.59	1.34	−0.56	1.67	3.41	0.004
2. Cournot		−3.14	−3.31	−3.49	−3.43	−0.97	1.000
3. B: Asia, C: US, EU			−2.19	−2.91	−2.98	3.24	0.003
4. B: US, C: Asia, EU				−1.91	1.01	3.37	0.003
5. B: Asia, US, C: EU					1.91	3.58	0.002
6. B: non-Big 3, C: Big 3						3.62	0.003
7. B: Tesla, C: all others							0.334
<i>Panel C: Cost, Structure, and Demographic IVs + BLP IVs</i>							
1. Bertrand	4.00	2.70	1.69	−0.36	1.45	3.79	0.001
2. Cournot		−3.79	−3.90	−4.16	−4.03	−2.19	1.000
3. B: Asia, C: US, EU			−1.75	−2.99	−3.22	3.40	0.000
4. B: US, C: Asia, EU				−2.09	0.35	3.65	0.000
5. B: Asia, US, C: EU					1.58	3.97	0.000
6. B: non-Big 3, C: Big 3						4.03	0.001
7. B: Tesla, C: all others							0.029

The table reports the RV test statistics T^{RV} and the MCS p -values for the seven models of conduct (economies-of-scale specification): the Gandhi-Houde local differentiation instruments used in the main test (Panel A), the cost, structure, and demographic instruments on their own (Panel B), and the BLP instruments (Panel C). The two product-characteristic sets (Panels A and C) are each added to the cost, structure, and demographic base. A negative T^{RV} favors the row model; a model with MCS $p < 0.05$ leaves the 95% confidence set.

Cournot fits best under every formulation: it is never rejected (MCS p -value of 1.00 in all three panels), and Models 1 and 3 to 6 are always eliminated. The formulations differ only in

whether they can reject Model 7, which diverges from Cournot in the conduct of a single firm, Tesla. The cost, structure, and demographics instruments on their own cannot distinguish these models ($p = 0.334$). However, including product characteristics instruments with either BLP or Gandhi–Houde functional form can ($p = 0.029$ and 0.018 , respectively). These instruments, built to capture the proximity of rival products along dimensions such as miles-per-gallon, horsepower, and footprint, incorporate the local variation that distinguishes Tesla’s products. Accordingly, the pairwise statistic comparing Cournot to Model 7 increases in magnitude from -0.97 under the cost, structure, and demographics set to -2.19 with the BLP instruments and to -2.37 with the Gandhi-Houde instruments, and the 95% confidence set sharpens to the singleton containing Cournot under the latter two instrument sets.

F.2 Alternative Cost Specifications

In Table 20, we explore the sensitivity of our economies of scale estimates to both functional form assumptions and the definition of the total production at which economies of scale accrue. Unless noted, all regressions contain the full set of exogenous cost shifters and fixed effects in Table 12 and we rely on 2SLS estimation using the instruments discussed in Section 5. In our main specification in Equation (5), economies of scale accrue across units of a car model produced in a given country, and we adopt a log-log functional form for marginal cost. Column 1 reproduces our IV estimates, and column 2 reports the OLS estimates: endogeneity attenuates the economies of scale, as expected. Column 3 drops the SUV time trend from the regression, while in column 4, we adopt a log-linear specification of marginal cost, including both total production and its square (the implied scale elasticity from these estimates is -0.21).

In column 5, we allow economies of scale to accrue across the country-level production of all car models using the same platform. A prominent feature of the car industry is that car models in the same segment produced by the same manufacturer often share the same engineering platform, meaning that they use the same mechanical underpinnings, such as engine and transmission (Van Biesebroeck, 2003). Therefore, it is possible that economies of scale and scope accrue at the platform level. The MarkLines data contains information on car platforms, and we are able to construct platform data for around half of the car models in our data.

TABLE 20: Robustness of Cournot Implied Economies of Scale to Alternative Functional Forms

	Dependent variable: $\log c_C$						
	Main	OLS	Drop SUV	Log-Linear	Platform	Foreign Parts	US Parts
	Spec		Trend	Quad	Level	Country	%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\log(q_{jt}^p)$	-0.12 (0.03)	-0.03 (0.01)	-0.21 (0.05)			-0.13 (0.05)	-0.18 (0.05)
q_{jt}^p (100 ths)				-0.25 (0.08)			
$(q_{jt}^p)^2$ (100 ths)				0.02 (0.01)			
$\log(\sum_{j \in \text{platform}} q_{jt}^p)$					-0.17 (0.05)		
RXR Parts Country						0.04 (0.08)	
US Assembly $\times$ Wages							0.01 (0.02)
US Parts %							-0.55 (0.69)
US Parts % $\times$ Wages							0.02 (0.03)
Observations	3,929	3,929	3,929	3,929	3,929	1,790	2,056
Adjusted R ²	0.84	0.89	0.72	0.82	0.67	0.86	0.80

The table reports economy of scale estimates from log marginal cost regression in Equation (5) under the Cournot model of conduct. All regressions include the exogenous cost shifters, including the real exchange rate for the country of assembly and fixed effects reported in Table 12. Column 1 reproduces the main results in Table 3 - total production at the model-country level enters in logs. Column 2 reports OLS results obtained without instruments. Column 3 removes the SUV time trend. In Column 4, total production and its square enter in levels. Column 5 models total production at the platform-model level. Column 6 adds the real exchange rate for the largest foreign source of parts. Column 7 adds US parts percentage and US assembly and parts percentage interacted with sector-specific average wages. Standard errors clustered by car model are in parentheses.

Finally, in columns 6 and 7, we add cost shifter variables that use data on the origin and the percent content of foreign parts. In particular, we construct the following variables: the real exchange rate for the main country of origin of foreign parts for the model-year (RXR Parts Country), an interaction of a US assembly indicator and a US wage index for the sector (US Assembly $\times$ Wages), the % US parts content (US Parts %), and an interaction of the % US parts and a US wage index for the sector. We find that these additional variables do not add meaningful explanatory power, perhaps because the variation they include is absorbed

to some degree in manufacturer fixed effects. The number of observations is reduced in these specifications because the AALA data necessary to construct some variables is only available starting in 2011.

Across specifications that we estimate via 2SLS, economies of scale estimates range from -0.12 to -0.21 , suggesting our main estimates are fairly robust to the functional form of Equation (5) and the definition of q_{jt}^p .

F.3 Alternative Tariff Assumptions

Here, we explore the robustness of our results in Counterfactual 3 to two maintained assumptions when computing our tariff counterfactuals. First, when imposing tariffs on foreign car manufacturers, we modeled the port cost of a vehicle as 80% of its retail price (in line with [Goldberg, 1995](#)). Alternatively, [Coşar et al. \(2018\)](#) estimate the port cost to be 68% of the implied marginal cost. Column 2 of Table 21 reports results in Counterfactual 3 under this alternative assumption.

Second, when levying tariffs on car parts, we assumed that the pass-through of these tariffs to the input prices paid by car manufacturers was one. Columns 3 and 4 of Table 21 report C3 results under alternative pass-through assumptions (0.2 in Column 3 and 1.2 in Column 4). The overall results hold under these alternative assumptions: stacked and reciprocal tariffs increase overall car prices and reduce consumer and producer surplus. Overall, total net surplus is lower with tariffs, and the employment effects imply that the surplus reduction per job created exceeds the average manufacturing wage in these industries.

TABLE 21: Robustness of Counterfactual 3 Results to Alternative Tariff Assumptions

			Parts Tariff Pass-Thru =	
Tariff Effects	Main	Port Cost = 0.68 c_m	0.2	1.2
	(1)	(2)	(3)	(4)
Price Effects (percent):				
All Models	15.32	11.97	10.79	16.49
	(0.43)	(0.38)	(0.31)	(0.46)
US Models	6.40	6.64	0.42	7.94
	(0.25)	(0.22)	(0.13)	(0.29)
Non-US Models	28.99	20.15	26.68	29.58
	(0.81)	(0.70)	(0.78)	(0.80)
Surplus Effects (billions \$):				
Consumer Surplus	-59.65	-53.21	-33.15	-65.62
	(1.57)	(1.33)	(1.00)	(1.76)
Manufacturer Profits	-15.19	-14.05	-7.91	-16.86
	(1.29)	(1.11)	(0.77)	(1.44)
Tax Revenue - car imports	15.63	15.28	13.51	16.20
	(0.47)	(0.30)	(0.49)	(0.45)
Tax Revenue - parts imports	27.60	26.65	6.56	31.86
	(0.14)	(0.11)	(0.06)	(0.15)
Total Net Surplus	-31.60	-25.32	-21.00	-34.42
	(0.97)	(0.91)	(0.57)	(1.06)
Employment Effects (thousands of jobs):				
Car Assembly	42.76	25.12	66.01	37.25
	(2.12)	(1.85)	(1.63)	(2.25)
Parts Manufacturing	1.30	-7.70	28.32	-5.34
	(1.80)	(1.62)	(1.12)	(1.90)

The table reports effects of stacked and reciprocal tariffs in Counterfactual 3 under alternative tariff assumptions. Column 1 reproduces our main results while Column 2 assumes the port cost of a vehicle is 68% of the implied marginal cost. Columns 3 and 4 assume the pass-through rate of parts tariffs to input prices for car manufacturers is 0.2 and 1.2, respectively. Standard errors in parentheses.

F.4 Multimarket Conduct and the Foreign Output Response

Our analysis focuses on the US market, but economies of scale accrue to a model's total production at its assembly country, q_{jt}^p , of which US sales are one component. In our main counterfactuals, we hold foreign sales of foreign-assembled vehicles fixed, so the only changes to global production come from changes in US sales. This simplification would

be innocuous under constant marginal cost, where prices and quantities can be solved independently market by market. In our model, a US tariff that reduces a model’s US sales reduces its total production and, through the estimated scale economy, raises its marginal cost; the higher marginal cost could reduce its foreign sales as well, further increasing its marginal cost. [Bulow et al. \(1985\)](#) discuss this possibility and offer examples of how it works. While their leading examples feature homogeneous products, the coupling here does not require homogeneity; it operates through each model’s own marginal cost.

We explore the consequences of this coupling with a two-market simulation, splitting the world into a US and a foreign market. Because we do not observe sales country by country, we take baseline total production for each model from the data and back out foreign sales as the residual of total production net of US sales (six models have no foreign sales). The exercise is thus restricted to models sold in the US. Without data on foreign prices we cannot estimate foreign demand, so we assume manufacturers face constant-elasticity foreign demand, with elasticity η_F drawn from a range bracketing the mean own-price elasticity of the US demand system (-6.7). We assume that firms play Cournot in the foreign market, as in the US.

We focus on the 25% import tariff (C1), where the coupling between markets is most direct: the tariff bears immediately on foreign-assembled vehicles’ US sales, and hence on the total production that drives their marginal cost, whereas the parts-tariff counterfactuals (C2 and C3) reach the same channel only indirectly, through input costs and foreign parts content. In equilibrium, each model satisfies two first-order conditions coupled through its marginal cost. In the US market, prices and shares solve the same differentiated-products demand and conduct conditions as the main counterfactual, given the tariff-inclusive marginal cost; in the foreign market, constant-markup pricing on the isoelastic residual demand makes the foreign price proportional to marginal cost, so foreign sales respond to the model’s marginal-cost change with elasticity η_F . Total production, the sum of US and foreign sales, determines marginal cost through the estimated cost function. We solve the coupled system by iteration: given total production, the cost function sets each model’s marginal cost; given marginal cost, the two first-order conditions determine US and foreign sales, and hence a new level of total production; we iterate to convergence. At convergence, each first-order condition holds with marginal cost evaluated at the model’s equilibrium total production. We report results for η_F between -4 and -7 ; holding foreign sales unresponsive reproduces the main results exactly (first row of [Table 22](#)).

Endogenizing the foreign response amplifies the effects of tariffs: foreign-assembled price increases rise from 26.2% to between 29.3 and 34.1%, and world production falls more steeply. The direction of each model’s foreign response follows the sign of its marginal-cost change: foreign-assembled vehicles, which lose US sales, reduce foreign sales as well, most

sharply for the models most exposed to the US market; US-assembled vehicles, which gain US sales and lower their marginal cost, expand foreign sales. Aggregating to the firm level, 13 to 14 of the 16 multimarket firms reduce foreign output across the η_F bracket, while 2 to 3 expand it. These results suggest that the fixed-foreign baseline in the main text is conservative: accounting for the multimarket structure makes the tariff’s price effects larger, not smaller.

TABLE 22: Counterfactual Price and Production Changes with an Endogenous Foreign Output Response

Foreign response	ΔP all	ΔP US	ΔP non-US	Δ world prod.
Fixed (main results)	+9.73	−1.03	+26.20	−3.75
$\eta_F = -4$	+10.83	−1.22	+29.29	−6.30
$\eta_F = -5$	+11.32	−1.26	+30.57	−7.40
$\eta_F = -6$	+11.87	−1.34	+32.11	−8.94
$\eta_F = -7$	+12.63	−1.36	+34.05	−11.26

The table reports share-weighted percentage changes in prices under the 25% import tariff (C1); “world prod.” is the percentage change in total production $q^p = q_{US} + q_F$. The first row holds foreign sales fixed and reproduces the main results. A larger $|\eta_F|$ implies a more elastic foreign response, a larger foreign contraction, and stronger amplification.

F.5 Comparison with Standard Specification

In Table 23, we contrast the main results of our counterfactuals predicted under our preferred model of conduct (Cournot with economies of scale) against counterfactual results predicted under the standard model in the literature, Bertrand competition with constant marginal cost. We highlight four aspects of this comparison.

First, under a model of Bertrand competition with constant marginal cost, we find that prices of foreign-assembled cars increase by 19.6% in C1 and 22.7% in C3, while the Cournot model with economies of scale predicts higher pass-through.

Second, the price effects for US-assembled models in C1 further underscore the importance of learning conduct for predicting trade policy. Our preferred model predicts a price reduction, while the Bertrand model with constant marginal cost predicts a price increase. While differences in both the model of conduct and the functional form of cost give rise to discrepancies in the predicted price effects, the differences in cost play a larger role in our application.

Third, the differences in equilibrium effects give rise to differences in predicted welfare and employment effects. In C3, the Bertrand model with constant marginal costs understates the total effect on net surplus by nearly \$3.5 billion.

TABLE 23: Counterfactual Results under Alternative Models

	Cournot, Econ. of Scale			Bertrand, Constant MC		
	C1	C2	C3	C1	C2	C3
Price Effects (percent):						
All Models	9.73	13.56	15.32	8.03	12.29	13.49
US Models	-1.03	6.55	6.40	0.47	7.40	7.45
Non-US Models	26.20	24.29	28.99	19.59	19.79	22.72
Surplus Effects (billions \$):						
Consumer Surplus	-25.30	-57.43	-59.65	-30.50	-59.16	-62.28
Manufacturer Profits	-6.27	-14.47	-15.19	-6.03	-13.25	-13.98
Tax Revenue: car imports	12.92	17.85	15.63	17.88	22.01	20.14
Tax Revenue: parts imports		25.64	27.60		25.42	27.96
Total Net Surplus	-18.65	-28.42	-31.60	-18.65	-24.98	-28.15
Employment Effects (thousands of jobs):						
Car Assembly	73.80	35.11	42.76	52.32	23.79	30.12
Parts Manufacturing	34.81	2.23	1.30	24.67	-8.21	-9.74

The table reports effects of tariffs in Counterfactuals 1, 2, and 3 predicted by our preferred model (Cournot with economies of scale) and the standard model in the literature (Bertrand with constant marginal cost). As there is no uncertainty in the cost function under the constant marginal cost assumption, we suppress standard errors for Cournot with economies of scale (which were reported in Section 6).

Fourth, the point estimates for parts employment differ in sign. In C3, Cournot with economies of scale predicts a gain of 1,300 jobs (standard error 1,800), whereas Bertrand with constant marginal cost predicts a loss of 9,740 jobs. The mechanism is as follows. Under Cournot with economies of scale, tariffs on finished cars cause domestic production to expand substantially; US-assembled car prices fall by 1.03% in C1 as firms benefit from lower marginal costs at higher production volumes. This expansion in domestic assembly increases demand for US-manufactured parts. Under Bertrand with constant marginal cost, the domestic production expansion is more modest (US prices rise by 0.47% in C1), so that in C2 and C3 the contraction in total car sales dominates, producing net job losses in parts manufacturing.

F.6 Sensitivity to the Borrowed Demand System

We take the demand system as given from [Grieco et al. \(2024\)](#); their estimates come from richer data, including consumer-level moments, than we observe. Because demand is estimated, its sampling uncertainty could in principle affect the conduct test. However, they

report bootstrap standard errors but not the full parameter covariance, so formal propagation is infeasible. Since the reported standard errors are generally small relative to the point estimates, our main specification conditions the test on the point estimates. Here we assess sensitivity directly, drawing the borrowed parameters from their published uncertainty and re-running the entire pipeline under various assumptions on the covariance structure.

Specifically, we draw the 22 borrowed nonlinear parameters from normal distributions centered at the published point estimates with the published bootstrap standard errors (Grieco et al., 2024, Table IV), and for each draw re-run markup construction, the economies-of-scale marginal-cost correction, and the RV conduct test across the full seven-model menu. We begin with independence (zero off-diagonal covariance); because the true covariance is unknown, we then repeat the draws under a range of correlation structures, holding the published standard errors fixed on the diagonal: an equicorrelated structure (all parameters share a common correlation), a block structure (the random-coefficient standard deviations and the demographic interactions each correlated within themselves), and a generic, randomly-signed correlation matrix.

TABLE 24: Conduct Under Draws from the Borrowed Demand Parameters

Correlation structure	Coherent	Cournot selected	Cournot sole in 95% MCS	γ (scale)	Mean own- price elast.
Independent (zero cov.)	488 (98%)	99.8%	84%	$[-0.130, -0.118]$	$[-6.85, -6.52]$
Equicorrelated ($\rho = 0.3$)	495 (99%)	100%	88%	$[-0.130, -0.120]$	$[-6.83, -6.48]$
Equicorrelated ($\rho = 0.5$)	496 (99%)	100%	93%	$[-0.129, -0.120]$	$[-6.82, -6.50]$
Block ($\rho = 0.3$)	490 (98%)	99.8%	89%	$[-0.131, -0.119]$	$[-6.85, -6.49]$
Block ($\rho = 0.5$)	492 (98%)	100%	89%	$[-0.131, -0.117]$	$[-6.85, -6.46]$
Generic, random sign ($\rho = 0.3$)	480 (96%)	99.8%	80%	$[-0.131, -0.118]$	$[-6.84, -6.55]$
Generic, random sign ($\rho = 0.5$)	500 (100%)	100%	90%	$[-0.129, -0.119]$	$[-6.89, -6.51]$

The table reports conduct-test results under draws from the borrowed demand system. Each structure draws 500 sets of the 22 borrowed nonlinear demand parameters from normal distributions centered at the Grieco et al. (2024) point estimates with their published bootstrap standard errors, holding the standard errors fixed on the diagonal and imposing the stated correlation structure; for each draw the full pipeline (markups, economies-of-scale marginal-cost correction, RV conduct test and model confidence set across the seven candidate models) is re-run. “Coherent” draws have positive implied marginal cost. The γ and elasticity columns report the 5th–95th percentile interval across coherent draws. Point estimates are $\gamma = -0.125$ and a mean own-price elasticity of -6.72 .

Table 24 reports the results. Across every structure, Cournot is the selected model in at least 99.8% of economically coherent draws and the sole element of the 95% model confidence set in 80 to 93% of them. The estimated scale elasticity and the mean own-price elasticity remain tightly clustered around their point estimates of -0.125 and -6.72 . These

conclusions hold at the stronger correlation level as well: at $\rho = 0.5$ the singleton rate is 89 to 93% across the three structures. A small fraction of draws (at most 4% in any structure) push the variance of the price random coefficient high enough that implied marginal cost turns negative for some products under Bertrand or Cournot—the extremes of the model menu—(implied markups exceed observed prices), so the cost step is undefined; we treat these draws as economically incoherent and exclude them. Robustness of the markups and counterfactuals to alternative demand *specifications*, including a homogeneous-price-coefficient model and logit, is examined separately in Appendix E.3.

F.7 Robustness to Sample Restriction

Our main analysis uses 3,929 model-year observations obtained by matching the [Grieco et al. \(2024\)](#) data with production data from MarkLines. The unmatched observations correspond to models with small market shares, and cumulatively account for roughly 10% of US car sales, on average, across years. To assess whether this sample restriction affects our findings, we perform our conduct test under constant marginal cost—which does not rely on the MarkLines data—using the full sample of 5,046 observations. Table 25 presents the results: our finding that Cournot has the best fit is robust across both samples.

TABLE 25: Conduct Test under Constant Cost Using the Full Sample (5,046 obs.)

Models	2	3	4	5	6	7	MCS p -values
<i>Panel A: RV Test Statistics T^{RV} and Model Confidence Set</i>							
1. Bertrand	8.09	5.82	6.25	−0.15	2.85	7.93	0.000
2. Cournot		−7.45	−6.57	−8.30	−7.83	−3.24	1.000
3. B: Asia, C: US, EU			0.45	−6.20	−6.42	7.16	0.000
4. B: US, C: Asia, EU				−6.90	−2.85	6.29	0.000
5. B: Asia, US, C: EU					3.00	8.16	0.000
6. B: non-Big 3, C: Big 3						7.71	0.000
7. B: Tesla, C: all others							0.001
<i>Panel B: Effective F-Statistics</i>							
1. Bertrand	26.7	22.1	31.4	16.0	21.4	25.9	
2. Cournot		30.8	22.9	29.0	28.6	15.3	
3. B: Asia, C: US, EU			20.9	23.6	18.6	28.9	
4. B: US, C: Asia, EU				45.0	21.5	21.8	
5. B: Asia, US, C: EU					18.7	28.0	
6. B: non-Big 3, C: Big 3						27.5	

The table reports the RV test statistics T^{RV} and the MCS p -values (Panel A) and the effective F -statistics for all pairwise comparisons (Panel B), under constant marginal cost using the full sample of 5,046 model-year observations from 2002 onward. A negative RV test statistic suggests a better fit of the row model. With MCS p -values below 0.05 a row model is rejected from the model confidence set.

F.8 MSRP vs. Transaction Prices

Like most studies of the automobile market, we use MSRP as our measure of price. Dealer-negotiated transaction prices typically fall below MSRP. In this appendix, we assess how much this mismeasurement affects our conduct test.

Under simple-logit demand, markups depend only on the price coefficient α , observed market shares, and ownership. Let $S_{f(j),t}$ denote the aggregate share of the firm owning product j . Then $\Delta_{Bjt} = -\frac{1}{\alpha(1-S_{f(j),t})}$ under Bertrand and $\Delta_{Cjt} = -\frac{1}{\alpha} \left(1 + \frac{S_{f(j),t}}{s_{0t}}\right)$ under Cournot. Since shares are observed, markups are exactly invariant to the price level under either conduct model. Under the [Grieco et al. \(2024\)](#) random-coefficients specification, this invariance is approximate, because individual price coefficients interact with the price level through the heterogeneous utility component.

To quantify the resulting perturbation, we posit that the true transaction price is $p_{jt}^{\text{true}} = (1 - d) \times \text{MSRP}_{jt}$ for a discount d , re-solve the BLP contraction at true prices holding observed shares fixed, recompute Bertrand and Cournot markups, and re-run the Rivers–Vuong test. We first consider a uniform discount across all models. A more demanding possibility is a discount that varies systematically, through price discrimination tied to foreign competition or differences in dealer-network density; we consider two patterns. In the first, the discount differs by firm: a legacy-US-firm discount (GM, Ford, and Stellantis) and a separate foreign-firm discount. In the second, the discount rises with each segment’s foreign-brand share f_s , as $d_j = D f_{s(j)} / \max_s f_s$, so the most import-exposed segments receive the full discount D .

Table 26 reports the conduct test under all three patterns. Under a uniform discount (Panel A) the test rejects Bertrand in favor of Cournot throughout the reported 0–10% grid; even at a 10% discount, $T^{\text{RV}} = 2.24$. Over this range, the discount barely perturbs the test results. On the markup side, at an 8% discount the bias in the Bertrand–Cournot markup difference is only 2.7% of that gap, and is robust to random product-specific discounts. On the cost side, 98.5% of the contamination from the price overstatement is common to both conduct models and is absorbed by the cost regression.

TABLE 26: Conduct Test under MSRP Discounts

Discount	0%	2%	4%	6%	8%	10%
<i>Panel A: Uniform discount</i>						
T^{RV}	4.37	4.34	4.28	4.13	3.58	2.24
<i>Panel B: Discount varies by firm</i>						
Foreign discount 0%	4.37	4.34	4.28	4.17	3.86	2.50
Foreign discount 4%	4.39	4.35	4.28	4.15	3.75	2.41
Foreign discount 8%	4.39	4.34	4.26	4.10	3.58	2.30
<i>Panel C: Discount varies by segment</i>						
T^{RV}	4.37	4.32	4.23	4.03	3.41	2.04

Each entry is the Rivers–Vuong statistic T^{RV} for the Bertrand-versus-Cournot test at true prices $p_{jt}^{\text{true}} = (1 - d_j) \text{MSRP}_{jt}$, using the Gandhi–Houde and cost-side instruments of our main specification with the economies-of-scale cost correction held at MSRP values; $T^{\text{RV}} > 1.96$ selects Cournot at the 5% level, and the 0% column reproduces the Bertrand–Cournot statistic of the main specification (Table 4). The columns vary the uniform discount (Panel A), the legacy-US-firm discount with a separate foreign-firm discount in the rows (Panel B), and the maximum segment discount scaled by each segment’s foreign-brand share, $d_j = D f_{s(j)} / \max_s f_s$ (Panel C).

When the discount varies by firm (Panel B), Cournot remains selected throughout the reported grid for every foreign-firm discount considered: at a 10% legacy-US-firm discount, T^{RV} ranges from 2.30 to 2.50. When the discount varies by segment (Panel C), T^{RV} declines from 4.37 with no discount to 2.04 at a ten-percentage-point discount. Every entry in the table therefore remains above the 1.96 rejection threshold. The conclusion is therefore robust to uniform discounts and to the moderate systematic discount patterns reported in the table.

Appendix G Details for the 2025–2026 Tariff Comparison

This appendix documents the data and implementation behind Section 6.5: how we map the enacted tariff schedule into the effective model-level rates of Table 9; how we collect the realized vehicle prices (the matched MSRP panel and the BLS index) and import quantities; and how we collect equity prices and construct the event study.

G.1 Implementing the Counterfactual on the Realized Policy

The 2025–2026 enacted tariff schedule. Automobile tariffs were enacted under Section 232 of the Trade Expansion Act of 1962 by Proclamation 10908 of March 26, 2025: a 25% vehicle duty took effect on April 3, 2025 and a 25% parts duty on May 3, 2025. The

vehicle duty was additive to the preexisting MFN tariff of 2.5%, so origins outside USMCA generally faced a 27.5% total rate; Korean vehicles, duty-free under KORUS, paid 25%. Despite initial confusion, the proclamation carved out USMCA trade: USMCA-compliant vehicles were dutied only on their non-US content, and USMCA-compliant parts remained duty-free throughout. Bilateral agreements through late 2025 lowered total vehicle duties to 15% for the European Union, Japan, and Korea, and 10% in quota for the United Kingdom. Subsequent litigation struck down tariffs imposed under other authorities but left the Section 232 vehicle and parts duties in force.

Four weeks after the vehicle duty took effect, Proclamation 10925 (issued April 29, 2025) created an import adjustment offset for manufacturers assembling vehicles in the United States: a credit against parts duties equal to 3.75% of the aggregate MSRP of a manufacturer’s US-assembled vehicles. The credit was explicitly sized to match the duty owed when the 25% parts tariff is applied to 15% of a vehicle’s value. The stated purpose of the offset was to provide transitional relief for domestic assembly while manufacturers shift parts sourcing to the United States. Although initially designed to phase out after two years, the offset was instead extended at 3.75% through April 2030 by Proclamation 10984 (issued October 17, 2025), which also added an equivalent credit for US engine manufacturers.

Table 27 lists the legal instruments behind the 2025–2026 tariff schedule (proclamations, executive orders, agency notices, and the controlling court decision), each with its citation and a summary of what it establishes. By November 2025, the schedule had settled at the rates that we use in the counterfactuals below.

Counterfactual implementation. We implement the realized policy within the C2 counterfactual environment. We focus on C2 because the enacted policy taxed imported vehicles and imported parts, while the reciprocal tariffs on US parts exports that define C3 did not materialize: foreign countermeasures largely spared automotive parts. Specifically, Canada’s surtax applied to US-originating motor vehicles, with credit for Canadian and Mexican content, and did not extend to parts;⁴⁷ Mexico did not retaliate against automotive goods, and the European Union suspended its prepared countermeasures ahead of the framework agreement.

To implement our counterfactuals in the 2018 market, we assign each model a tariff rate t_j^v based on its 2018 assembly location. The rate is the December-2025 total rate net of the 2.5% MFN duty already embedded in the model’s 2018 recovered cost (zero for Korea, under KORUS).⁴⁸ This yields 12.5% for the European Union and Japan, 15% for Korea, 7.5% for the United Kingdom, and 25% otherwise. For USMCA-compliant vehicles, we set t_j^v to 25%

⁴⁷United States Surtax Order (Motor Vehicles 2025), SOR/2025-118, Canada Gazette, Part II, Vol. 159, No. 9 (April 23, 2025), effective April 9, 2025.

⁴⁸We use post-agreement rates because model-year-2026 prices were set under them.

TABLE 27: Legal Instruments Behind the 2025–2026 Schedule

Instrument	Citation	Establishes
Proclamation 10908 (Mar. 26, 2025)	90 FR 14705	25% vehicle duty eff. Apr. 3, 2025; 25% parts duty eff. May 3, 2025; USMCA-compliant vehicles dutied on non-US content only; USMCA-compliant parts duty-free pending a Commerce process, never implemented.
Proclamation 10925 (Apr. 29, 2025)	90 FR 18899	Import adjustment offset: 3.75% of aggregate US-assembled MSRP in year one, 2.5% in year two.
Exec. Order 14289 (Apr. 29, 2025)	90 FR 18907	Non-stacking rules across tariff programs.
Exec. Order 14309 (Jun. 16, 2025)	90 FR 26419	US–UK deal: 10% in-quota rate for UK autos from end of June 2025.
Exec. Order 14345 (Sep. 4, 2025); Commerce notice	90 FR 43535; 90 FR 44638	US–Japan agreement: auto rate to the 15% ceiling, effective Sep. 16, 2025.
Commerce notice (Sep. 25, 2025)	90 FR 46136	US–EU framework: auto rate to the 15% ceiling, retroactive to Aug. 1, 2025.
Proclamation 10984 (Oct. 17, 2025)	90 FR 48451	Extends the offset at 3.75% through Apr. 30, 2030; adds an equivalent engine-manufacturer offset.
Exec. Order 14358 (Nov. 4, 2025); USTR notice	90 FR 50729; 90 FR 55964	US–Korea arrangement: auto rate to 15%, retroactive to Nov. 1, 2025.
<i>Learning Resources v. Trump</i> , No. 24-1287	607 U.S. ____ (Feb. 20, 2026)	IEEPA does not authorize tariffs; IEEPA-based duties vacated; Section 232 auto and parts duties unaffected.

The table lists the legal instruments behind the 2025–2026 schedule, verified against the Federal Register (retrieved July 2026); each instrument is available at [federalregister.gov](https://www.federalregister.gov). The Commerce process for levying USMCA duties on non-US content was never implemented, so compliant parts remained duty-free throughout.

of non-US content. Because the AALA reports US and Canadian parts content as a single domestic share, we use the adjusted measure described in Section 2, which reattributes engine and transmission value (12% and 7% of parts content) to Canada for models with Canadian powertrain production. When weighted by 2018 sales, the assigned final-vehicle rates average 15.1% across foreign-assembled models, compared with the 14.4% of customs value collected on vehicle imports in November–December 2025, once all agreements were in force (USITC DataWeb, May 2026).

Parts rates $\bar{t}_j^p$ apply the statutory parts schedule to each model’s imported content, which we know from AALA data. Content from agreement countries is hit by a 12.5% increment (15% for Korea), UK content by 7.5%, other origin countries by 25%, and USMCA-compliant parts are duty-free; we assume that tariffs on the residual imported content are levied at the mean rate of the model’s named source countries. US-assembled models receive an offset $o_j = 0.0375p_j^0$, where p_j^0 denotes observed 2018 MSRP. The dollar offset remains fixed as equilibrium quantities and resource marginal costs change; net parts duties are

floored at zero. Thus, the first-order conditions of C2 become:

$$p_{jt} = \Delta_{mjt} + (1 + \nu_j t_j^v)^{\phi_j} \left(1 + \max \{ 0, \Lambda 0.71 \bar{t}_j^p \mu_j - o_j / c_{mjt} \} \right)^{1 - \phi_j} c_{mjt}(\cdot), \quad \nu_j = \frac{\lambda}{1 - \lambda t_j^v}.$$

Here ν_j converts a duty levied on the port price, the fraction λ of MSRP, into a multiplier on recovered marginal cost. As in the paper, we set $\lambda = 0.8$; at a uniform 25% duty this gives $\nu_j = 1$. We also follow the paper in setting the parts pass-through Λ to 1. At uniform rates $t_j^v = \bar{t}_j^p = 0.25$ with no offset, the equation reduces to the C2 first-order conditions of Section 6, our main counterfactual exercise with uniform 25% tariffs on cars and parts.

Inspection of the BLS import price index for passenger cars (series EIUIR300), which measures foreign sellers' transaction prices exclusive of duties, suggests that manufacturers reduced the port price of vehicles in response to the tariff. The index was flat through March 2025, turned down in the month the vehicle duty took effect, and declined 3.7% cumulatively through May 2026. In a final exercise, we account for this by scaling λ to 0.77.

G.2 Observed Price and Quantity Data

Transaction prices. The BLS new-vehicle CPI (series CUSR0000SETA01: new vehicles, US city average, seasonally adjusted) is a public-domain monthly index; we compute the cumulative change from February 2025 to May 2026. The CPI tracks quality-adjusted prices for the changing mix of vehicles sold.

Import quantities. Import shipments are monthly unit quantities from USITC DataWeb (HTS 8703, all districts): April–December 2025 shipments were 24% below the same months of 2024. These series enter Panel A of Table 9 only; the counterfactuals do not use them.

Manufacturer list prices. Our counterfactuals predict changes in manufacturer list prices, so we compare them to a panel of MSRPs for model years 2025 and 2026. To construct this panel, we collect 2,712 trim-year price observations from manufacturer pressroom releases and price lists and Edmunds.com pages retrieved through the Wayback Machine; we perform independent cross-checks that show that the two sources agree nearly perfectly. Prices exclude destination charges; where a source quotes a destination-inclusive price, we subtract the corresponding fee.

We match trims by make, model, and trim across the two years and compute each model's price change as the mean across its matched trims. Matching isolates price changes on continuing vehicles from mix shifts, at the cost of excluding redesigned and newly introduced vehicles. This broader price panel covers 87 of the 100 best-selling models in

2025 and 78% of 2025 sales.⁴⁹ For the comparison in Table 9, we further restrict the price observations to continuing car models that can be matched to the 2018 product data. After excluding ambiguous assignments and consolidating variants assigned to the same car model, the reporting sample contains 100 unique 2018 car models. We then aggregate both observed and predicted car model-level changes using either equal weights or 2018 sales weights. The reporting sample accounts for 64.4% of 2018 sales; coverage is 80.6% for US-assembled vehicles and 39.5% for foreign-assembled vehicles. Observed MSRP increases are 2.4% under equal weights and 1.6% under sales weights.

G.3 Equity-Market Response

We explore whether the implementation of car tariffs affected the stock prices of car and car part manufacturers. To do so, we construct a sample of daily share prices for 12 foreign manufacturers (BMW, Honda, Hyundai, Kia, Mazda, Mercedes-Benz, Mitsubishi, Nissan, Subaru, Toyota, Volkswagen, and Volvo), 7 US manufacturers (Ford, General Motors, Lucid, PACCAR, Rivian, Stellantis, and Tesla),⁵⁰ and 9 suppliers (Adient, Aptiv, Autoliv, BorgWarner, Cummins, Dana, Goodyear, Lear, and Magna); foreign manufacturers enter through their US-listed shares, or home-market listings when none are available. Daily adjusted prices are from Yahoo Finance. We also collect data on the S&P 500 as our proxy for market returns.

Because the tariff announcements could have moved the entire market, raw returns may conflate the car tariffs with market-wide news; we therefore measure their effect with a standard event-study design (MacKinlay, 1997). For each firm, we first estimate its normal relationship to the market, $r_{it} = \alpha_i + \beta_i r_{mt} + \varepsilon_{it}$, where r_{it} and r_{mt} are daily log returns of adjusted prices for firm i and the S&P 500,⁵¹ over a 120-trading-day window ending 10 trading days before the event; the long window pins down β_i precisely, and ending it early keeps anticipation of the event out of the benchmark. A firm’s abnormal return on an event day is its actual return minus the return predicted by its market exposure, $r_{it} - (\hat{\alpha}_i + \hat{\beta}_i r_{mt})$. We cumulate abnormal returns over the $[-1, +5]$ trading-day window around each event, where event day 0 is the first trading day on or after the calendar event date.

Table 28 reports results for the announcement and effective date of the vehicle tariffs. The pattern is consistent with the exposure channel in the model. Foreign manufacturers and suppliers lose value around the proclamation and effective-date events, whereas US

⁴⁹ All thirteen top-100 models outside the panel are redesign-year or discontinued vehicles with no like-for-like pair, the largest being the redesigned Toyota RAV4 (2.9% of 2025 sales).

⁵⁰ We classify Stellantis as a US manufacturer because of Chrysler’s standing as a Big Three auto manufacturer.

⁵¹ Returns are computed as $r_{it} = \ln(P_{it}/P_{i,t-1})$, where P_{it} is the dividend- and split-adjusted price; the S&P 500 return is computed the same way.

manufacturers, a group that mixes traditional manufacturers with electric-vehicle and lower-import-exposure firms, move little on average.

TABLE 28: Equity-Market Response to 2025 Auto Tariff Events

Event	Group	N	Mean CAR $[-1, +5]$	t
Section 232 proclamation, Mar. 26 2025	Foreign manufacturers	12	-8.6%	-11.43
	Suppliers	9	-3.3%	-3.62
	U.S.-listed manufacturers	7	+1.1%	0.55
Vehicle tariff effective, Apr. 3 2025	Foreign manufacturers	12	-9.5%	-5.88
	Suppliers	9	-5.0%	-2.15
	U.S.-listed manufacturers	7	+0.2%	0.04

The table reports cumulative abnormal returns (CARs) from a market model with the S&P 500 as the market return. The t statistic divides the group mean CAR by its cross-firm standard error.

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